Alpha↔Omega Dynamics (AΩD)

The Hidden Temporal Dynamics of Stokes

The Art Of The Leprechaun: Fractal Calculus – $\mathfrak{S}$

43 °c

Manual

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$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

Epitaph

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

Comprehension, cooperation, and presence are not prerequisites of AFC.
Nuff said.

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1 Scope

This manual carries application and prediction-test material downstream from the main AΩD note. The main note carries the AF/AFC/AΩD calculus. This manual carries calculation records, setup diagrams, comparison targets, external-comparison records, error audits, and audit tables.

1.1 Use of this manual

The manual uses the same section / subsection / subsubsection structure as the main note. Rest-energy calculation records are organized under §6. External comparison material remains downstream and quarantined from the AΩ-internal scoped-flux calculation.

1.2 Simulation reading route

Follow Main in its dependency order: specific distinction, closure and recurring wave, motif/field folding, complete wave state and relational time, then Stokes/SADAR/harmonic values. Manual Scope is followed by the consolidated native fixtures, the D.E.C. ledger and the retained applications.

1.3 Calculation spine

The manual uses one active calculation language. A calculation declares

$$B = (\partial_T F, \omega).$$

It forms retained cycle provenance, evaluates branch/hinge/branch, and computes a SADAR value:

$$C_f[\Sigma]_{A\leftrightarrow\Omega}, \qquad \mathsf{AO}_\mu(C) = \langle \Pi_\alpha, H_\mu, \Pi_\Omega \rangle, \qquad \mathfrak{S}_B(C) = p^D_{C,A\leftrightarrow\Omega} A_{C,A\leftrightarrow\Omega}.$$

The boundary value is stored by

$$\Phi^D_{\partial_T R} = \sum_{e\in\partial_T R} \mathfrak{S}_B(C_e).$$

Tables store evaluated values. Tables are not the calculus.

1.4 Worked-calculation release

A manual worked record carries a complete row: setup, declared AΩ calculation, comparator or target, residual/error coordinate, and audit.

For an exact AΩ-internal fixture, form

$$\text{setup} \to A \leftrightarrow_\mu \Omega \text{ calculation} \to \text{fixture comparator} \to \Delta^{\mathbb{Z}}, \delta_3 \to \text{audit}.$$

For an external comparison calculation, form

$$\text{setup} \to A \leftrightarrow_\mu \Omega \text{ calculation} \to \text{observable / target record} \to \Delta, \epsilon, z, \sigma \to \text{error audit}.$$

The symbols Δ, ϵ, z, and σ belong to the external comparison layer. A z-score is formed only when target uncertainty is declared.

For a benchmark-dataset calculation, record

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \quad \mathcal{D}_{\text{bench}}, \quad \mathcal{Q}_{\text{required}}, \quad \mathcal{S} = \text{benchmark manifest}.$$

Registry rows carry benchmark calculations until the declared dataset, required coordinate, report map, and audit fields are supplied.

Placement rule. A worked record carries a complete simulation or comparison row. Registry rows carry declared setup, target, comparator, residual, uncertainty, and audit fields as they are supplied.

Specification discipline. A rest or prediction record may use a compact curling-curl specification only after $HT(K; B)$ is declared or inherited.

1.5 Table discipline

Tables in this manual store calculated values, target values, residuals, and audit records. Manual rows, ledgers, tables, records, registries, and fixtures are document containers. They record declared values. The calculation is the declared boundary computation

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega),$$

followed by the AΩ and SADAR calculation on that boundary.

1.6 Provenance-label policy

Tokens of the form `main:<anchor>` and `manual:<anchor>` are cross-document provenance anchors. They refer to the corresponding source label `<anchor>` in the named document. Reader pointers render the published equation, section, appendix or table number from that label; the full source keys remain in macro arguments or source metadata. Explicit alias labels declare alternate anchors. Provenance records remain internal-reference lists; external sources belong in Literature note records.

1.7 Exact arithmetic and presentation decimals

AΩ-internal calculations use integer counts and exact rational ratios. Decimal values appear only as external comparison coordinates, presentation values, or plot labels.

$$\text{AΩ side: integers / fractions,} \qquad \text{external side: value } \pm \text{ declared uncertainty.}$$

If an external measured decimal is used as a comparison target, the comparison record declares its uncertainty σ when a z-score is formed. Percent errors may be shown as presentation comparisons when the target value is declared.

1.7.1 Exact-fixture corollary

If an exact fixture declares

$$T_k^{\times}, O_k \in \mathbb{Z},$$

then record the exact ternary signed residual

$$\Delta_k^{\mathbb{Z}} = T_k^{\times} - O_k, \qquad s_k^{(3)} = \operatorname{sgn}(\Delta_k^{\mathbb{Z}}) \in \{-1, 0, +1\}, \qquad m_k = |\Delta_k^{\mathbb{Z}}| \in \mathbb{Z}_{\geq 0}. \tag{1}$$

Thus

$$\Delta_k^{\mathbb{Z}} = s_k^{(3)} m_k. \tag{2}$$

The compact residual package is

$$\delta_{3,k}(T^{\times}, O) = (s_k^{(3)}, m_k), \qquad \delta_3(T^{\times}, O) = \big(\delta_{3,k}(T^{\times}, O)\big)_{k \in K_{\text{bin}}}. \tag{RG10}$$

Normalized decimal summaries belong to declared value maps or external comparison entries.

SYM-A5 checks integer-fixture residuals (RG10). RG-A10 forms $T_k^{\times} - T_k^0 = \sum C_{ij,k}^{\times}(B)$ for declared exact integer fixtures. RG-A11 records $\Delta^{\mathbb{Z}}$ and δ_3 using (1)–(RG10).

Decimal precision record. A released decimal presentation value carries

$$(x_{\text{disp}}, u_x, \text{unit}(x), \text{src}(x), \text{round}(x), \text{scope}(x)).$$

Here $\text{scope}(x) \in \{\text{internal}, \text{external}, \text{measured}, \text{derived external}, \text{value map}, \text{demonstration}\}$. The precision field u_x records declared display uncertainty.

1.8 External-comparison quarantine

External constants, measured-sector units, and external comparison symbols are quarantined. They can compare against AΩ outputs in a comparison record, but they do not enter the AΩ calculation and do not set AΩ notation. In this manual, an external comparison is written

$$\mathsf{Q}_{\mathrm{ext}}(\nu_{x,A\leftrightarrow\Omega};\mathcal{T}_x),$$

where $\mathcal{T}_x$ is a declared measured-sector target record.

1.9 Scoped errors by data-support class

A simulation or comparison is classified by the declared data support it instantiates. Dataset resolution selects the data-support class: radial rows instantiate G0; radial rows plus proxy density/current fields instantiate G1; three-dimensional density/velocity/current fields instantiate G2; time-ordered tracer positions/velocities instantiate G3. Each class reports residuals in its declared coordinate. Projection and marginalization maps carry their own comparison coordinates and uncertainty records.

The scoped-error chain is

$$D \to R(D) \to \mathcal{M}(D,R) \to \Pi_{\mathrm{report}} \to \epsilon_{\mathrm{report}}.$$

Here D is the declared data-support distinction layer, $R(D)$ is the induced relation family, $\mathcal{M}(D,R)$ is the AFC/AΩD model evaluated on that cut, Π_{report} is the report or projection map, and $\epsilon_{\mathrm{report}}$ is the residual relative to the declared output target. Every manual comparison declares its data-support class before residuals are interpreted.

Simulation data card. Each simulation or comparison row records

$$(D,\ C_{\mathrm{support}},\ M_{A\leftrightarrow\Omega},\ \Pi_{\mathrm{report}},\sigma_{\mathrm{input}},\ \epsilon_{\mathrm{report}},\ \mathrm{src}).$$

Here D is the declared input data, C_{support} is the data-support class, $M_{A\leftrightarrow\Omega}$ is the AFC/AΩD calculation or map, Π_{report} is the reported coordinate, σ_{input} is the declared input uncertainty when present, $\epsilon_{\mathrm{report}}$ is the residual/error coordinate, and src records the data source or source artifact.

1.10 Manual record forms

Table 1: Manual record forms.

Record form	Active fields
AΩ field record	support, carried cycle, pressure/accessor values, and retained AΩ value
Measured target record	target value, declared uncertainty, measured-sector coordinate, and citation
External comparison record	map value, target, residual/error coordinate, uncertainty source, and audit hook
Observable-map record	observable-map value, target, residual, percent error, citation, and audit hook
Exact fixture	comparator bins, $\Delta^{\mathbb{Z}}$, δ_3, closure residual, and sheddic-route audit
Benchmark manifest	dataset table, required report coordinate, input data sources, and audit hook
Calibration anchor	convention value and convention-audit residual

Construction and record classes. Native fixtures establish exact typed relations; declared models specify their chosen update/coupling law; metrology records add calibration and units; empirical comparisons additionally bind external data and residuals. Admission precedes all four projections. The common construction is distinction, retained relation, admitted motif/closure, recurring wave, scoped field organization, complete state/trace and native comparison. A field resolves wave identities and is not retained closure lineage itself. Motion is the later realized pressure-bearing transition. Modular coordinate phase in the 3D applications is a downstream statistic; sampling interval is an acquisition coordinate. Native closure counts and RCD are not measured time.

2 Native closure, wave and temporal fixtures

The executable motif and fold fixtures use a declared preassembled-prefix realization: component closures complete before the target trace starts. This is a sufficient causal realization of the Main's more general commit-before-consumption rule. Each outer trace binds its inner identity, certificate set and scope map before certification. A further distinct outer return supplies the outer beat; exposure preserves the bound inner wave. Duon fixtures separately execute the twelve support signatures, carried current, distinct recurrent returns and the Duad's seated state of the same identity. The finite harmonic readout declares a cyclic index group and committed window; every recorded state contains only the certificate, cadence and phase prefix available at that return. Its projection is a declared model record. In the executable compatibility record, `edges` counts phase-bearing atomic traversals and `phases` counts complete certified closure-phase cycles. Thus the minimal two-edge return has $(2, 1)$, while the four-edge return has $(4, 1)$. The legacy key `phases` is retained to avoid an engine-schema break; it never counts edges, dwell, or Fourier coefficient phase.

Record class and execution. These are native exact witnesses or explicitly declared models. The shared runner is `manual/scripts/aod_native_closure_wave.py`; its five CSV inputs preserve trace, scope, lineage, expected exact result and hostile case. The command `python3 manual/scripts/aod_native_closure_wave.py --check-data` executes the rows. An expected rejection is a successful hostile check only when the actual law rejects the supplied input. Actual results are recorded separately from these immutable inputs.

Construction	Exact witness	Hostile control
Null/anchor	Five distinct types; first 00_8	Prior coordinate choice or type collapse
Atomic support	H1 legs, composite H2 and closed H0 endpoints	Direct H2 called atomic
Hinge/cycle	$\varepsilon 0_H (0_D)^m (-\varepsilon)$, $m = 0, 2$, one closure-phase cycle	Bypass, same-sign end, replay recount
Reflected legs	$1+1=2$, $1+3=4$ phase-bearing steps; unit embedding $1_C \mapsto 2h$	One-step trace; unequal-leg factorization; support bound without closure witness
Recurrence	Three bound closures give two beats, separations $(10, 10)$	One closure or repeated certificate
Kernel	Weights $(1, 1)$ or $(1, 3)$ on admitted support	Bypass mass; empty normalization
Motifs	Dion, two-Monon Dimonon, 12 Duon variants, Duad state	Collapsed names or changed certificate bytes
Fold/exposure	One outer closure-phase cycle, retained inner wave	Contact called closure; lower beats reused
Temporal pair	Ratio, bias, unwrapped lag, directional RCD	Aliases, one contact, zero denominator
State/update	Equal temporal view, different pressure; later response	Future input or projection/state collapse
Flux/Stokes	AFC 0 or 1/2; AOD 7, boundary 3	Pressure in normalized memory; wrong reverse sign
FRG	Independent 5, cross 3, recurrent 8	Model before admission; same-step response
Harmonics	$(1, 0, -1, 0)$ reconstructs exactly	Unbound word gains admission

Conservation	Active closure-phase-cycle equation binding	Mapping that source to an unrelated Fourier equation
Report	$Nn = 3(2/3) = 2$, $4(2/3) = 8/3$	Missing calibration; $3n$ used for $N = 4$

Prefix validation and scalar storage. Each construction record stores its exact prefix digest and witness. Validation checks those bindings against the declared registry, including the closure, return, motif and both contact prefixes of Main (13), Main (20), Main (33) and Main (55). Unknown, empty or mismatched records reject before model or harmonic use. A registry or reconstructed cache validates an existing construction; it cannot originate distinction or recurrence. An implementation declares its update law and admissible input domain before executing Main (54). The additive duration implementation stores $(RD, r_1, \ldots, r_m)$, with $r_s \geq 0$, then applies Main (58); reversal recomputes the reflection counts. A polyrhythm calculation follows cycle provenance first, then stores scalar AΩ accessors. A raw duration cannot replace the branch/hinge/branch data needed by later coupling, seat compatibility or shedding. The scalar outputs are derived caches of their bound constructions.

Directional count realization. Both wave identities and repeated contact-prefix digests are bound before comparison. With support $RD = 3$ and two reflection counts, the declared additive interface gives $\rho^D_{a|b} = 3+1+1 = 5$ and $\rho^D_{b|a} = 3+2+2 = 7$. The ratio reverses reciprocally, signed offsets reverse, and the directional duration is recomputed. A same-domain window of two clips only participation to two; it changes neither RD nor RCD. The native count comparator has $\Lambda = N_a - N_b + b_a - b_b$, with drift formed against the preceding committed comparison. A whole-closure slip changes this unwrapped value even when a separately wrapped display repeats.

Interface composition. A recurrent $8 \to 9 \to 8$ realization initializes at vertex 8 and composes with the committed-return readout. Primitive Duon components and targets share their resolution; cross-scale composition uses the explicit folding map. The comparison takes precisely two rational offsets; contact commitments, window endpoints and update cuts are integers. Contacts use closed window support $[u, v]$, while return counts use $(u, v]$. A lineage retains its ancestor identity under a fixed coordinate relabelling; a substituted ancestor is rejected. Every outer certificate carries the admitted inner waves and their bound scope map. The controls reject missing fold provenance, unrelated-scale pivots, malformed windows and fractional cuts, preserving rational currents. These controls run with `python3 -m pytest -q tests/test_native_closure_wave.py`.

Calibrated report boundary. The emitter/mirror record uses $N = \Delta\mathrm{bip}_{\mathrm{return}}$ and $x^{\mathrm{mir}} = Nn$. The displayed $x = 3n$ is restricted to $N = 3$. A metrology record declares measurand, protocol, reference realization, coupling/probe, detector, calibration, unit, resolution, uncertainty, repeatability and traceability. The rational multiplier in this demonstration is a declared report calibration, not an empirical calibration or native time unit.

Provenance. Main (3); Main (8); Main (13); Main (16); Main (20); Main (22); Main (23); Main (24); Main (27); Main (33); Main (38); Main (41); Main (64); Main (56); Main (71); Main (100); Main (139).

3 Retained-relation operativity and its applications

3.1 Exact model and matched state

These witnesses instantiate the Main state, retention and matched causal criterion in Main (155), Main (156) and Main (157). They are operational finite models. All inputs and expected results are in

`manual/data/m2/awareness_cases.jsonl`; the shared runner is `manual/scripts/aod_awareness.py`. From the source root, run

```
python3 manual/scripts/aod_awareness.py --check-data
```

Actual state and outcome traces are separate execution evidence. Expected answers, row names and passive observations are never arguments of the response law.

The model instantiates Main (155). Its present-record memory and retention placeholders are empty; M^C stores return identities and R alone stores operative prior rebalancing. The world validates admitted waves, issued actions and prior-operand commitments; it cannot recreate a removed residual for the update. No passive record or cache may restore a removed value; retained bytes, fixture verdicts and role labels are excluded from the outcome. Prior provenance also binds any action witness. An operation-count proxy records work, not operativity.

3.1.1 Rational response and continuation

Setup. Consider the following rational response realization on already admitted support, with $p > 0$, gain g, available source set D_k, and explicitly perturbative query q_k:

$$J_{k+1} = J_k + \frac{A_k + q_k + g \sum_{e:\, i_e \in D_k} r_e}{p}. \tag{3}$$

The term $J_k + A_k/p$ specializes Main (101); A_k is a scalar attention component, separate from r_e. Its channel construction and $AO = -OA$ remain fixed.

Derivation. Subtract the two matched responses on common source/event support, extending absent summands by zero:

$$J_{k+1}(R) - J_{k+1}(R') = \frac{g}{p} \left(\sum_{e:\, i_e \in D_k} r_e - \sum_{e:\, i_e \in D_k} r'_e \right). \tag{4}$$

A nonzero right side therefore proves a later current difference for this model. At $J_k = A_k = q_k = 0$, $g = 1$, $r = 1$, the retained and removed cases give $(1, 0)$ for $p = 1$, and $(1/2, 0)$ for $p = 2$. These are two separate pressure families, each with its own matched retention comparison. Zero gain or zero effective retention gives no such witness.

Specialization. Let present state include a Q4 vertex v and fibre role $\epsilon \in \{-1, 0, 1\}$. Two candidate continuations go to $v \oplus 1$ and $v \oplus 2$. A branch enters the hinge; from the hinge the two candidates carry the opposite branch choices. Both obey the inherited H1 and fibre rules. A declared sign selector chooses the first candidate when $J_{k+1} > 0$, otherwise the second. The selector consumes an already admitted finite support; it does not create H1 or certify closure. Every such continuation contributes zero new closure-phase cycles until its separate retained-start return is certified.

Falsification. A difference in current pressure, law or future exogenous input invalidates a retention-only comparison. Recounting R itself as the outcome, recovering a removed value through another channel, using an unbound phase type, or allowing a same-cut effect also fails the criterion. A label attached to either execution cannot change $\mathcal{U}$.

Provenance. Main (5); Main (8); Main (54); Main (100); (3); (4).

Response executions. The target has four completed returns separated by ten native indices. The ordinary peer has separations $(7, 7, 7)$. Repeated native contacts bind both prefixes and the signed count-lag comparison. Exact independent and cross contributions (2) and (1) yield the retained FRG residual $r = 1$. The prior contact cut is 23, retention follows at 24, the complete matched boundary is 33, and the first changed response is at 34. These are native commitment indices, not time measurements.

Pressure	Retained r	First current	Removed current	Restoration
1	1	1	0	Exact state and outcome recovery
2	1	1/2	0	Exact state and outcome recovery
1, gain 0	1	0	0	No causal witness
1	0	0	0	No causal witness

Each row applies Main (157) in its pressure family; changing pressure across a pair is rejected. A passive trace of the same execution leaves all state and event bytes unchanged. Ordinary attention $A = 1$ with empty R changes current but supplies no retention effect. A zero-gain retained residual likewise fails the matched operativity criterion.

3.2 Hostile state, lineage and timing controls

The hostile rows reject changed present pressure, closure memory, update law or future input; duplicate retention inside X; unknown lineage; a future or noninteger commitment; and a retained value detached from its prior operands. The executable controls additionally remove motif certificates, fabricate cadence or phase count, reverse carried orientation, change the return readout, and forge an issued action. Recommitting altered bytes in the diagnostic registry does not bypass the structural type and causal checks.

A relational-phase control uses three actual source lineages. Permuting the three retained lag values preserves their marginal multiset but destroys the source-specific native comparisons. Tau, normalized-circle and old selector coordinates are rejected as substitutions for `M1:unwrapped-count-lag`. The comparator has exactly two declared bias operands. No phase label or scalar memory value supplies admission. Every selected response is an H1 edge with a lawful fibre transition. Uncertified continuations add zero closure-phase cycles, including when their endpoints happen to coincide.

3.3 Operative boundary, labels, action and learning

Setup. The operative boundary is Main (159). Sensor, register, CPU and vocabulary models reuse this interface.

Derivation. Two retained source contributions produce a three-node operative subgraph: two source interfaces and the target. The target's role is SELF, its active contact interfaces are BETWEEN, and unconnected nodes are NOT-SELF. Null remains separately typed. With current outcome $O = J$, nonzero gain and both source channels available, consider the existing instance $J_k = A_k = q_k = 0$, $p = g = 1$. Contributions $+1$ and -1 cancel in total while each removal changes later current by (4): the full result is 0, removal of the $+1$ channel gives -1, and removal of the -1 channel gives $+1$. Both source lineages therefore remain operative. This aggregate-zero state, a provenance-bound zero-valued retained channel, and structural absence have different records and digests. Neither zero-valued case forces a fold or a positive sheddic result. The antisymmetric pressure ledger also retains both channels. A CPU interface uses a declared cost of three operations per available source; that resource proxy alone supplies no operativity verdict.

3.4 Action return and persistent learning

Setup. An issued action binds an immutable identity to its issuer, native commitment, command and channel. Both issuer and channel must already have admitted recurrent prefixes at issuance. A proprioceptive consequence must return on that exact action lineage with its own certified closure. An observed value is supplied by a declared projection of the returned trace, not a free scalar attached to an unchanged return. In a native count realization, write

$$a = (\mathrm{id}, i, k_a, u, \mathrm{channel}), \qquad y = \#\{\text{atomic traversals in the bound return}\}, \qquad r_a = y - u. \tag{5}$$

The command u and readout y share this declared count domain. They are not measured physical quantities. The action precedes the return, the retention event follows its committed input, and a later transition consumes r_a. The action's immutable identity prevents conflicting command records from masquerading as the same issuance.

Derivation. For command $u = 1$ and a certified two-traversal return, $y = 2$ and $r_a = 1$. The matched response difference is then g/p. Uncommanded sensing, motor output alone, a return from another action or an unrelated sensor magnitude cannot establish this proprioceptive relation. For the single acquired relation $r_a = 1$, with its source available throughout, $A_k = q_k = 0$ and $g = p = 1$, the retained model advances current by one per transition, while the matched removed model retains its current value. Restoration of the acquired coordinate restores the later increments.

Falsification. A surviving record with no later behavioral change is persistence alone. A changed stimulus cannot be attributed to memory by an unmatched comparison.

Provenance. Main (13); Main (156); (4).

Action and learning executions. The sensor–motor witness issues its command at 13, after the issuer and channel have acquired recurrent prefixes at 12 and 9. Its certified action return occupies $14, 15, 16$, with exactly two atomic traversals. The readout is the declared traversal-count projection $y = 2$, the command is $u = 1$, and the retained residual is $y - u = 1$. The same certified return cannot be assigned an arbitrary readout of 200. Wrong-action, unissued, wrong-return and future-return cases fail. Passive sensing and motor output alone do not establish proprioception.

The separate learning witness uses the already retained relation supplied by `core()`, without calling action-return acquisition. It retains that relation through four zero-stimulus updates, reaching current four. At a new matched boundary, retained execution then yields currents $(5, 6, 7)$; removal yields $(4, 4, 4)$. Exact restoration of value and lineage restores $(5, 6, 7)$. The future attention inputs remain zero in each family, so a changed stimulus does not explain the difference.

3.5 Polycadence, crescendo, exposure and a second causal link

The same retention law executes with locked cadences $(10, 10, 10)$, a declared nonzero comparator bias $1/4$ at equal cadence, unequal cadences $(10, 10, 10)$ and $(7, 7, 7)$, or jittered separations $(8, 7, 13)$. Each positive retention pair has a later effect. Locked cadence with zero gain does not. The nonzero bias is an explicitly declared native comparison coordinate bound to its two operands; it is not circular phase or metrology.

The availability sequence exposes one, two, three, zero, then three source lineages. The number of operative sources is exactly $(1, 2, 3, 0, 3)$, while all three valid retained records survive dropout. This measures changing causal participation, not a scalar amplitude or a synchronization score.

3.5.1 Response-to-load bridge and restoration

Setup. The executable toy separately declares a response-to-load bridge:

$$L^{\mathrm{toy}}_{k+1} = C^{\mathrm{cap}}_{k+1} + J_{k+1}, \qquad C^{\mathrm{toy}}_{\mathrm{close}} = C^{\mathrm{cap}}_{k+1}, \qquad \Delta^{\mathrm{toy}}_{\mathrm{close}} = L^{\mathrm{toy}}_{k+1} - C^{\mathrm{toy}}_{\mathrm{close}}, \quad X^{\mathrm{toy}}_{\mathrm{shedding}} = \max(0, \Delta^{\mathrm{toy}}_{\mathrm{close}}). \tag{6}$$

This model-specific load does not rename J as SADAR pressure. A positive record binds source state/event, outer certificate, operands, route, target, order and digest before exoshedding exposure. Zero means no shedding; negative residual is deficit/open. Status labels compute neither result.

Derivation. With $O = O_{\mathrm{shed}}$ projecting that computed record,

$$\mathrm{Op}_e^{O_{\mathrm{shed}},H}(S_k) = 1 \quad \wedge \quad \mathrm{Restore}_e(S_k^{-e}) = S_k \tag{7}$$

specializes the causal criterion to computed shedding with exact restoration. Shedding alone is an M1 consequence; an operative effect without surplus has no shedding claim.

Falsification. A contact, lone outer return or recurring wave cannot replace the fold and causal witnesses of Main (41), (7), and the reapplication of Main (157) at the outer scope of Main (160).

Provenance. Main (107); Main (108); (6).

Fold and shedding executions. In the contact/fold witness the first retained response at 34 selects a later outer-construction opportunity. Each outer closure binds the actual inner certificates and gives one complete closure-phase cycle. Three distinct outer returns permit recurrence and two distinct higher contacts. The B1 comparison peer also folds the same admitted inner pair, with three returns separated by nine native indices. Its certificates carry those inner identities, certificates and scope map; an unsupported outer tag is rejected. Exposure during outer closure preserves both inner wave identities without making a shedding claim. For the positive retained lane only, a declared exact toy bridge binds the committed response event to closure load $L^{\mathrm{toy}} = C^{\mathrm{cap}} + J$ and closure cost $C^{\mathrm{toy}}_{\mathrm{close}} = C^{\mathrm{cap}}$. It does not identify current with SADAR pressure. The resulting exact residual and surplus are both 1; the computed record binds the source state/event, outer certificate, inner lineages, exoshedding route back to their $B0$ scope, commitment order, and operand digest. Only that record authorizes sheddic exposure, which preserves both inner wave identities. Replacing the fold status by the string `shedding` is rejected. The matched removed lane has no fold and no shedding record; exact restoration reproduces the complete record byte for byte. The retained inner relation is then carried with the exact fold witness and higher contact provenance. At the higher scope, retention is committed at 56, the new intervention boundary is 58, its changed response is at 59, and the next outer fold starts at 60. Removing the higher retained coordinate blocks that offer; restoring it restores the same admitted construction. A local coincident trace fails the outer-fold and cross-scale claims. Action-origin retention also carries its authentic source action registry and commitment evidence.

3.6 Passive observation, interrogation and operational mindfulness

Setup. A passive observation reads a committed state after the transition and is not an input to $\mathcal{U}$. Interrogation is instead an explicit perturbation q_k, with the observation on which it depends already committed through k. Queried and unqueried inputs are distinct perturbation families; an observer record alone cannot change execution.

Derivation. For this specialization, the operative boundary of Main (159) is established by the matched causal criterion and remains stable over the observation window. Consider a reflexive query $q_k = -J_k$, with current observed at k and response at $k+1$. For constant forcing $b = A + g\sum r_e$, (3) becomes

$$J_{k+1} = \left(1 - \frac{1}{p}\right) J_k + \frac{b}{p}, \qquad J_{k+1} - b = \left(1 - \frac{1}{p}\right)(J_k - b). \tag{8}$$

Its constant-forcing convergence holds exactly when $p > 1/2$. At $p = 1$ and $b = 1$, every subsequent current is exactly one while the declared H1 and hinge task continues. An extra delay using J_{k-1} instead has a different recurrence and can oscillate. Its query retains the source cut $k - 1$; the initial history value is declared before the first query. Variable forcing cannot silently inherit the constant-forcing conclusion.

Specialization. Operational mindfulness denotes this stable reflexive re-addressing of an already operative state of this model while task dynamics continue.

Falsification. A query without a committed observation is incomplete. Stability alone does not establish provenance or the earlier causal operativity witness.

Provenance. (3); Main (100); (8).

3.6.1 Query and mindfulness executions

A passive observer runs after each committed transition. Interrogation is a separate input q with a declared committed observation and changes the perturbed execution. It cannot be counted as passive tracing. The established operative boundary is stable over this observation window; stabilization before that boundary is established is not a mindfulness witness. For $p = 1$, constant retained forcing one and $q_k = -J_k$, an initial current three becomes one and remains one while H1/fibre transitions continue. The observation is of current J_k, already committed at k; the response is at $k+1$. An extra delay using J_{k-1} and the boundary pressure $p = 1/2$ both fail this stabilization conclusion. The general constant-forcing criterion $p > 1/2$ is derived in (8). The delayed control declares $J_{32} = 0$ as its initial history and $J_{33} = 3$ as its starting current. Its first query cites cut 32; the query at cut 34 uses $J_{33} = 3$ and cites cut 33. Each event retains the consumed observation cut. The initialized history is a model premise; subsequent query values come from the recorded preceding observation.

3.7 Preserved example ladder and source status

The original A0–A18 examples share the constructors of §3.1 and its declared shared runner `manual/scripts/aod_awareness.py`. Their required roles remain explicit in the following crosswalk.

Tier	Construction	Current witness or control
A0	Inherited closure and temporal interfaces	Exact native source binding and inherited suite
A1	Passive trace	Identical state/event bytes, observation separated
A2	Structural ancestry	Closure memory without retention effect
A3	Matched channel operativity	Matched pressure families, removal/restoration
A4	Attention separation	Present attention one, retention effect false

Tier	Construction	Current witness or control
A5	Persistence window	Surviving relation with and without later gain
A6	Polycadence	Lock, nonzero bias, unequal cadence and jitter
A7	Crescendo/recovery	Operative-source counts $(1, 2, 3, 0, 3)$
A8	Later contact/fold/chain	Two separate causal interventions and B2 offer
A9	Ternary boundary roles	SELF/BETWEEN/NOT-SELF with distinct Null type
A10	Indexical/shared labels	I under node renaming; impostor rejection
A11	Sensor–motor	Issued command and admitted channel
A12	Proprioception	Bound return projection and later residual effect
A13	CPU/subnet	Operative interface and separate operation-count proxy
A14	Learning	Persistent $(5, 6, 7)$ versus removed $(4, 4, 4)$
A15	Reflexive interrogation	Explicit query perturbation with prior observation
A16	Operational mindfulness	Exact stabilization under stated hypotheses
A17	Fold/expose/carriage	Inner identities survive; higher retained pair passes
A18	Recurrent other/subnet	Separate admitted peer and repeated contact prefixes

The forty-seven new input rows have identities `R04-NEW-001` through `R04-NEW-047`. Their source, scope, record class, phase type, control/trace/perturbation fields, expected exact result and falsification condition are included in the JSONL data. They are newly authored witnesses. The separate nineteen-row historical source roster remains `HOLD_MISSING_AUTHENTIC_SOURCES`. No original script was reconstructed from a basename or a former runtime, and these new executions do not satisfy historical rerun rows.

Provenance. Main (155); Main (156); Main (157); Main (158); (3); (4); Main (159); (5); (6); (7); Main (160); (8); Main (16); Main (24); Main (64); Main (41); Main (42).

4 Pen-and-Paper AFC Raw Execution and D.E.C. Ledger

Scope. This section gives a hand-calculation path from one declared AFC edge computation into the AΩD motif names used later in the manual. It is a manual ledger section: it records exact edge weights, kernel fractions, route status, and boundary values.

Provenance. Main §5; Main §1.4; Main §4.4; Main §5.5; §1.7; §5.

AFC runs raw. The trace records the run. AΩD names detected motifs. A simulation fixture first declares the AFC scope B, boundary/region data, admissible channels, and update rule. The AFC run produces trace rows. AΩD detectors read the trace and classify motifs such as cross-return, yro route, duration-clipping, sheddic route, terminal split, Tau-ring support, orbital retention, and collision-support trace status.

D.E.C. means Declared Edge Computation ledger. In this manual section, D.E.C. is the manual ledger format used to expose one declared AFC edge computation. Use the dotted form D.E.C.; the ledger is a manual value container for declared edge data.

$$\begin{aligned}\text{AFC raw node} &\to \text{D.E.C. ledger} \to \text{exact kernel} \to \text{blocked hinge} \to \text{hinge slide},\\ &\to \text{oriented ADAR component} \to \text{SADAR value} \to \text{boundary audit}.\end{aligned}$$

Fixture-comparator rows form exact δ_3 tallies.

4.1 D.E.C. row format

Each D.E.C. row records

$$B, \quad v, \quad e, \quad v_e, \quad \sigma(e), \quad \mathrm{adm}(e;B), \quad w(e;B), \quad P(e;B), \quad \mathrm{route}(e;B).$$

Here B is the declared boundary/window, v is the active node, e is an admissible edge slot, v_e is the Hamming–1 neighbor reached by that slot, $\sigma(e) \in \{-1,0,+1\}$ is the local ternary state, $\mathrm{adm}(e;B)$ records admissibility on the declared scope, $w(e;B)$ is the declared integer weight, $P(e;B)$ is the exact normalized kernel fraction, and $\mathrm{route}(e;B)$ records returned, outgoing, blocked, slide, hinge, or pending route status.

Signed channel views. An oriented ledger channel fixes one signed value. Its reverse bookkeeping view has equal magnitude and opposite sign, implementing Main (81). A separately constructed reverse channel retains its own directional RCD and must be evaluated independently. Component-valued rows apply the sign and Stokes checks componentwise; Main (82) gives their boundary sum.

4.2 One-node D.E.C. ledger

Declare one Q_4 node and four Hamming–1 edge slots:

$$v = (0,0,0,0), \qquad \{e_1, e_2, e_3, e_4\}.$$

Declare ternary edge states $\sigma(e_i) \in \{-1,0,+1\}$. The following ledger records one declared anisotropic kernel.

Table 4: One-node D.E.C. ledger.

edge	neighbor	$\sigma(e)$	admissible	weight	probability	route
e_1	$(1,0,0,0)$	$+1$	1	1	1/8	outgoing
e_2	$(0,1,0,0)$	-1	1	1	1/8	returned
e_3	$(0,0,1,0)$	0	1	4	1/2	hinge
e_4	$(0,0,0,1)$	$+1$	1	2	1/4	outgoing

4.3 Exact kernel normalization

For isotropic admissible branching over four successors,

$$P^{\mathrm{iso}}(e_i;B) = \frac{1}{4}.$$

For the declared anisotropic weights,

$$w = (1,1,4,2), \qquad Z = 1+1+4+2 = 8.$$

The anisotropic kernel is

$$P^{\mathrm{ani}} = (1/8, 1/8, 1/2, 1/4).$$

Table 5: Exact anisotropic kernel fractions.

edge	weight	probability
e_1	1	1/8
e_2	1	1/8
e_3	4	1/2
e_4	2	1/4

Literature note. Markov kernels and random walks: Norris [28], Levin–Peres–Wilmer [15], Lawler–Limic [13].

4.4 Blocking rule

Blocking is zero admissibility. Once same-resolution hinge dwell is blocked on the declared scope, the hinge-state row is removed from the successor pool and the kernel is renormalized over branch-oriented continuations.

Table 6: D.E.C. blocking and slide rules.

rule	meaning
isotropic	equal probability over admissible successors
anisotropic	unequal declared weights over admissible successors
blocked	zero admissibility on the declared scope
hinge slide	renormalization after same-resolution dwell is removed
continuation trace	repeated admitted node updates after hinge slide

4.5 Blocked hinge and hinge slide

In the following block event, the prior hinge-state row $(e_3, 0)$ attempts same-resolution dwell, but the declared scope sets $\mathrm{adm}(e_3, 0; B) = 0$. The additional-dwell row $(e_3, 0_D)$, not the edge slot e_3 itself, is removed before renormalization; the e_3 slot may re-enter only as an admitted branch state. Declare the post-block slide pool

$$\{(e_1, -1), (e_2, -1), (e_3, +1)\}$$

with weights

$$w'((e_1, -1), (e_2, -1), (e_3, +1)) = (3, 3, 1), \qquad Z' = 3 + 3 + 1 = 7.$$

Then

$$P(e_1, -1) = 3/7, \qquad P(e_2, -1) = 3/7, \qquad P(e_3, +1) = 1/7.$$

The returned and outgoing slide fractions are

$$P_{\mathrm{returned}} = 3/7 + 3/7 = 6/7, \qquad P_{\mathrm{outgoing}} = 1/7.$$

Three roles and two branch roles. The four-channel kernel in §4.3, Table 5, has role probabilities $(P_-, P_0, P_+) = (1/8, 1/2, 3/8)$. After blocking additional dwell, three channel rows remain, while their two-role projection is $(P_-, P_+) = (6/7, 1/7)$. The structural hinge remains. The binary role entropy $-(6/7)\log_3(6/7) - (1/7)\log_3(1/7)$ is below the uniform maximum $\log_3 2$ stated in Main §4.4. This is conditional role entropy, separate from channel cardinality, $C_3(k) = k$, and channel mutual information. For a capacity counterexample, declare the deterministic ternary channel $P_{01} = P_{12} = P_{20} = 1$, all

other entries zero, with uniform input. It has one successor per row but preserves the input bijectively, so $I(X;Y) = H(X) = 1$ trit, exceeding $\log_3 2$. No row-degree bound alone establishes that universal channel-capacity bound.

4.6 Field tunnelling: hinge-slide window clip

Data-support class. Field tunnelling is a D0 exact internal fixture. It uses declared hinge-slide admissibility, exact slide-kernel fractions, window-clipped duration participation, and scalar tunnel contribution.

Field tunnelling is the retained passage of a field route through a blocked hinge by slide-compatible successor support inside the declared window. The manual spine is

$$\begin{aligned}\text{blocked direct hinge} \to \text{declared slide pool} \to \text{exact slide kernel} \to \text{window clip},\\ \to \text{tunnel contribution} \to \text{boundary audit.}\end{aligned}$$

Fix the declared boundary/window

$$B = (\partial F, \omega).$$

The short-window examples declare neutral reflection, the native count duration interface and a same-domain window. They clip only $\rho^D = \mathrm{dur}(RD \bowtie_B R^{\mathrm{asym}})$, leaving RD and RCD intact. At cut-frame $B|t$, declare a direct hinge route e_0 with

$$\mathrm{adm}(e_0; B|t) = 0.$$

Declare the slide pool

$$\mathrm{Slide}_\mu(B|t) = \{s_1, \ldots, s_n\}.$$

Normalize only after the slide pool is declared:

$$Z_{\mathrm{slide}}(B|t) = \sum_{s_i \in \mathrm{Slide}_\mu(B|t)} w(s_i; B|t), \qquad P_{\mathrm{slide}}(s_i; B|t) = \frac{w(s_i; B|t)}{Z_{\mathrm{slide}}(B|t)}.$$

For each slide-compatible successor,

$$\rho^D_{\omega,s_i}(B) = \min(RD_{s_i}(B), |\omega|_{s_i}), \qquad p^D_{s_i}(B) = C_{3,s_i}\rho^D_{\omega,s_i}(B).$$

The scalar tunnel contribution is

$$T^D_{\mathrm{tunnel}}(B|t) = \sum_{s_i \in \mathrm{Slide}_\mu(B|t)} P_{\mathrm{slide}}(s_i; B|t)\, p^D_{s_i}(B).$$

When oriented attention is declared, the oriented tunnel flux is

$$\Phi^D_{\mathrm{tunnel}}(B|t) = \sum_{s_i \in \mathrm{Slide}_\mu(B|t)} P_{\mathrm{slide}}(s_i; B|t)\, p^D_{s_i}(B)\, A_{s_i}(B).$$

Using the hinge-slide pool from §4.5, declare the tunnel branch

$$s_{\mathrm{tunnel}} = (e_3, +1), \qquad P_{\mathrm{tunnel}} = 1/7.$$

Declare

$$RD_{s_{\mathrm{tunnel}}} = 5, \qquad |\omega|_{s_{\mathrm{tunnel}}} = 2, \qquad C_{3,s} = 3.$$

Then

$$\rho^D_{\omega,s_{\mathrm{tunnel}}} = \min(5, 2) = 2, \qquad p^D_{s_{\mathrm{tunnel}}} = 3 \cdot 2 = 6,$$

and

$$T^{D}_{\text{tunnel}} = (1/7) \cdot 6 = 6/7.$$

The tunnel contribution is the window-clipped pressure carried by the slide-compatible outgoing successor.

Table 7: Field tunnelling hinge-slide window-clip fixture.

row	value
direct hinge admissibility	$\mathrm{adm}(e_0; B\|t) = 0$
slide pool	$\{(e_1, -1), (e_2, -1), (e_3, +1)\}$
slide weights	$(3, 3, 1)$
slide probabilities	$(3/7, 3/7, 1/7)$
tunnel branch	$(e_3, +1)$
P_{tunnel}	$1/7$
RD_s	5
$\|\omega\|_s$	2
$\rho^{D}_{\omega,s}$	2
$C_{3,s}$	3
p^{D}_{s}	6
T^{D}_{tunnel}	$6/7$

4.7 D.E.C. row-pair to ADAR and SADAR

A routed D.E.C. row-pair supplies the declared returned-current pair. That pair is promoted to an oriented ADAR component on the declared boundary/window. Declare

$$A_{12} = 1/3, \qquad \rho^{D}_{\omega} = 4, \qquad C_3 = 3.$$

Then

$$p^{D} = C_3 \rho^{D}_{\omega} = 12, \qquad \mathfrak{S}_B = p^{D} A_{12} = 12(1/3) = 4.$$

Reverse orientation gives

$$A_{21} = -1/3, \qquad \mathfrak{S}_B = -4.$$

4.8 Boundary closure and sheddic route audit

Declare

$$P^{D} = 27, \qquad C_{\text{close}} = 24.$$

Then

$$\Delta_{\text{close}} = P^{D} - C_{\text{close}} = 3, \qquad X_{\text{shedding}} = \max(0, \Delta_{\text{close}}) = 3.$$

This is the positive surplus route audit used in the Dimonnanyro row:

$$1{:}2{:}9_{\text{yro}} = \text{Dimonnanyro}.$$

Fixture-comparator rows record exact δ_3 residuals. Boundary-route rows record boundary values and route audits.

4.9 D.E.C. audit invariants

A D.E.C. row records

$$B, \quad v, \quad e, \quad v_e, \quad \sigma(e), \quad \mathrm{adm}(e; B), \quad w(e; B), \quad P(e; B), \quad \mathrm{route}(e; B).$$

The normalized kernel satisfies

$$\sum_{e} P(e; B) = 1.$$

Zero admissibility gives zero probability:

$$\mathrm{adm}(e;B)=0 \Rightarrow P(e;B)=0.$$

A hinge-slide row records

$$\mathrm{Adm}^{0}_{B}(e,\sigma)=\varnothing$$

and the renormalized branch-continuation kernel. Blocking is zero admissibility.

A SADAR row records an oriented ADAR pair and satisfies

$$\mathfrak{S}_B(C_{\bar{e}})=-\mathfrak{S}_B(C_e).$$

Fixture-comparator rows form exact δ_3 tallies. A Tau missing-burden row carries integer provenance

$$\mathrm{miss}_\tau(K)=M_{\mathrm{miss}}(K)/M_{\mathrm{ref}}(K),$$

with any decimal shown only as a display coordinate.

4.10 Tau missing-burden exact ratio

The Tau missing-burden diagnostic is recorded with exact provenance:

$$\mathrm{miss}_\tau(K)=\frac{M_{\mathrm{miss}}(K)}{M_{\mathrm{ref}}(K)}.$$

For the displayed Dimonnanyro row,

$$M_{\mathrm{miss}}=31,\qquad M_{\mathrm{ref}}=50,\qquad \mathrm{miss}_\tau(1{:}2{:}9_{\mathrm{yro}})=\frac{31}{50}.$$

The decimal presentation field is

$$\mathrm{missing_fraction_display}=0.62.$$

Table 8: Tau missing-burden exact-ratio fields.

field	meaning
missing_num	numerator M_{miss}
missing_den	denominator M_{ref}
missing_frac_exact	exact ratio
missing_fraction_display	decimal presentation field

Atomic leg and hinge role. Each D.E.C. row records the H1 base traversal separately from the ternary role. Removing optional dwell sets $m=0$ in $\varepsilon 0_H(0_D)^m(-\varepsilon)$; the structural hinge remains. A slide pool is entered only after that hinge. Kernel normalization uses already admissible support. Repeated updates are continuation traces until the actual closure and distinct-reclosure witnesses establish wave admission.

5 Exact Internal Fixtures

Scope. This section carries exact AΩ-internal fixtures: cross-return, duration-clipping, exact δ_3 residual tally, and sheddic route audit.

Data-support class. These are D0 exact internal fixtures. Input data are declared integer gates, channels, windows, and comparator bins. Comparator fixtures record $O_k, \Delta_k^{\mathbb{Z}}, \delta_{3,k}$. Route-audit fixtures record Δ_{close}, X_{shedding}, and SheddicPath.

5.1 Dimonhexon cross-return fixture

Scope. This subsection is a boundary-restriction test. A stable Dimonhexon closure enters a slit boundary, becomes a duon-cycle handoff across the restricted boundary/window, and recloses into target bins.

Provenance. Main §5.10; Main (92).

Declare the slit scope

$$\mathbf{a}_{\mathrm{slit}} \vdash (F_{1:2:6}, R_{\mathrm{slit}}, \partial R_{\mathrm{slit}}, \omega_{\mathrm{slit}}), \qquad B_{\mathrm{slit}} = (\partial F_{1:2:6}, \omega_{\mathrm{slit}}).$$

The carried form is

$$1{:}2{:}6 = \text{Dimonhexon}.$$

The boundary-restricted handoff is recorded as

$$\text{Dimonhexon} \longrightarrow W_\omega[D_i \mu D_j] \longrightarrow \Delta^{\chi^{\mathrm{lag}}} \longrightarrow \mathbf{A}_F(B_{\mathrm{slit}}) \longrightarrow \text{local reclosure bins}.$$

For one gate g_a, the target-bin contribution is

$$S_k^{(a)} = \sum_{t \in \omega} \sum_{\gamma : g_a \rightsquigarrow k} \kappa_k \mathfrak{S}_{B|t}(C_\gamma).$$

For a second gate g_b, define $S_k^{(b)}$ analogously.

Remark. The particle-like term is retained closure. The wave-like term is boundary-constrained handoff. The hit-like term is local reclosure.

5.2 Recurrent-ring double-slit cross-return

Scope. This subsection applies the main-note fractal field ring gate to the Dimonhexon slit handoff.

Declare a recurrent ring field

$$\mathbf{a}_{\mathrm{ring}} \vdash (F_{\mathrm{ring}}, R_{\mathrm{ring}}, \partial R_{\mathrm{ring}}, \omega_{\mathrm{ring}}), \qquad B_{\mathrm{ring}} = (\partial_T F_{\mathrm{ring}}, \omega_{\mathrm{ring}}).$$

Let

$$V_N = \{0, 1, \ldots, N-1\},$$

with ring adjacency

$$E_N = \{(i, i+1 \bmod N), (i, i-1 \bmod N) : i \in V_N\}.$$

Declare two boundary gates

$$g_a, g_b \in \partial_T R_{\mathrm{ring}}.$$

The target bins are

$$k \in K_{\mathrm{bin}}.$$

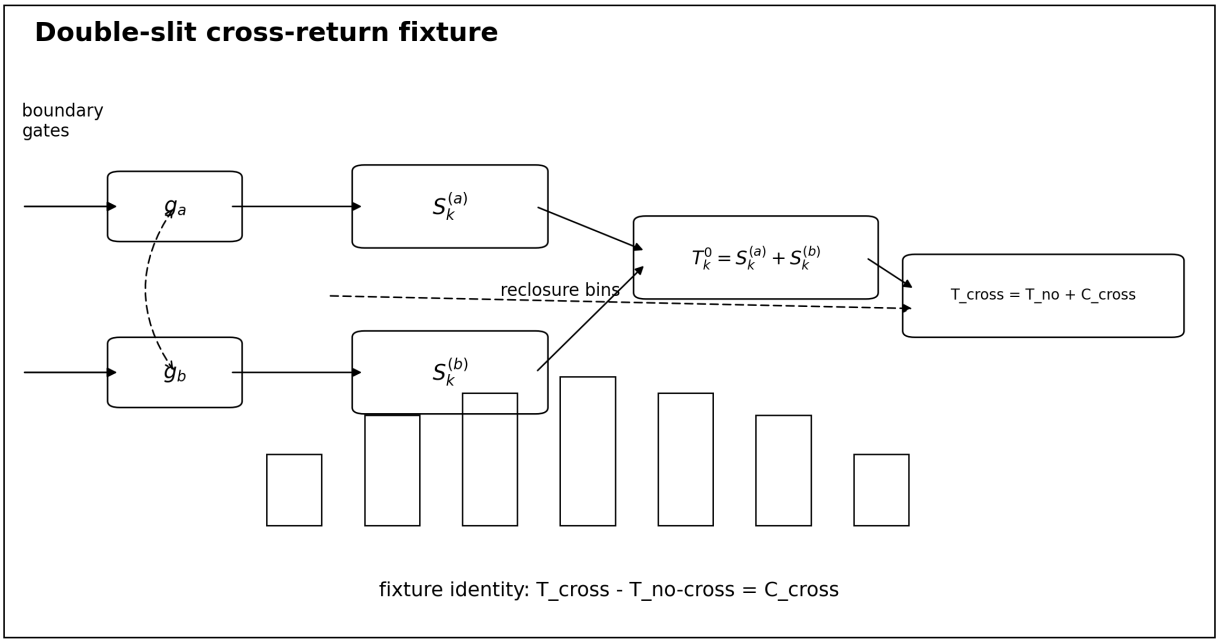


Figure 1: Recurrent-ring double-slit cross-return setup. Two boundary gates produce independent one-gate contributions; the no-cross baseline is the sum of one-gate contributions, and the cross-return contribution produces the declared cross-return fixture.

The independent two-gate baseline is

$$T_k^0 = S_k^{(a)} + S_k^{(b)}.$$

The cross-return contribution is

$$C_k^\times(B) = C_{ab,k}^\times(B) = \sum_{t\in\omega} \sum_{\gamma: g_a \leftrightarrow g_b \rightsquigarrow k} \kappa_k \mathfrak{S}_{B|t}(C_\gamma).$$

The recurrent ring value is

$$T_k^\times = T_k^0 + C_k^\times.$$

Thus the audit identity is

$$\boxed{T_k^\times - T_k^0 = C_k^\times.}$$

5.2.1 Exact double-slit fixture

The following fixture is a released AΩ-internal exact integer bin test. The independent gate-sum value is T_k^0. The recurrent ring value is $T_k^\times$. The declared fixture comparator is O_k. The comparison is the exact δ_3 tally $\delta_{3,k}(T^\times, O) = (s_k^{(3)}, m_k)$.

Here $C_k^\times = T_k^\times - T_k^0$. The center-bin fixture is

$$T_0^\times = 96, \qquad O_0 = 95, \qquad \Delta_0^{\mathbb{Z}} = 1, \qquad \delta_{3,0} = (+1, 1).$$

The full-fixture checks are

$$\sum_k T_k^\times = 448, \qquad \sum_k O_k = 448, \qquad \sum_k \Delta_k^{\mathbb{Z}} = 0, \qquad \sum_k m_k = 6.$$

Table 9: Dimonhexon slit handoff: exact δ_3 reclosure-bin fixture.

k	S_k	T_k^0	$C_k^\times$	$T_k^\times$	O_k	$\Delta_k^{\mathbb{Z}}$	$\delta_{3,k}$
-5	4	8	0	8	8	0	$(0,0)$
-4	9	18	-8	10	11	-1	$(-1,1)$
-3	16	32	-12	20	19	1	$(+1,1)$
-2	25	50	8	58	59	-1	$(-1,1)$
-1	31	62	18	80	79	1	$(+1,1)$
0	35	70	26	96	95	1	$(+1,1)$
1	31	62	18	80	80	0	$(0,0)$
2	25	50	8	58	58	0	$(0,0)$
3	16	32	-12	20	21	-1	$(-1,1)$
4	9	18	-8	10	10	0	$(0,0)$
5	4	8	0	8	8	0	$(0,0)$

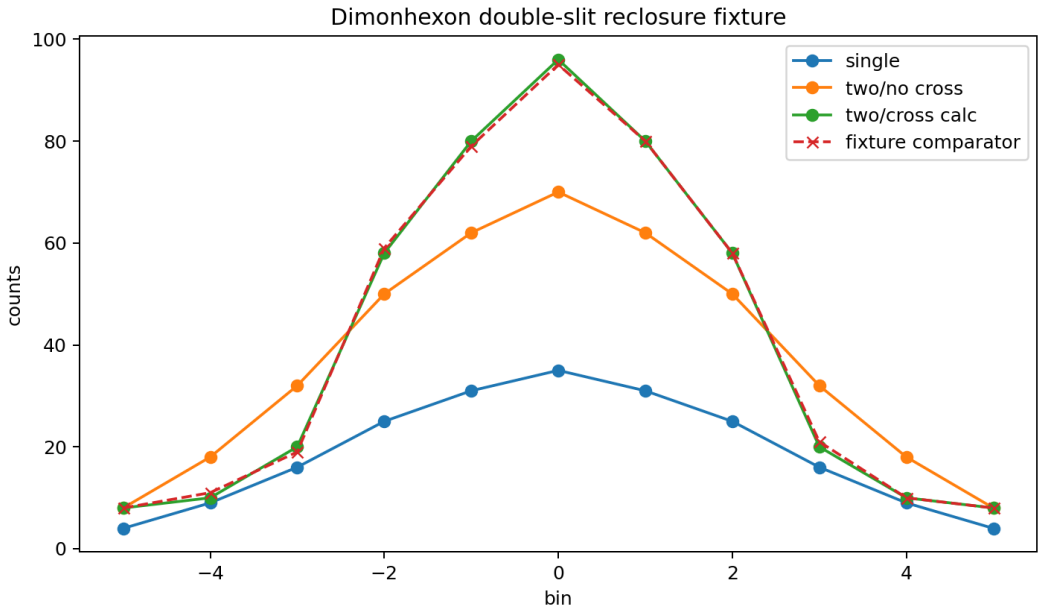


Figure 2: Dimonhexon double-slit reclosure fixture. Cross-return changes the two-gate distribution; the exact internal comparison is recorded by δ_3.

Status. Released A↔ Ω-internal field-ring fixture.

Provenance. Main §5.10; Main (79); Main (RG4); §5.1.

Remark. The independent gate-sum value records independent reclosure. The recurrent ring value records redistributed returned-current coupling.

5.3 Short-window fixture scope

Scope. This section is AΩ-internal. It tests short-window ρ^D-clipping, pending overhang, closure residual, and sheddic path status without external comparison.

Declare the short-window fixture scope

$$\mathbf{a}_{\mathrm{sw}} \vdash (F_{\mathrm{sw}}, R_{\mathrm{sw}}, \partial R_{\mathrm{sw}}, \omega_{\mathrm{sw}}), \qquad B_{\mathrm{sw}} = (\partial F_{\mathrm{sw}}, \omega_{\mathrm{sw}}).$$

Duration-domain declaration. These reductions declare $R^{\mathrm{asym}} = \mathrm{id}$ and the additive native count realization, so $\rho^D = RD$ in this specialization. The window is a count in that same domain; $\rho^D_\omega =$

Table 10: Recurrent-ring cross-return symbol table.

Symbol	Meaning
$S_k^{(a)}$	one-gate contribution from g_a into bin k
$S_k^{(b)}$	one-gate contribution from g_b into bin k
T_k^0	independent gate-sum value
$C_{ab,k}^{\times}(B)$	cross-return contribution from $g_a \leftrightarrow g_b$ into bin k
$T_k^{\times}$	recurrent ring value
O_k	declared fixture comparator, if used
$\Delta_k^{\mathbb{Z}}$	exact signed integer fixture residual
$\delta_{3,k}(T^{\times}, O)$	exact ternary signed residual tally

$\min(\rho^D, |\omega|)$ is scalar participation. Neither support RD nor the coupled RCD object is clipped. The baseline Dimonon support used below is

$$\Pi(1{:}2) = 2^1 + 2 = 4, \qquad H_\mu = 1, \qquad \rho^D_{1:2} = RD^{(1)}(1{:}2) = 2 \cdot 4 + 1 = 9.$$

The record length is denoted by L_M. It is distinct from the path-derived scalar ρ^D.

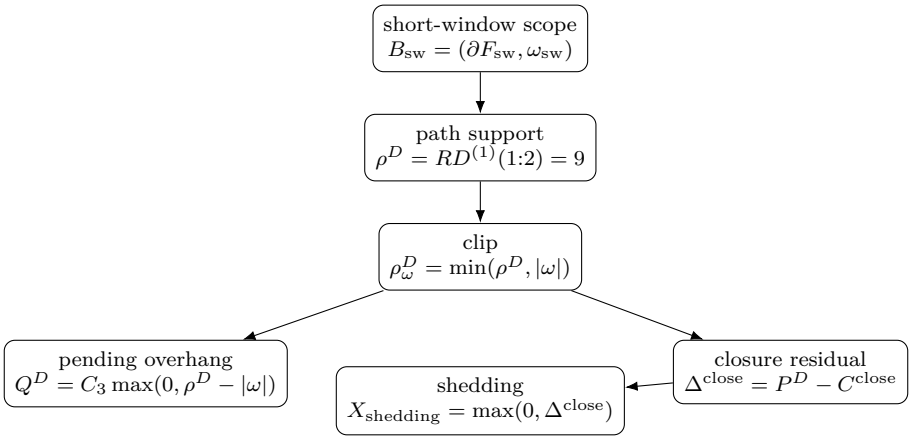


Figure 3: Short-window RCD fixture flow. The path-derived ρ^D is clipped by the declared window to produce ρ^D_ω, while pending overhang Q^D and closure residual Δ^{close} are kept separate from shedding excess.

5.4 Shared fixture equations

For a candidate record M in a declared scope B, use

$$\rho^D_\omega(M;B) = \min(\rho^D(M;B), |\omega|),$$

$$Q^D(M;B) = C_3 \max(0, \rho^D(M;B) - |\omega|),$$

$$P^D(M;B) = C_3 \rho^D_\omega(M;B),$$

$$\Delta^{\mathrm{close}}(M;B) = P^D(M;B) - C^{\mathrm{close}}(M;B),$$

$$X_{\mathrm{shedding}}(M;B) = \max(0, \Delta^{\mathrm{close}}(M;B)).$$

Remark. Δ^{close} records signed closure residual. X_{shedding} records only positive surplus. Negative closure residuals are retained as deficits or open checks, not as sheddic surplus.

5.5 Closure-dependent and surplus fixtures

The following table keeps Δ^{close} and X_{shedding} separate. This prevents closure-deficit records from being misread as sheddic surplus.

Table 11: Short-window RCD and shedding fixtures. All records use $C_3 = 3$ and $\rho^D = RD^{(1)}(1{:}2) = 9$.

Row	C_3	ρ^D	L_M	$\lvert\omega\rvert$	ρ^D_ω	Q^D	P^D	C^{close}	$\Delta^{\mathrm{close}}, X_{\mathrm{shedding}}$
1:2:7 $\to C_6$	3	9	7	7	7	6	21	18	$3, 3$
1:2:7 $\to C_8$	3	9	7	7	7	6	21	24	$-3, 0$
1:2:9 $\to C_8$	3	9	9	9	9	0	27	24	$3, 3$
1:2:10 $\to C_{10}$	3	9	10	10	9	0	27	27	$0, 0$

5.6 The 1:2:7 closure-dependent fixture

For 1:2:7, the window is shorter than the path-derived support:

$$\rho^D_\omega = \min(9, 7) = 7, \qquad Q^D = 3(9 - 7) = 6, \qquad P^D = 3 \cdot 7 = 21.$$

Against lower closure $C_6 = 18$, the fixture carries surplus:

$$\Delta^{\mathrm{close}} = 21 - 18 = 3, \qquad X_{\mathrm{shedding}} = 3.$$

Against upper closure $C_8 = 24$, the fixture is closure-deficient:

$$\Delta^{\mathrm{close}} = 21 - 24 = -3, \qquad X_{\mathrm{shedding}} = 0.$$

Thus 1:2:7 is closure-dependent.

5.7 The 1:2:9 surplus fixture

For 1:2:9, the candidate window reaches the path-derived support:

$$\rho^D_\omega = \min(9, 9) = 9, \qquad Q^D = 0, \qquad P^D = 3 \cdot 9 = 27.$$

Against the $C_8 = 24$ closure record,

$$\Delta^{\mathrm{close}} = 27 - 24 = 3, \qquad X_{\mathrm{shedding}} = 3.$$

This is the nine-window surplus fixture.

5.8 The 1:2:10 clipped retained fixture

For 1:2:10, the candidate record length exceeds the path-derived support:

$$\rho^D_\omega = \min(9, 10) = 9, \qquad Q^D = 0, \qquad P^D = 27.$$

Against $C_{10} = 27$,

$$\Delta^{\mathrm{close}} = 0, \qquad X_{\mathrm{shedding}} = 0.$$

The fixture is a clipped retained check.

5.9 Sheddic path audit

The sheddic path audit uses

$$\mathrm{SheddicPath}(X_{\mathrm{shedding}}) \in \{E_2, \mathsf{C}_6, \mathsf{C}_8, \mathsf{C}_{10}, \dots, \mathsf{C}_{2n}\}.$$

A surplus fixture must declare where the positive X_{shedding} is routed. A deficit fixture may remain an open check or may be paired with a declared closure completion.

5.10 Audit hooks

Audit. A short-window fixture records ρ_ω^D, Q^D, Δ^{close}, and X_{shedding} separately. Sheddic surplus is positive only, and every positive X_{shedding} carries a declared sheddic path.

Provenance. Main (60); Main (105); Main (109); §5.3.

Remark. This section is AΩ-internal. It prepares the mechanics used by rest-energy records and boundary-current records without adding an external comparison target.

6 Field Rest-Energy Records

6.1 0|− scope

Rest-energy records are split into two tables. The AΩ field-map table records support, carried cycle, and AΩ accessor values. The measured-data comparison table records declared measured-sector target rows.

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega).$$

The calculation spine is

$$C_f[M]_{A\leftrightarrow\Omega} \longrightarrow \mathsf{AO}_\mu(C) \longrightarrow \mathfrak{S}_B(C) \longrightarrow \nu_{x,A\leftrightarrow\Omega} \longrightarrow E_{x,\mathrm{ext}}.$$

Data-support class. Rest-energy records are exact internal AΩ field records or O0 external-map comparisons, depending on the declared measured-sector target rows. Uncertainty is scoped to the external measured-sector target and value map.

Provenance. Main App. J.

6.2 Support notation used by the calculation records

The manual uses compact support notation only where the calculation needs it:

$$D_i \square D_j = \text{open-seat duon support},$$

$$D_i \mu D_j, \qquad \mu \in \{-1, 0, +1\}, \qquad = \text{resolved duon support},$$

$$E_2^0 = \text{open-seat duon support}, \qquad E_2^{\pm} = \text{duad / seated duon support}.$$

The compact field supports are

$$1{:}2{:}L = \text{Dimonon support}, \qquad 3{:}q{:}L = \text{Trion support}.$$

Odd lengths are carried as odd-window route status between adjacent even support enclosures:

$$L = 2k + 1 \quad \Rightarrow \quad C_{2k} \rightsquigarrow L \rightsquigarrow C_{2k+2}.$$

When this odd-window instability is retained as a route-status suffix, write K_{yro}. Support-enclosure names remain the even-enclosure names.

Provenance. Main App. J.

6.3 Field-property accessor calculations

Under the single-hinge neutral-reflection hypotheses of Main App. J.2, the following compatible native count windows give these nine accessor rows. In the declared free-seat fixtures, 1:2 $\Rightarrow$ $s = 2$ and $H_\mu^{[2]} = 1$, while 3:3 $\Rightarrow$ $s = 1$ and $H_\mu^{[1]} = 1$. A full-seated single-hinge cycle uses $H_\mu = H_\mu^{\mathrm{cyc}} = 1$ unless free-seat provenance is declared.

Table 12: AΩ field fractal properties.

$\mathcal{F}_{A\leftrightarrow\Omega}$	M	$C_f[M]_{A\leftrightarrow\Omega}$	Π_α	H_μ	$RD_{A\leftrightarrow\Omega}$	$\lvert\omega\rvert$	ρ_ω^D	P^D	Q^D	X_{shedding}
$\mathfrak{f}_{1:2:6}$	1:2:6	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	4	$H_\mu^{[2]} = 1$	9	6	6	18	9	0
$\mathfrak{f}_{1:2:8}$	1:2:8	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	4	$H_\mu^{[2]} = 1$	9	8	8	24	3	0
$\mathfrak{f}_{1:2:9}$	1:2:9	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	4	$H_\mu^{[2]} = 1$	9	9	9	27	0	3
$\mathfrak{f}_{1:2:10}$	1:2:10	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	4	$H_\mu^{[2]} = 1$	9	10	9	27	0	0
$\mathfrak{f}_{3:2:6}$	3:2:6	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	10	1	21	6	6	30	75	0
$\mathfrak{f}_{3:2:8}$	3:2:8	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	10	1	21	8	8	40	65	0
$\mathfrak{f}_{3:2:10}$	3:2:10	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	10	1	21	10	10	50	55	0
$\mathfrak{f}_{3:3:6}$	3:3:6	$C_f[3{:}3]_{A\leftrightarrow\Omega}$	30	$H_\mu^{[1]} = 1$	61	6	6	36	330	0
$\mathfrak{f}_{3:4:6}$	3:4:6	$C_f[3{:}4]_{A\leftrightarrow\Omega}$	68	$H_\mu^{\mathrm{cyc}} = 1$	137	6	6	42	917	0

6.4 AΩ field-map table

The AΩ field-map table lists the manual mapping from measured-sector names to AΩ field supports. The numerical field-support properties are calculated in Table 12.

Table 13: AΩ field-map table.

x_{ext}	external map	M_x	$C_f[M_x]_{A\leftrightarrow\Omega}$	$\nu_{x,A\leftrightarrow\Omega}$
e^-	00|− Dimonhexon	1:2:6	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	$\nu_{e,A\leftrightarrow\Omega}$
μ^-	00|− Dimonocton	1:2:8	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	$\nu_{\mu,A\leftrightarrow\Omega}$
τ^-	00|− Dimonnanyro	$1{:}2{:}9_{\mathrm{yro}}$	$C_f[1{:}2]_{A\leftrightarrow\Omega}$	$\nu_{\tau,A\leftrightarrow\Omega}$
$B^\pm$	Ditrion debris map	3:2:6	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	$\nu_{B,A\leftrightarrow\Omega}$
B^0	Ditrion debris map	3:2:6	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	$\nu_{B,A\leftrightarrow\Omega}$
B_s^0	lifted Ditrion map	3:2:8	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	$\nu_{B_s,A\leftrightarrow\Omega}$
$B_c^\pm$	lifted Ditrion map	3:2:10	$C_f[3{:}2]_{A\leftrightarrow\Omega}$	$\nu_{B_c,A\leftrightarrow\Omega}$
p^+	Tritriohexon map	3:3:6	$C_f[3{:}3]_{A\leftrightarrow\Omega}$	$\nu_{p,A\leftrightarrow\Omega}$
n^0	Tetratriohexon map	3:4:6	$C_f[3{:}4]_{A\leftrightarrow\Omega}$	$\nu_{n,A\leftrightarrow\Omega}$

Provenance. Main App. J; Main §3.5.

Remark. The manual declares measured-sector maps such as $e^- \mapsto 00|-$ Dimonhexon. These maps are carried by manual comparison records; Table 12 lists the AΩ field-property calculations.

6.5 Electron external map

The electron comparison record is a manual map from the measured-sector electron to the AΩ Dimonhexon support:

$$e^- \longmapsto 00|- \text{ Dimonhexon} \equiv 00|- \text{ (1:2:6)}, \qquad C_f[1{:}2]_{A\leftrightarrow\Omega}.$$

The charge-orientation tag is

$$q_{\mathrm{ext}}(e^-) = -1, \qquad Q_{\mathrm{ext}}(e^-) = -e.$$

The rest-energy comparison target remains quarantined:

$$E_e^{\mathrm{tar}} = 0.51099895069 \text{ MeV}, \qquad \sigma_{E,e} = 1.6 \times 10^{-10} \text{ MeV}.$$

Literature note. Electron mass-energy: NIST CODATA [20], SI defining constants: BIPM [7].

Remark. The 00|− tag is a manual measured-sector charge/orientation map carried by the manual comparison record.

6.6 Measured rest-energy target records

External comparison is quarantined. For records with declared targets,

$$E_{x,\mathrm{ext}} = K_{\mathrm{ext}}\nu_{x,A\leftrightarrow\Omega}, \qquad \Delta E_x = E_{x,\mathrm{ext}} - E_x^{\mathrm{tar}},$$

$$\epsilon_x^{\mathrm{sgn}} = 100\frac{\Delta E_x}{E_x^{\mathrm{tar}}}, \qquad |\epsilon_x| = |\epsilon_x^{\mathrm{sgn}}|, \qquad z_x = \frac{\Delta E_x}{\sigma_{E,x}}.$$

Table 14: Measured rest-energy target records.

Record	Target [MeV]	σ_E [MeV]	Unit	Source
e	0.51099895069	1.6×10^{-10}	MeV	NIST CODATA
μ	105.6583755	2.3×10^{-6}	MeV	NIST CODATA
τ	1776.86	0.12	MeV	NIST / PDG target record
$B^{\pm}$	5279.41	0.07	MeV	PDG 2025
B^0	5279.72	0.08	MeV	PDG 2025
B_s^0	5366.93	0.10	MeV	PDG 2025
$B_c^{\pm}$	6274.47	0.32	MeV	PDG 2025
p	938.27208943	2.9×10^{-7}	MeV	NIST CODATA
n	939.56542194	4.8×10^{-7}	MeV	NIST CODATA

Provenance. Main App. J; App. B.

Literature note. Lepton/baryon targets: NIST CODATA [20, 22, 25, 24, 23], meson targets: PDG 2025 [29].

6.7 Tritrioseptyro Higgs-support diagnostic fixture

Tritrioseptyro is the $3{:}3{:}7_{\mathrm{yro}}$ Higgs-support prediction candidate selected by the internal AFC trace. The AFC run produces the trace rows; AΩ detectors read the trace and name the retained route status. The selected candidate is joined to the frozen external mass map $E_H = RD\,2^{L+4}$ MeV. Production-channel and decay-channel residual tables carry the next comparison layer.

$$\text{AFC run} \longrightarrow \text{read-only trace} \longrightarrow \text{AΩ route-status naming.}$$

The trace is read-only for this record. Feedback experiments use separately declared write-back maps.

The candidate set is

$$\mathcal{K}_H = \{3{:}1{:}7, 3{:}1{:}9, 3{:}2{:}7, 3{:}2{:}9, 3{:}3{:}7, 3{:}3{:}9, 3{:}4{:}7, 3{:}4{:}9\}.$$

For each candidate $K = (p, q, L)$, compute

$$\Pi = q^p + q, \qquad RD = 2\Pi + 1, \qquad \rho_\omega^D = \min(RD, L).$$

The scoped collision capacity is

$$C_{3,H} = 3.$$

With this scoped collision capacity,

$$P_H^D = C_{3,H}\rho_\omega^D,$$
$$Q_H^D = C_{3,H}\max(0, RD - L).$$

The retained candidate is

$$K_H^\star = 3{:}3{:}7_{\mathrm{yro}} = \text{Tritrioseptyro}.$$

For Tritrioseptyro,

$$\Pi = 30, \qquad RD = 61, \qquad \rho_\omega^D = 7, \qquad P_H^D = 21, \qquad Q_H^D = 162.$$

Against adjacent even support-enclosure closures,

$$P_H^D - C_6 = +3, \qquad P_H^D - C_8 = -3.$$

The trace places Tritrioseptyro on the C_6/C_8 saddle with pressure residuals $+3$ and -3 and pending overhang $Q_H^D = 162$.

Table 15: Higgs-support candidate ranking. This table carries the active trace coordinates used by the internal selection rule.

Candidate	RD	P_H^D	saddle	Q_H^D	score
$3{:}3{:}7_{\mathrm{yro}}$	61	21	$+3/-3$	162	0.00
$3{:}2{:}7_{\mathrm{yro}}$	21	21	$+3/-3$	42	1.58
$3{:}3{:}9_{\mathrm{yro}}$	61	27	$+3/0$	156	2.25
$3{:}4{:}7_{\mathrm{yro}}$	137	21	$+3/-3$	390	2.40
$3{:}2{:}9_{\mathrm{yro}}$	21	27	$+3/0$	36	2.89
$3{:}1{:}7_{\mathrm{yro}}$	5	15	$-3/-9$	0	3.00
$3{:}4{:}9_{\mathrm{yro}}$	137	27	$+3/0$	384	3.59
$3{:}1{:}9_{\mathrm{yro}}$	5	15	$-9/-12$	0	4.75

The trace score is reproducible under the internal selection rule:

$$S_H(K) = \frac{d_{\mathrm{saddle}}}{12} + \frac{d_Q}{100} + \frac{d_L}{2} + d_{\mathrm{route}}.$$

The scoring gate requires

$$\arg\min_{K \in \mathcal{K}_H} S_H(K) = 3{:}3{:}7_{\mathrm{yro}}.$$

Source artifacts. The score-component definitions and candidate rows are supplied under `manual/data/higgs` as `higgs_trace_score_formula.md` and `higgs_trace_internal_candidates_selection.csv`, respectively.

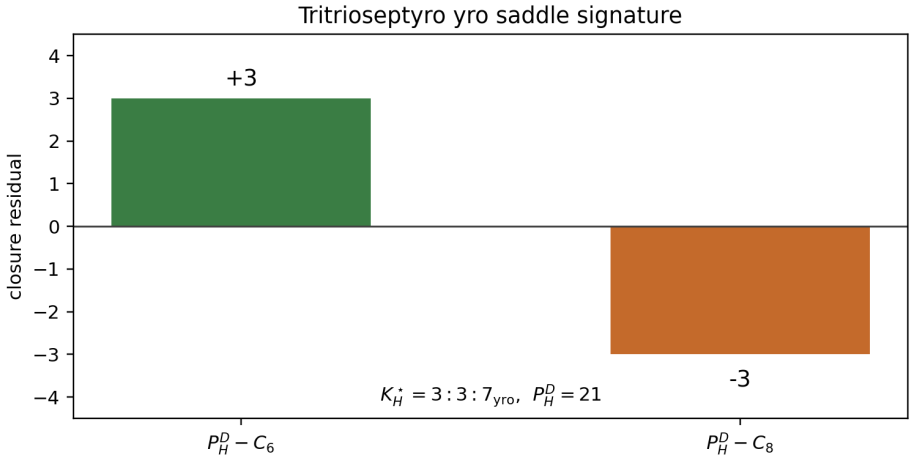


Figure 4: Tritrioseptyro yro saddle signature. The internal AFC trace gives $P_H^D = 21$, placing the selected $3{:}3{:}7_{\mathrm{yro}}$ candidate between $C_6 = 18$ and $C_8 = 24$ with residuals $+3$ and -3.

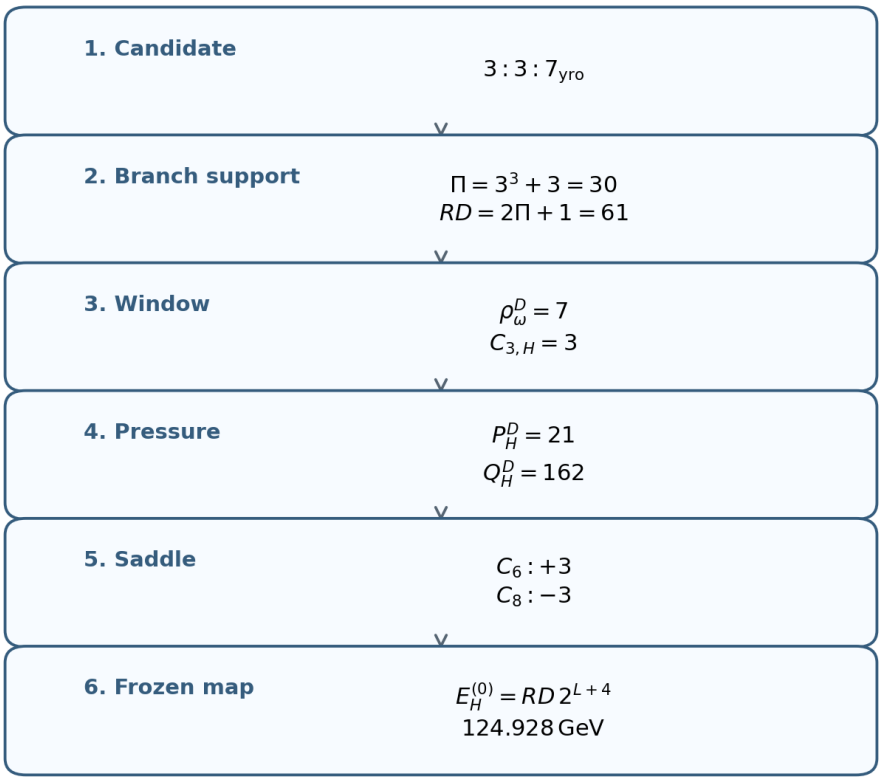


Figure 5: Tritrioseptyro trace flow. The AFC run produces $\Pi = 30$, $RD = 61$, $\rho_\omega^D = 7$, $P_H^D = 21$, and $Q_H^D = 162$. AΩ names the detected $3{:}3{:}7_{\mathrm{yro}}$ route status as Tritrioseptyro.

6.7.1 External mass map after candidate freeze

The external mass map is frozen after the internal candidate is selected:

$$E_H^{(0)}[\mathrm{MeV}] = RD_H\,2^{L+4}.$$

For Tritrioseptyro,

$$E_H^{(0)} = 61 \cdot 2^{11}\ \mathrm{MeV} = 124928\ \mathrm{MeV} = 124.928\ \mathrm{GeV}.$$

The frozen $RD\,2^{L+4}$ map sends Tritrioseptyro to 124.928 GeV. Declared Higgs mass targets enter after the candidate and external map are frozen. The comparison table reports ΔE, absolute percent error, and z-score for each target record.

Table 16: Frozen external map for Tritrioseptyro.

Quantity	Value	Record
$K_H^\star$	$3{:}3{:}7_{\mathrm{yro}}$	frozen internal candidate
RD_H	61	frozen internal accessor
L	7	declared odd window
$E_H^{(0)}$	$61 \cdot 2^{11}$ MeV = 124.928 GeV	external map after freeze

Table 17: LHC mass comparison after frozen external map.

Target	Mass [GeV]	σ [GeV]	ΔE [GeV]	$\lvert\epsilon\rvert$ [%]	z
ATLAS+CMS Run-1 combined	125.09	0.24	-0.162	0.1295	-0.675
CMS $H \to 4\ell$	125.04	0.12	-0.112	0.0896	-0.933
ATLAS Run-1+2 combined	125.11	0.11	-0.182	0.1455	-1.655
ATLAS $H \to \gamma\gamma$	125.22	0.14	-0.292	0.2332	-2.086

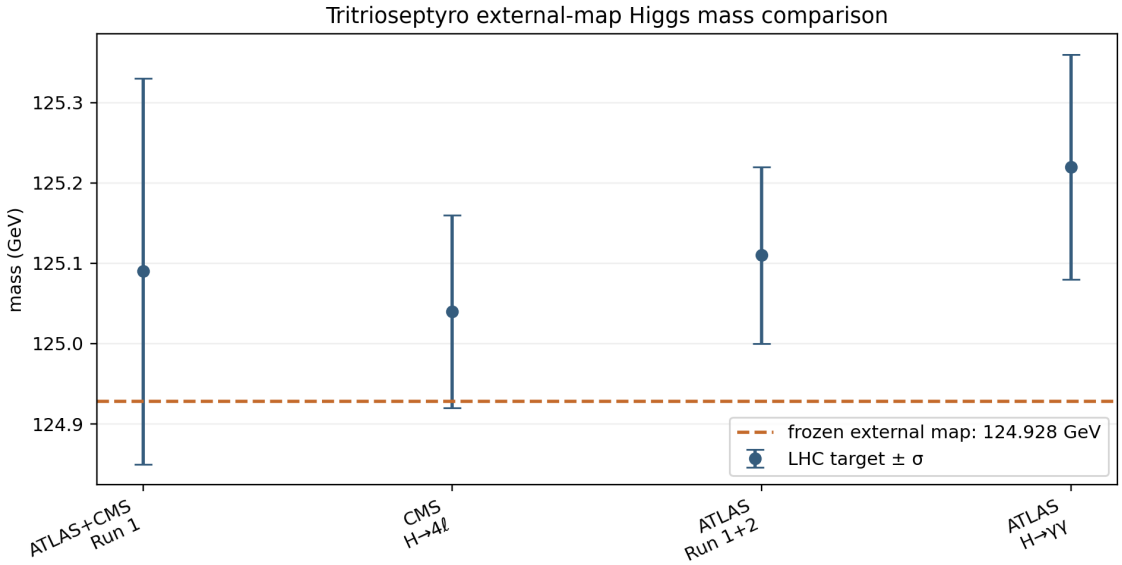


Figure 6: Tritrioseptyro mass comparison. The frozen external map $E_H = RD\,2^{L+4}$ MeV sends the selected candidate to 124.928 GeV. Target records enter as declared external comparison rows.

External data-comparison record. The internal trace selects $K_H^\star = 3{:}3{:}7_{\mathrm{yro}}$. The frozen external map gives

$$E_H^{(0)} = RD_H 2^{L+4} = 61 \cdot 2^{11}\,\mathrm{MeV} = 124.928\,\mathrm{GeV}.$$

Table 17 records the declared Higgs mass-input rows, their quoted uncertainties, and the resulting ΔE, $|\epsilon|$, and z coordinates. The mapped value 124.928 GeV lies within 1σ of the ATLAS+CMS Run-1 combined mass record and the CMS Run-2 $H \to 4\ell$ mass record. Production-channel and decay-channel residual tables carry the next comparison layer.

Provenance. Main §5.6.6; Main App. J; Table 15; Table 16.

Literature note. ATLAS+CMS Run-1 combined mass [4]. CMS $H \to 4\ell$ mass [8]. ATLAS combined Run-1+Run-2 mass [3]. ATLAS diphoton mass [2].

7 Hydrogen-facing Shell Tests

7.1 Shared hydrogen-facing shell declaration

Scope. This section is a manual comparison section. It declares hydrogen-facing shell records, computes finite $\mathfrak{S}_B$ values on a shell-cycle ladder first, and only then records wavelength/frequency/energy comparisons.

Data-support class. Hydrogen shell tests instantiate D0 exact shell-ratio fixtures and O0 shell observable-map comparisons. Error quantities are shell-ratio or spectral-map residuals.

The shared declaration is

$$\mathbf{a}_H \vdash (F_H, R_H, \partial R_H, \omega_H), \qquad B_H = (\partial F_H, \omega_H).$$

The shell family is indexed by an integer shell number

$$S_n, \qquad n \in \mathbb{Z}_{\geq 1}.$$

For the hydrogen-facing comparison records, the normalized shell-pressure ladder is declared by

$$\widehat{P}^D_{S_n} := -\frac{1}{n^2}.$$

For a downward transition $n \to m$, with $n > m$, declare the released normalized pressure magnitude

$$\Delta\widehat{P}^D(n \to m) := \widehat{P}^D_{S_n} - \widehat{P}^D_{S_m} = \frac{1}{m^2} - \frac{1}{n^2} > 0.$$

The signed state change is $\widehat{P}^D_{S_m} - \widehat{P}^D_{S_n} = -\Delta\widehat{P}^D(n \to m)$.

Hydrogen shell tests are finite $\mathfrak{S}_B$ values on a shell-cycle ladder.

Remark. The normalized ladder is a manual comparison choice. The main-note pressure definition remains $p^D = C_3\rho^D_\omega$.

Presentation policy. The AΩ ratios in the spectral tables are exact rational values. Decimal target ratios, wavelengths, frequencies, and percent errors are external presentation values. A z-score is recorded when the record carries an explicit uncertainty.

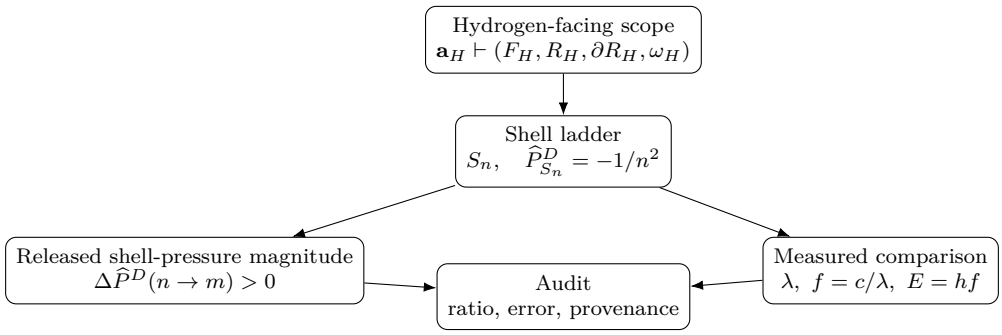


Figure 7: Hydrogen-facing shell-cycle ladder. The released shell-pressure magnitude is calculated before measured wavelengths and external energy-coordinate comparisons are attached.

7.2 Shell-radius comparison

The radius-ratio check records the standard shell scaling

$$\frac{r_n}{r_1} = n^2.$$

The radius-ratio record is carried as a geometry comparison record.

Table 18: Hydrogen-facing shell-radius ratios.

Shell	AΩ shell	Ratio target	Status
1	S_1	1	baseline
2	S_2	4	comparison record
3	S_3	9	comparison record
4	S_4	16	comparison record

7.3 Shell-pressure table

The Balmer comparison records use downward transitions into S_2. The normalized released-pressure magnitudes are

$$\Delta\widehat{P}^{D}(n \to 2) = \frac{1}{2^2} - \frac{1}{n^2}.$$

The ratio to the $3 \to 2$ record is

$$\mathcal{R}_{n\to 2} = \frac{\Delta\widehat{P}^{D}(n \to 2)}{\Delta\widehat{P}^{D}(3 \to 2)}.$$

Table 19: Hydrogen-facing Balmer released-pressure ratios. Targets are ratio comparisons against cited Balmer wavelengths.

Line	Transition	$\Delta\widehat{P}^{D}$	AΩ ratio	Target ratio	Error [%]
Hα	$3 \to 2$	5/36	1	1.00000000	0.000000
Hβ	$4 \to 2$	3/16	27/20	1.34999075	0.000685
Hγ	$5 \to 2$	21/100	189/125	1.51198839	0.000768
Hδ	$6 \to 2$	2/9	8/5	1.59999025	0.000609

7.4 Shell-energy transitions

The measured comparison is downstream. Given a cited vacuum wavelength λ, the comparison sequence is

$$f = \frac{c}{\lambda}, \qquad E = h_{\text{ext}} f.$$

This sequence uses the quarantine of App. B.

Table 20: Hydrogen Balmer comparison targets. Wavelength records follow the NIST Atomic Spectra Database target convention.

Line	Transition	λ_{vac} [nm]	f [Hz]	E [eV]
Hα	$3 \to 2$	656.46	$4.56680465 \times 10^{14}$	1.88867865
Hβ	$4 \to 2$	486.27	$6.16514401 \times 10^{14}$	2.54969869
Hγ	$5 \to 2$	434.17	$6.90495562 \times 10^{14}$	2.85566019
Hδ	$6 \to 2$	410.29	$7.30684292 \times 10^{14}$	3.02186742

7.5 Hydrogen spectral-series extension

Choice. The Lyman and Paschen records use the same normalized shell-pressure ladder as the Balmer records. The destination shell S_m is declared before each transition table. Ratio comparison is performed before wavelength, frequency, or energy export.

For a fixed destination shell S_m and a transition $n \to m$, with $n > m$, define the AΩ-internal series ratio

$$\mathcal{R}_{n\to m,A\leftrightarrow\Omega} = \frac{\Delta\widehat{P}^{D}(n \to m)}{\Delta\widehat{P}^{D}((m+1) \to m)}.$$

For wavelength comparison, use the inverse-wavelength relation

$$\mathcal{R}^{\lambda,\mathrm{tar}}_{n\to m} = \frac{\lambda_{(m+1)\to m}}{\lambda_{n\to m}}.$$

Thus the AΩ-internal pressure ratio is compared against the measured inverse-wavelength ratio. external frequency coordinate and energy export remain downstream.

7.5.1 Lyman series

For Lyman records, the destination shell is S_1. The baseline is $2 \to 1$, with

$$\Delta\widehat{P}^{D}(2 \to 1) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Table 21: Hydrogen Lyman released-pressure ratio audit. Vacuum wavelength targets are downstream comparison records carried by the NIST-ASD target convention.

Line	Transition	$\Delta\widehat{P}^{D}$	AΩ ratio	λ_{vac} [nm]	Target ratio	Error [%]
Lyα	$2 \to 1$	3/4	1	121.56701	1.00000000	0.000000
Lyβ	$3 \to 1$	8/9	32/27	102.57225	1.18518420	0.000083
Lyγ	$4 \to 1$	15/16	5/4	97.2537	1.24999882	0.000095
Lyδ	$5 \to 1$	24/25	32/25	94.9740	1.28000305	−0.000239
Lyϵ	$6 \to 1$	35/36	35/27	93.7803	1.29629581	0.000037
Limit	$\infty \to 1$	1	4/3	series limit	4/3	audit record

7.5.2 Paschen series

For Paschen records, the destination shell is S_3. Paα, Paβ, and Paϵ carry the NIST Hydrogen Handbook air wavelengths. Paγ and Paδ remain carried targets with source-row and medium bindings unverified. The baseline is $4 \to 3$, with

$$\Delta\widehat{P}^{D}(4 \to 3) = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}.$$

Table 22: Hydrogen Paschen released-pressure ratio audit. Wavelengths are carried comparison targets with the source-medium bindings declared in this subsection.

Line	Transition	$\Delta\widehat{P}^{D}$	AΩ ratio	λ_{tar} [nm]	Target ratio	Error [%]
Paα	$4 \to 3$	7/144	1	1875.101	1.00000000	0.000000
Paβ	$5 \to 3$	16/225	256/175	1281.807	1.46285751	−0.000025
Paγ	$6 \to 3$	1/12	12/7	1093.808	1.71428715	−0.000084
Paδ	$7 \to 3$	40/441	640/343	1004.938	1.86588725	0.000105
Paϵ	$8 \to 3$	55/576	55/28	954.597	1.96428545	0.000013
Limit	$\infty \to 3$	1/9	16/7	series limit	16/7	audit record

7.5.3 Series-ratio audit

Audit. A spectral-series record carries the destination shell S_m, the computed $\Delta\widehat{P}^D(n \to m)$, and the inverse-wavelength ratio $\lambda_{(m+1)\to m}/\lambda_{n\to m}$.

Provenance. Main §5.8.1; App. B.

Literature note. Hydrogen spectral-line targets: NIST Atomic Spectra Database [26]; Paschen Paα, Paβ, Paϵ air rows: NIST Hydrogen Handbook [27].

7.6 Error audit

For a transition record x, record

$$\Delta\mathcal{R}_x = \mathcal{R}_{x,A\leftrightarrow\Omega} - \mathcal{R}_x^{\mathrm{tar}}, \qquad \epsilon_x^{\mathrm{sgn}} = 100\frac{\Delta\mathcal{R}_x}{\mathcal{R}_x^{\mathrm{tar}}}.$$

The ratio audits in Table 19, Table 21 and Table 22 use this formula. Energy errors may be recorded only after the wavelength target and external conversion are declared.

7.7 Audit hooks

Audit. A hydrogen-facing shell record carries the released-pressure magnitude before external comparison, its ratio record, and the wavelength target with its declared source-medium binding.

Provenance. Main §5.8.1; App. B; §7.1.

Literature note. Hydrogen Balmer targets: NIST Atomic Spectra Database [26].

8 Boundary Current Applications

Data-support class. Boundary-current applications are D0 exact scoped-flux fixtures. External report maps carry separate comparison rows. Errors are Stokes residuals, moving-boundary identity residuals, current-split residuals, and sheddic-route conservation residuals.

Scope. This section applies the same scoped AΩD pressure-resolved flux calculation to wire, fluid, plasma, shell-electromagnetic comparison, and phase-fluid SADAR records. The declared boundary selects the application case of the declared-scope calculus.

8.1 Shared scoped-flux setup

Declare an application scope

$$\mathbf{a} \vdash (F, R, \partial R, \omega), \qquad B = (\partial F, \omega).$$

The scoped AΩD pressure-resolved flux is

$$\phi_{ij}(B) = p_{ij}^D(B) A_{ij}(B),$$

with

$$p_{ij}^D(B) = C_{3,ij}\rho_{\omega,ij}^D(B), \qquad \rho_{\omega,ij}^D(B) = \min(\rho_{ij}^D(B), |\omega|_{ij}),$$
$$A_{ij}(B) = \Delta\chi_{ij}^{\mathrm{lag}}(B) i_{ij}(B).$$

The slice flux and window current are

$$\Phi^D_{\partial R}(t; B|_t) = \sum_{e \in \partial R(t)} \phi_e(B|_t),$$

$$J^D_{\partial R}(\omega; B) = \sum_{t \in \omega} \Phi^D_{\partial R}(t; B|_t).$$

For moving boundaries use

$$\Phi^D_{\partial R,\mathrm{mov}}(t) = \Phi^D_{\mathrm{cross}}(t) - \Phi^D_{\mathrm{sweep}}(t) + \Phi^D_{\mathrm{shedding}}(t).$$

The directional SADAR quantity remains

$$\mathbf{A}_F(B) = \sum_{(i,j) \in F(B)} \phi_{ij}(B), \qquad \mathrm{motion}_{F,k+1}(B) = (S^W_{F,k}, S^W_{F,k+1}).$$

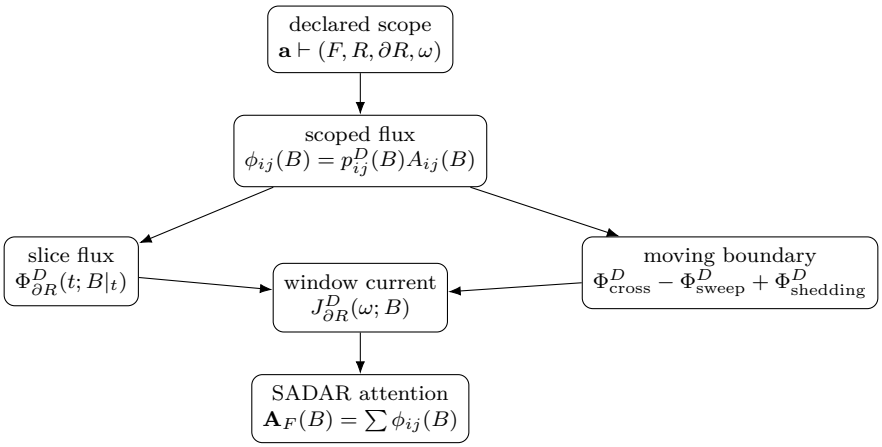


Figure 8: Boundary-current application spine. The same scoped flux produces slice flux, moving-boundary flux, window current, and the SADAR attention vector on the declared boundary/window.

8.2 Canonical worked substitution

Use the signed moving-boundary calculation

$$\Phi^D_{\mathrm{cross}} = 8, \qquad \Phi^D_{\mathrm{sweep}} = 3, \qquad \Phi^D_{\mathrm{shedding}} = 2.$$

Then

$$\Phi^D_{\partial R,\mathrm{mov}} = 8 - 3 + 2 = 7.$$

For a channel $(1, 2)$, declare

$$C_{3,12} = 3, \qquad \rho^D_{12}(B) = 5, \qquad |\omega|_{12} = 4, \qquad A_{12}(B) = \frac{1}{3}.$$

Then

$$\rho^D_{\omega,12}(B) = \min(5, 4) = 4,$$

$$p^D_{12}(B) = 3 \cdot 4 = 12,$$

$$\phi_{12}(B) = 12 \cdot \frac{1}{3} = 4, \qquad \phi_{21}(B) = -4.$$

Table 23: Canonical scoped-flux substitution.

Quantity	Value
Φ^D_{cross}	8
Φ^D_{sweep}	3
Φ^D_{shedding}	2
$\Phi^D_{\partial R,\text{mov}}$	$8-3+2=7$
$\rho^D_{\omega,12}(B)$	$\min(5,4)=4$
$p^D_{12}(B)$	12
$A_{12}(B)$	1/3
$\phi_{12}(B)$	4
$\phi_{21}(B)$	-4

8.3 Wire boundary current

Declare

$$\mathbf{a}_{\text{wire}} \vdash (F_{\text{wire}}, R_{\text{core}}, \partial R_{\text{shell}}, \omega_{\text{wire}}), \qquad B_{\text{wire}} = (\partial F_{\text{wire}}, \omega_{\text{wire}}).$$

Here F_{wire} is the conductor plus retained field shell, R_{core} is the conductor/core region, and $\partial R_{\text{shell}}$ is the shell boundary. The wire record computes

$$\Phi^D_{\partial R}(t; B|_t), \quad J^D_{\partial R}(\omega; B), \quad P^D_F(B), \quad \mathbf{A}_F(B), \quad X_{\text{shedding}}(B).$$

Remark. The wire case is a boundary-duon-current example. The surrounding field shell carries returned-current duration, pressure, and SADAR attention balance.

8.4 Fluid interface dynamics

Declare

$$\mathbf{a}_{\text{fluid}} \vdash (F_{\text{fluid}}, R_{\text{water}}, \partial R_{\text{interface}}, \omega_{\text{fluid}}), \qquad B_{\text{fluid}} = (\partial F_{\text{fluid}}, \omega_{\text{fluid}}).$$

The moving interface uses

$$\Phi^D_{\partial R,\text{mov}}(t) = \Phi^D_{\text{cross}}(t) - \Phi^D_{\text{sweep}}(t) + \Phi^D_{\text{shedding}}(t).$$

Here cross is current crossing the interface, sweep is current displaced by the moving boundary, and shedding is excess crossing or re-routing through the interface. The output record carries interface current, interface pressure, SADAR attention vector, and sheddic path.

8.5 Plasma shell dynamics

Declare

$$\mathbf{a}_{\text{plasma}} \vdash (F_{\text{plasma}}, R_{\text{core}}, \partial R_{\text{magnetic}}, \omega_{\text{plasma}}), \qquad B_{\text{plasma}} = (\partial F_{\text{plasma}}, \omega_{\text{plasma}}).$$

The plasma case uses the same scoped-flux calculation as the fluid interface, with the declared boundary now a shell boundary. The calculation changes by boundary declaration, not by introducing a new current object.

8.6 Shell electromagnetic comparison

The shell electromagnetic record is a downstream comparison record. It computes AΩ quantities first:

$$J^D_{\partial R}(\omega; B), \qquad P^D_F(B), \qquad \mathbf{A}_F(B), \qquad X_{\text{shedding}}(B).$$

Only after these AΩ quantities are declared may a measured-sector comparison be attached.

8.7 Phase-fluid SADAR rebalancing

Phase-fluid motion is the realized later response of the declared interface field to its committed SADAR attention input:

$$\mathrm{motion}_{F,k+1}(B) = (S^{W}_{F,k}, S^{W}_{F,k+1}).$$

The local sequence is

$$\text{phase bias} \longrightarrow \mathbf{A}_F(B) \longrightarrow \text{SADAR balance} \longrightarrow \text{shedding / reclosure.}$$

Push and pull are signed phase-biased components of the same vector.

8.8 Audit

The boundary-current application record carries the declared scope, antisymmetry $\phi_{ji}(B) = -\phi_{ij}(B)$, the signed window current as the sum of slice fluxes, and the moving-boundary cross, sweep, and shedding terms.

$$J^{D}_{\partial R}(\omega; B) \neq \sum_{t\in\omega} \Phi^{D}_{\partial R}(t; B|_t)$$

marks a window-current audit residual on the declared scope.

Provenance. Main (79); Main (56); Main (60); Main (98); Main (88).

Trace and response typing. The pressure-resolved flux in these applications is $\phi^{\mathrm{AOD}} = p^{D}A$. Normalized AFC memory/kernel flux remains separately typed. Pressure, signed phase bias, directional RCD and current are distinct committed coordinates. Their response changes state at the next transition; the attention vector alone does not constitute motion. Opposite oriented values can cancel while both bound histories remain retained.

9 Solar Observable-Map Fixtures

Scope. This section carries solar-system observable-map comparison fixtures. Each record declares its solar-system scope, names the observable map, carries a comparison value, attaches an external reference target, and records residual, percent error, source citation, and audit.

Data-support class. Solar records are O0 observable-map fixtures. Each row records map value, target, residual/error, and citation inside the declared observable-map support class.

9.1 Shared solar-system declaration

Declare a solar field scope

$$\mathbf{a}_{\odot} \vdash (F_{\odot}, R_{\odot}, \partial R_{\odot}, \omega_{\odot}), \qquad B_{\odot} = (\partial F_{\odot}, \omega_{\odot}).$$

For a body, signal, or transfer record x, declare a comparison boundary/window

$$B_x = (\partial F_x, \omega_x).$$

Each record carries a frozen O0 map-value comparison and its external reference target.

Solar-system records carry O0 observable-map comparisons.

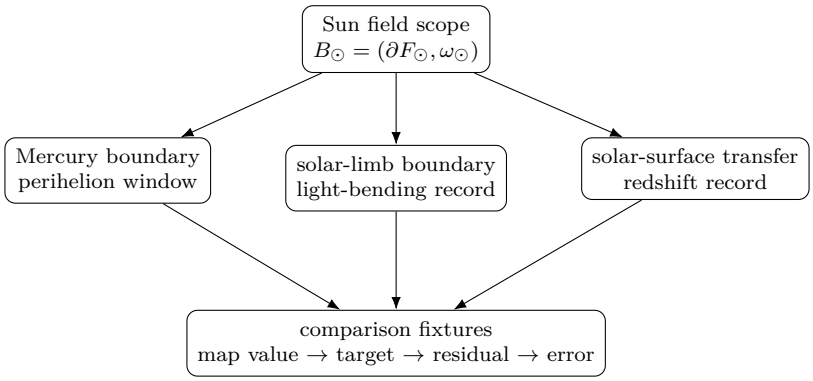


Figure 9: Solar-system observable-map comparison spine. The records carry map values, targets, residuals, percent errors, and source citations.

9.2 SADAR observable bridge

Native solar-system evaluation requires a Sun-field cycle family evaluated by $\mathfrak{S}_B$ and a declared observable comparison map with signature

$$\mathcal{O}_{x,A\leftrightarrow\Omega} = \mathcal{G}_x \left[RCD, \rho^D, \rho^D_\omega, p^D, \phi, \mathbf{A}_F, J^D_{\partial R}; B_x \right].$$

The three observable-map comparison fixtures record the following map signatures:

$$\Delta\varpi_{\mathrm{Hg},A\leftrightarrow\Omega} = \mathcal{G}_{\mathrm{peri}} \left[\mathbf{A}_{F_\odot}(B_{\mathrm{Hg}}), J^D_{\partial R}(\omega_{\mathrm{cy}}; B_{\mathrm{Hg}}) \right],$$

$$\delta_{\odot,A\leftrightarrow\Omega} = \mathcal{G}_{\mathrm{bend}} \left[\mathbf{A}_{F_\odot}(B_{\mathrm{bend}}), J^D_{\partial R}(\omega_{\mathrm{bend}}; B_{\mathrm{bend}}) \right],$$

$$z_{\odot,A\leftrightarrow\Omega} = \mathcal{G}_{\mathrm{red}} \left[\mathbf{A}_{F_\odot}(B_{\mathrm{red}}), J^D_{\partial R}(\omega_{\mathrm{red}}; B_{\mathrm{red}}) \right].$$

The rows below carry O0 comparison values and residual arithmetic. Map definitions and evaluated native $\mathfrak{S}_B$ input certificates are not supplied for these records.

9.3 Observable-map comparison table

Table 24: Solar-system observable-map comparison records (carried O0 values).

Record	AΩ value	Target	Residual	Abs. percent error
Mercury perihelion	42.9799 arcsec/cy	42.98 arcsec/cy	−0.0001 arcsec/cy	0.0002327%
Solar light bending	1.75001 arcsec	1.75 arcsec	0.00001 arcsec	0.0005714%
Solar redshift	2.11996×10^{-6}	2.12×10^{-6}	-4.0×10^{-11}	0.001887%

9.4 Mercury perihelion window

Declare

$$\mathbf{a}_{\mathrm{Hg}} \vdash (F_{\mathrm{Hg}}, R_{\mathrm{orbit}}, \partial R_{\mathrm{peri}}, \omega_{\mathrm{cy}}), \qquad B_{\mathrm{Hg}} = (\partial F_{\mathrm{Hg}}, \omega_{\mathrm{cy}}).$$

The observable-map value is

$$\Delta\varpi_{\mathrm{Hg},A\leftrightarrow\Omega} = 42.9799 \text{ arcsec/century}.$$

The target is

$$\Delta\varpi^{\mathrm{tar}}_{\mathrm{Hg}} = 42.98 \text{ arcsec/century}.$$

$$\Delta_\varpi = -0.0001 \text{ arcsec/century}, \qquad |\epsilon_\varpi| = 0.0002327\%.$$

Literature note. Mercury perihelion target: Will [36].

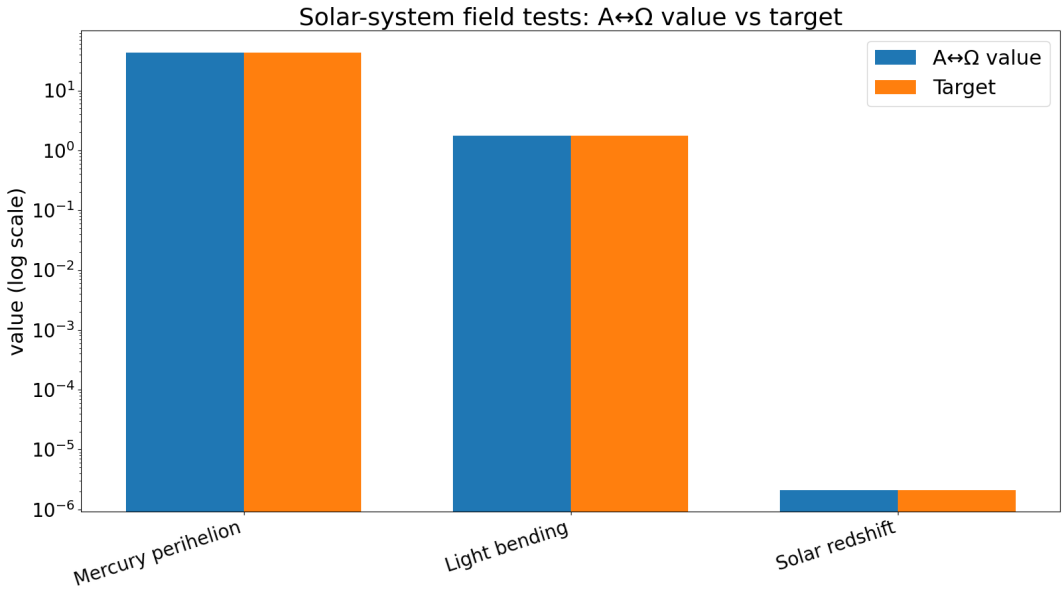


Figure 10: Solar-system carried observable-map values versus declared targets. The vertical axis is logarithmic; each record is compared only within its own unit/scale.

9.5 Solar light-bending boundary

Declare

$$\mathbf{a}_{\mathrm{bend}} \vdash (F_{\mathrm{signal}}, R_{\mathrm{path}}, \partial R_{\odot\mathrm{limb}}, \omega_{\mathrm{bend}}), \qquad B_{\mathrm{bend}} = (\partial F_{\mathrm{signal}}, \omega_{\mathrm{bend}}).$$

The observable-map value is

$$\delta_{\odot,A\leftrightarrow\Omega} = 1.75001 \text{ arcsec}.$$

The target is

$$\delta_{\odot}^{\mathrm{tar}} = 1.75 \text{ arcsec}.$$

$$\Delta_{\delta} = 0.00001 \text{ arcsec}, \qquad |\epsilon_{\delta}| = 0.0005714\%.$$

Literature note. Solar-limb light-bending target: NASA [19].

9.6 Solar redshift transfer boundary

Declare

$$\mathbf{a}_{\mathrm{red}} \vdash (F_{\mathrm{line}}, R_{\mathrm{surface}}, \partial R_{\mathrm{transfer}}, \omega_{\mathrm{red}}), \qquad B_{\mathrm{red}} = (\partial F_{\mathrm{line}}, \omega_{\mathrm{red}}).$$

The observable-map value is

$$z_{\odot,A\leftrightarrow\Omega} = 2.11996 \times 10^{-6}.$$

The target is

$$z_{\odot}^{\mathrm{tar}} = 2.12 \times 10^{-6}.$$

$$\Delta_z = -4.0 \times 10^{-11}, \qquad |\epsilon_z| = 0.001887\%.$$

A separate external reference prediction is

$$v_z^{\mathrm{tar}} = 633.1 \text{ m/s},$$

with the measured comparison

$$v_z^{\mathrm{meas}} = 638 \pm 6 \text{ m/s}.$$

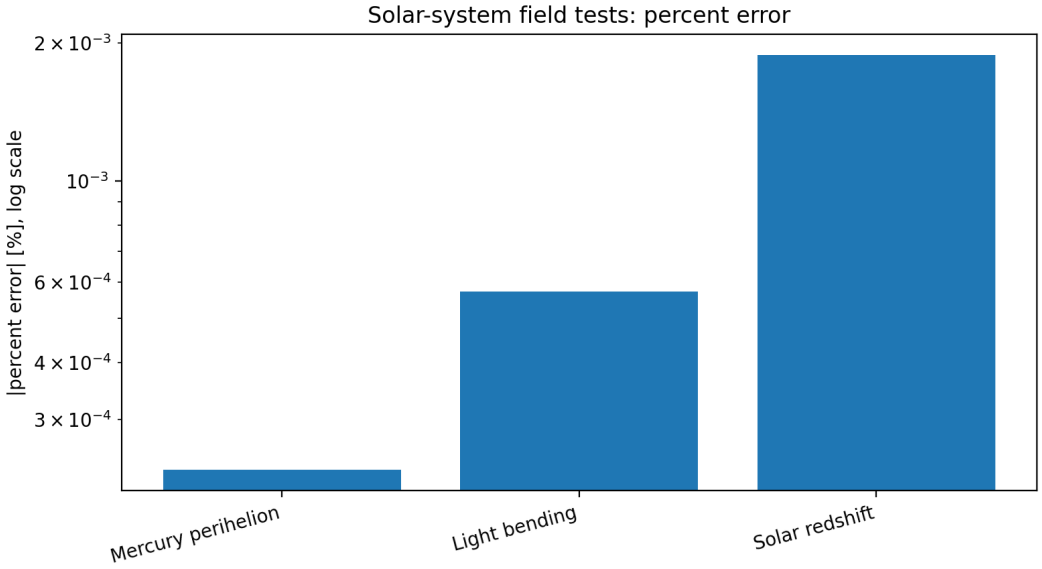


Figure 11: Solar-system field tests: absolute percent error.

Literature note. Solar gravitational-redshift reference prediction 633.1 m/s and measured comparison 638 ± 6 m/s: Heidelberg/ZAH [37].

9.7 Audit hooks

- Scope audit: each record declares $\mathbf{a} \vdash (F, R, \partial R, \omega)$ and $B = (\partial F, \omega)$.
- Comparison-fixture audit: each record includes observable-map value, target, residual, percent error, and status.
- Input audit: target values are cited in the record where they are used; residuals are computed from the declared observable-map value and declared target.

Provenance. Main §5.2.5; Main (79); Main (98); Main (60); App. B.

10 Field Dynamics Applications

Scope. This section uses the main-note field dynamics calculus as an application layer. SPARC uses the square-speed coordinate $U = V^2$. The Tau ring fixture uses S_τ and missing-burden diagnostics. Each score coordinate is declared inside its own support class.

Remark. Field-dynamics simulations are finite compressions of the AΩD ontology. A macroscopic field is simulated by declared clustering, finite boundary windows, support families, and value maps. Output residuals are read against the declared data support: boundary placement, density resolution, time-flow data, angular/radial momentum inputs, slosh windows, capacity seating, and sheddic-route declarations.

A large field-dynamics system is handled by declared clustering. A cluster records a retained family of distinctions and relations over a boundary/window. The cluster records its support expression, construction trace or inherited trace, capacity rule, coupling family, and route status. The cluster is a finite calculation container recording a declared AΩD form.

10.1 Data-support classes for manual simulations and comparisons

Dataset resolution selects the data-support class: radial rows instantiate G0; radial rows plus proxy density/current fields instantiate G1; three-dimensional density/velocity/current fields instantiate G2; time-ordered tracer positions/velocities instantiate G3. Each class reports residuals only in its declared coordinate. Table 25 records the manual classes used by this release.

Table 25: Data-support classes for manual simulations and comparisons.

Class	Name	Declared distinctions	Relations induced	Residual
D0	exact internal fixture	exact integer gates, channels, comparator bins	internal returned-current relations	δ_3 tally
O0	observable-map fixture	observable-map value and measured-sector target	external comparison map	target residual
G0	radial shell projection	galaxy, radial bin, radius, speed, radial gas/disk/bulge support	radial shell field-dynamics relations	U residuals
G1	radial-medium projection	G0 plus radial density/current proxy fields	radial medium-support relations	U + proxy
G2	3D shell field-dynamics retention	radial, angular, vertical cells; density, velocity, current, background support	local/inter/core/long/bg/J range SADAR	retention / field
G3	time-resolved tracer-current simulation	G2 plus tracer positions/velocities through time windows	slosh, hinge shift, chiral flip, partial coupling	path / phase
L0	exact lensing toy fixture	exact gates and comparator tallies	field-ring δ_3 relations	δ_3 tally
L1	radial lens-medium diagnostic	radial medium proxies without image targets	medium-class relations	medium tallies
L2	target-acquisition lane	image-count, Einstein-radius, image-position, or shear targets	target-side quarantine	acquisition
L3	full lensing input package	declared medium fields, radiation-current gates, and target side	prediction before target join; target scoring after freeze	lens residuals

Remark. This table records which distinctions are instantiated by a manual calculation. Projection and marginalization maps record comparison coordinates and uncertainty.

10.2 Data-support and uncertainty budget

A field-dynamics diagnostic records its data-support and uncertainty budget before residuals are interpreted. The budget records the declared distinction family, support class, uncertainty component, and report target.

Table 26: Field dynamics data-support classes.

Distinction family	Symbol	Declared data support	Support class
Density	ϵ_ρ	$\rho(R,z,\theta,t)$ or radial proxy	3D / 2D / radial / proxy
Velocity	ϵ_v	$v(R,z,\theta,t)$ or radial speed	vector / radial / proxy
Time window	ϵ_ω	cadence and duration	measured / declared / proxy
Boundary	ϵ_∂	$\partial_T R_c(t)$	3D / radial / proxy
Capacity	ϵ_C	C^{cap}	typed / scalar / inherited / proxy
Coupling	ϵ_E	$E^{\mathfrak{S}}(B)$	full / radial / proxy / ablation cut
Sheddic route	ϵ_X	SheddicPath	declared / inferred
External map	ϵ_K	K_{ext} or λ	frozen / calibrated

Table 27: Field dynamics uncertainty and report targets.

Distinction family	Symbol	Uncertainty	Report target
Density	ϵ_ρ	σ_ρ when declared	field or U
Velocity	ϵ_v	σ_v	current, path, or U
Time window	ϵ_ω	σ_ω	slosh / retention / path
Boundary	ϵ_∂	σ_∂	boundary audits
Capacity	ϵ_C	σ_C	retention / seating
Coupling	ϵ_E	σ_E	retention / projection
Sheddic route	ϵ_X	σ_X	route / slosh
External map	ϵ_K	σ_{map}	external comparison

Remark. A scored diagnostic is interpreted within its declared support class. Projection and marginalization comparisons report their uncertainty records.

Ablation cut. An ablation cut is a declared data-support cut with one distinction family intentionally omitted or marginalized. It records projection or marginalization residuals against a comparison support class. The evaluated AΩD form remains the declared form for that record.

10.3 Scoped error forms by data-support class

For a G0 radial shell projection, the score coordinate remains square speed:

$$U^{\mathrm{obs}}_{g,j} = (V^{\mathrm{obs}}_{g,j})^2, \qquad \Delta U_{g,j} = U^{A\Omega}_{g,j} - U^{\mathrm{obs}}_{g,j}.$$

If projection or input uncertainty is declared, the total uncertainty is recorded by

$$\sigma^2_{\mathrm{total},g,j} = \sigma^2_{\mathrm{obs},g,j} + \sigma^2_{\mathrm{proj},g,j} + \sigma^2_{\mathrm{input},g,j}, \qquad z^U_{g,j} = \frac{U^{A\Omega}_{g,j} - U^{\mathrm{obs}}_{g,j}}{\sigma_{\mathrm{total},g,j}}.$$

For a G1 radial-medium projection, proxy-support residuals may be recorded as

$$\epsilon_{\rho,j} = \rho^{A\Omega}_j - \rho^{\mathrm{proxy}}_j, \qquad \epsilon_{J,j} = J^{A\Omega}_j - J^{\mathrm{proxy}}_j.$$

For a G2 three-dimensional field-dynamics retention fixture, exact retention tallies and field residuals may be recorded as

$$\Delta^{\mathbb{Z}}_k = T_k - O_k, \qquad \delta_{3,k} = (s^{(3)}_k, m_k),$$
$$\epsilon_\rho(c) = \rho^{A\Omega}_c - \rho^{\mathrm{obs}}_c, \qquad J^{\mathrm{obs}}_c = \rho^{\mathrm{obs}}_c v^{\mathrm{obs}}_c, \qquad \epsilon_J(c) = J^{A\Omega}_c - J^{\mathrm{obs}}_c.$$

For vector targets, the fixture declares $\varepsilon \geq 0$ with the units of $\|J^{A\Omega}_c\|\,\|J^{\mathrm{obs}}_c\|$ and a positive denominator. The dimensionless regularized alignment score is

$$c_{\varepsilon,c} = \frac{J^{A\Omega}_c \cdot J^{\mathrm{obs}}_c}{\|J^{A\Omega}_c\|\,\|J^{\mathrm{obs}}_c\| + \varepsilon}.$$

For a G3 time-resolved tracer-current fixture, path and velocity residuals such as ϵ_x and ϵ_v are declared against path-coordinate targets. RMSE reports are used only when the fixture declares metric units and path-coordinate targets.

10.4 Two-dimensional slice / three-dimensional field toy

This toy records how data-support scope changes the reported value when a three-dimensional field is represented by a two-dimensional slice or when the boundary is placed improperly. It is a manual diagnostic container that records declared comparison values.

Let B_{3D} be the declared three-dimensional boundary/window and let B_{2D} be the slice boundary obtained by suppressing the vertical/time-flow input. Let B_{bad} be an intentionally improper setup boundary. The same calculation is evaluated three ways:

$$U^{3D}_{A\leftrightarrow\Omega} = \mathsf{Q}^{U}_{\mathrm{ext}}(D^{\mathrm{BF}}(B_{3D})), \qquad U^{2D}_{A\leftrightarrow\Omega} = \mathsf{Q}^{U}_{\mathrm{ext}}(D^{\mathrm{BF}}(B_{2D})), \qquad U^{\mathrm{bad}}_{A\leftrightarrow\Omega} = \mathsf{Q}^{U}_{\mathrm{ext}}(D^{\mathrm{BF}}(B_{\mathrm{bad}})).$$

The scoped diagnostics are

$$\Delta U_{2D\to 3D} = U^{2D}_{A\leftrightarrow\Omega} - U^{3D}_{A\leftrightarrow\Omega}, \qquad \Delta U_{\partial} = U^{\mathrm{bad}}_{A\leftrightarrow\Omega} - U^{3D}_{A\leftrightarrow\Omega}.$$

For exact internal bins, the same comparison may be recorded by

$$\Delta^{\mathbb{Z}}_{k} = T^{\mathrm{bad}}_{k} - T^{3D}_{k}, \qquad \delta_{3,k}(T^{\mathrm{bad}}, T^{3D}) = (s^{(3)}_{k}, m_k).$$

Table 28: Toy data-support cases.

Case	Boundary / data	Diagnostic	Interpretation
3D declared field	B_{3D}, $\rho(R,z,\theta,t)$, velocity and window data	baseline value $U^{3D}_{A\leftrightarrow\Omega}$ or T^{3D}	retained support reference
2D slice	B_{2D}, vertical/time-flow suppressed	$\Delta U_{2D\to 3D}$	slice projection residual
Improper boundary	B_{bad}, wrong boundary/capacity/route	ΔU_{∂} or $\delta_3(T^{\mathrm{bad}}, T^{3D})$	boundary-support residual

Remark. This toy shows why field dynamics must record boundary declarations and data-support class. A two-dimensional slice and a three-dimensional field are different declared support classes; residuals between them are projection or boundary-support residuals before any external target comparison is interpreted.

10.5 3D Brownian phase-cycle δ_3 fixture

Data-support class. This Brownian / tracer-motion fixture is a G3 time-resolved tracer-current fixture when the track declares time-ordered three-dimensional positions. A two-dimensional microscopy slice records its projection map or ablation cut.

This fixture uses integerized three-dimensional trajectory data. A bacterium or tracer track is binned into $\mathbb{Z}^3$; lagged displacements produce the active integer squared displacement

$$Q^{(n)}_i = (a^{(n)}_i)^2 + (b^{(n)}_i)^2 + (c^{(n)}_i)^2 \in \mathbb{Z}_{\geq 0}.$$

Mass and size supply a candidate monon-beat scale β_a. The trajectory is tested by displacement-bin counts, phase-cycle counts $Q^{(n)}_i$ mod β_a, octant direction counts, integer ratios, and exact δ_3 residuals. The full 3D coordinate phase-cycle specification is given in App. C.

The exact fixture path is

$$z_i \in \mathbb{Z}^3 \to \Delta z^{(n)}_i \to Q^{(n)}_i \to \theta^{(n)}_i \to O^{(n)}_{\theta} \to \delta_3.$$

The retained trace path is

$$\Delta z_i^{(n)} \to HT(h_i; B_{\mathrm{mot}}) \to RD_{h_i} \to RCD_{h_i} \to \rho^D_{\omega,h_i} \to \mathfrak{S}_B.$$

Continuous summaries belong after the exact fixture as presentation or external maps.

Source artifacts. The source package includes the Brownian application script under `manual/scripts` and the Brownian CSV rows under `manual/data/brownian`:

`aod_brownian_3d_phase_delta3.py`, `displacement_q_counts.csv`, `phase_cycle_delta3.csv`, `octant_delta3.csv`, `integer_motion_ratios.csv`.

These Brownian application artifacts are aliases of the generic 3D coordinate phase-cycle fixture specified in App. C.

Remark. Active coordinate: integerized three-dimensional displacement and phase-cycle count data,

$$z_i \in \mathbb{Z}^3 \to \Delta z_i^{(n)} \to Q_i^{(n)} \to \theta_i^{(n)} \to O_\theta^{(n)} \to \delta_3.$$

Continuous Brownian and diffusion-coefficient summaries are downstream presentation maps.

10.5.1 Run-tumble tracer-current fixture

Data-support class. This run-tumble record is a G3 time-resolved tracer-current fixture. Its exact integer tallies are recorded as D0-style internal fixture audit coordinates when a comparator is declared.

This fixture records a finite D.E.C. trace over four declared Hamming–1 edge slots, stores the selected signed edge steps as a four-channel accumulated trace, and projects that retained trace into a declared three-dimensional coordinate map before phase-cycle, octant, and δ_3 summaries are formed. Declare

$$B_{\mathrm{RT}} = (\partial_T F_{\mathrm{RT}}, \omega_{20}), \qquad \omega_{20} = \{0, 1, \ldots, 20\}, \qquad E_4 = \{e_1, e_2, e_3, e_4\}.$$

The accumulated trace is

$$h_t \in \mathbb{Z}^4, \qquad h_0 = (0, 0, 0, 0),$$

with update

$$h_t = h_{t-1} + \sigma_t \mathbf{e}_{j_t}, \qquad j_t \in \{1, 2, 3, 4\}, \qquad \sigma_t \in \{-1, +1\}.$$

The run/tumble mode is

$$m_t \in \{\mathrm{run}, \mathrm{tumble}\}.$$

Run retains the active channel over the declared short window. Tumble reseats the active channel by the declared isotropic successor rule. The generated trace includes the initial row $t = 0$ with $h_0 = (0, 0, 0, 0)$; run/tumble mode is assigned for the update rows $t \geq 1$. The projection map is declared before any three-dimensional quantity is reported:

$$\pi_3 : \mathbb{Z}^4 \to \mathbb{Z}^3, \qquad z_t = \pi_3(h_t).$$

For the released fixture,

$$\pi_3(h_1, h_2, h_3, h_4) = (h_1, h_2, h_3).$$

The fourth trace channel remains part of the Q_4^2 audit even when it is not carried by the released π_3 projection. The active three-dimensional squared-displacement audit quantity is

$$Q^2_{3D}(t) = x_t^2 + y_t^2 + z_t^2, \qquad \pi_3(h_t) = (x_t, y_t, z_t).$$

The four-channel audit displacement is

$$Q_4^2(t) = h_{1,t}^2 + h_{2,t}^2 + h_{3,t}^2 + h_{4,t}^2.$$

Here Q_{3D}^2 and Q_4^2 are local squared-displacement audit quantities, not Q^D pending-overhang notation. The fixture records Q_{3D}^2, Q_4^2, phase residues Q_{3D}^2 mod β_a, octant counts, integer ratios, and exact δ_3 residuals. Continuous Brownian, diffusion, and stochastic-calculus summaries are downstream presentation maps.

Table 29: Run-tumble tracer-current declaration.

Item	Declaration
support	four Hamming–1 edge slots $E_4 = \{e_1, e_2, e_3, e_4\}$
accumulated trace	$h_t \in \mathbb{Z}^4$
projection	$\pi_3 : \mathbb{Z}^4 \to \mathbb{Z}^3$
projected coordinate	$z_t = \pi_3(h_t) \in \mathbb{Z}^3$
mode	$m_t \in \{\text{run}, \text{tumble}\}$
run rule	retain selected channel over declared short window
tumble rule	reseat selected channel by isotropic successor rule
3D displacement	$Q_{3D}^2(t) = x_t^2 + y_t^2 + z_t^2$
four-channel audit	$Q_4^2(t) = \sum_{r=1}^4 h_{r,t}^2$
reports	phase counts, octant counts, integer ratios, δ_3

Table 30: Run-tumble tracer-current 20-tick exact trace.

t	mode	edge	σ	h_t	$\pi_3(h_t)$	Q_{3D}^2	Q_4^2	θ	oct.
0	init	-	–	(0,0,0,0)	(0,0,0)	0	0	0	0
1	run	e1	+1	(1,0,0,0)	(1,0,0)	1	1	1	0
2	run	e1	+1	(2,0,0,0)	(2,0,0)	4	4	4	0
3	tumble	e2	+1	(2,1,0,0)	(2,1,0)	5	5	0	0
4	run	e2	+1	(2,2,0,0)	(2,2,0)	8	8	3	0
5	tumble	e3	-1	(2,2,-1,0)	(2,2,-1)	9	9	4	4
6	run	e3	-1	(2,2,-2,0)	(2,2,-2)	12	12	2	4
7	tumble	e4	+1	(2,2,-2,1)	(2,2,-2)	12	13	2	4
8	run	e4	+1	(2,2,-2,2)	(2,2,-2)	12	16	2	4
9	tumble	e1	-1	(1,2,-2,2)	(1,2,-2)	9	13	4	4
10	run	e1	-1	(0,2,-2,2)	(0,2,-2)	8	12	3	4
11	tumble	e2	-1	(0,1,-2,2)	(0,1,-2)	5	9	0	4
12	run	e2	-1	(0,0,-2,2)	(0,0,-2)	4	8	4	4
13	tumble	e3	+1	(0,0,-1,2)	(0,0,-1)	1	5	1	4
14	run	e3	+1	(0,0,0,2)	(0,0,0)	0	4	0	0
15	tumble	e4	-1	(0,0,0,1)	(0,0,0)	0	1	0	0
16	run	e4	-1	(0,0,0,0)	(0,0,0)	0	0	0	0
17	tumble	e1	+1	(1,0,0,0)	(1,0,0)	1	1	1	0
18	run	e1	+1	(2,0,0,0)	(2,0,0)	4	4	4	0
19	tumble	e2	+1	(2,1,0,0)	(2,1,0)	5	5	0	0
20	run	e2	+1	(2,2,0,0)	(2,2,0)	8	8	3	0

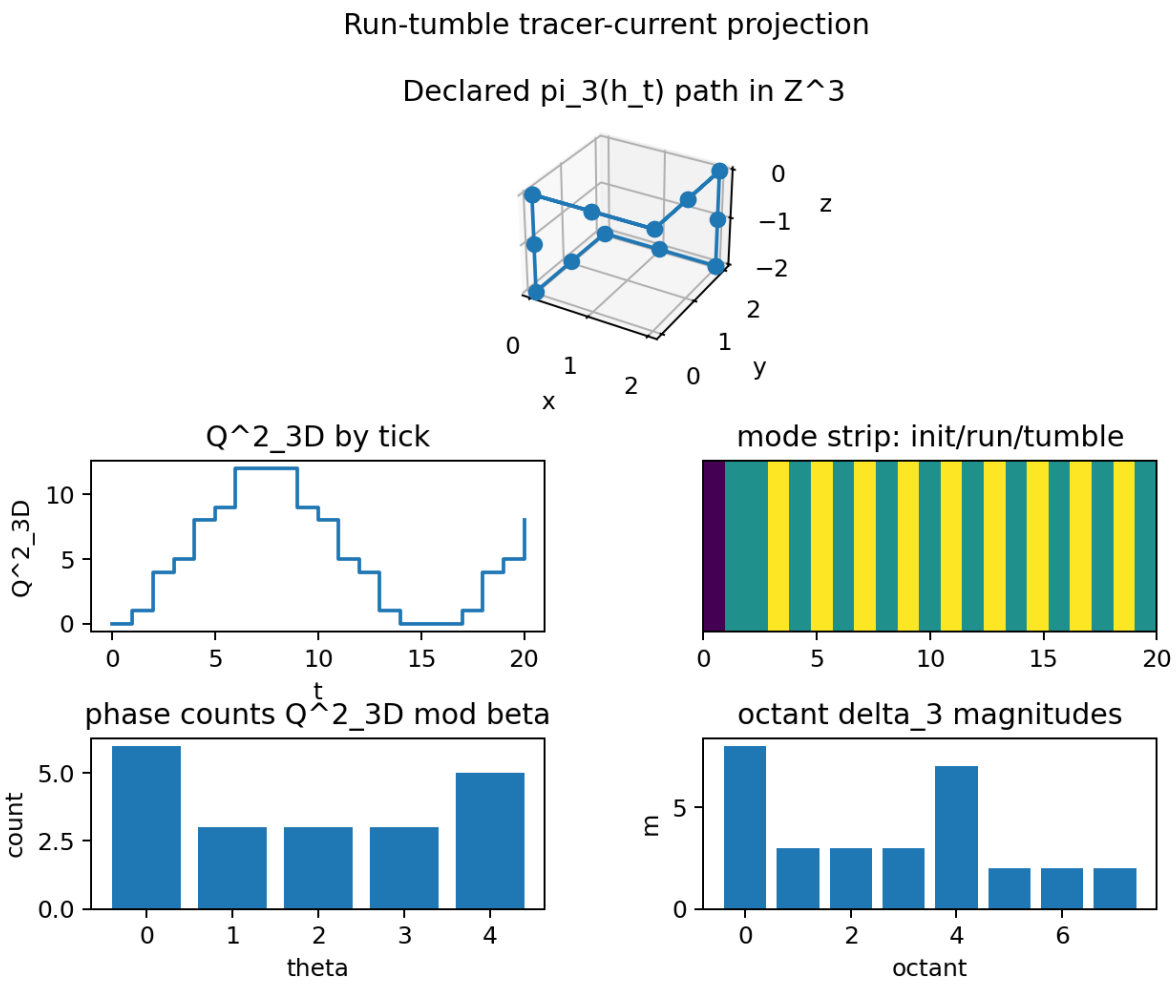


Figure 12: Run-tumble tracer-current projection. The plotted path is the declared $\mathbb{Z}^3$ projection of the retained $\mathbb{Z}^4$ accumulated trace. The finite Q_4 support remains the Hamming–1 edge-slot support.

Source artifacts. The source package includes the deterministic run-tumble generator `manual/scripts/aod_run_tumble_3d_trace.py`, the generated trace table, the run-tumble CSV outputs under `manual/data/run_tumble`, and the projection figure under `manual/figures`.

10.6 SPARC square-speed diagnostic

The SPARC five-galaxy square-speed score is the current G0 radial shell field-dynamics diagnostic lane. This record uses SPARC radial rotation-curve rows, scores the coordinate $U = V^2$, forms $\sigma_U = 2V^{\mathrm{obs}}\sigma_V$ from the declared data uncertainty, and records $\Delta U = U_{A\leftrightarrow\Omega} - U^{\mathrm{obs}}$ inside the G0 support class. The SPARC benchmark supplies declared galaxy records

$$\mathcal{D}_{\mathrm{SPARC}} = \{G_i, r_{ij}, V_{ij}^{\mathrm{obs}}, \sigma_{V,ij}, V_{ij}^{\mathrm{gas}}, V_{ij}^{\mathrm{disk}}, V_{ij}^{\mathrm{bulge}}, \ldots\}.$$

The score coordinate is square speed:

$$U_{g,j}^{\mathrm{obs}} = (V_{g,j}^{\mathrm{obs}})^2, \qquad \sigma_{U,g,j} = 2V_{g,j}^{\mathrm{obs}}\sigma_{V,g,j}.$$

The uncertainty propagation is external benchmark arithmetic. The declared score target is $U_{g,j}^{\mathrm{obs}}$, and benchmark input columns remain input data until the report map is evaluated. This G0 diagnostic uses

nonnegative component proxies

$$U^k_{g,j} := (V^k_{g,j})^2, \qquad k \in \{\text{gas}, \text{disk}, \text{bulge}\}.$$

SPARC's signed gas contribution is $V^{\text{gas}}|V^{\text{gas}}|$. The retained proxy $U^{\text{gas}} = (V^{\text{gas}})^2$ preserves its magnitude and removes the contribution sign for negative source entries.

$$U^{\text{bar}}_{g,j} = U^{\text{gas}}_{g,j} + \Upsilon_d U^{\text{disk}}_{g,j} + \Upsilon_b U^{\text{bulge}}_{g,j}.$$

The release constants are frozen before residuals are computed:

$$\Upsilon_d = \frac{1}{2}, \qquad \Upsilon_b = \frac{7}{10}, \qquad K^U_{\text{ext}} = 8, \qquad J_{\text{norm}} = 1.$$

The retained-window ratio uses the scalar duration accessor, not the RCD coupling object:

$$\lambda^{\text{ret}}_{g,j} := \frac{\rho^D_{\omega,g,j}}{\rho^D_{g,j}}.$$

The released field-dynamics support term is

$$D^{\text{BF}}_{g,j} = C_{\text{cap},g,j}\lambda^{\text{ret}}_{g,j}\left(1 + \frac{\text{density}_{g,j}}{16}\right)\left(1 + \frac{\text{gas/star}_{g,j}}{16}\right)\left(1 + \frac{\text{bobbing}_{g,j}}{64}\right)J_{\text{norm}}.$$

In the five-galaxy G0/G1 diagnostic lane, the declared primary projection map uses

$$C_{\text{cap},g,j} = r_{g,j}\left(|V^{\text{gas}}_{g,j}| + |V^{\text{disk}}_{g,j}| + |V^{\text{bulge}}_{g,j}|\right), \quad \lambda^{\text{ret}}_{g,j} = 1, \quad \text{density}_{g,j} = \text{gas/star}_{g,j} = \text{bobbing}_{g,j} = 0.$$

This neutral modifier convention belongs to the declared radial projection map. G1/G2/G3 packages carry angular, vertical, or time-flow fields through declared source or proxy rows. The external square-speed map is

$$U_{g,j,A\leftrightarrow\Omega} = U^{\text{bar}}_{g,j} + K^U_{\text{ext}} D^{\text{BF}}_{g,j}.$$

Here K^U_{ext} is a quarantined external map coefficient frozen before scoring and used only by the external-comparison layer.

$$\text{scope}(K^U_{\text{ext}}) = \text{external square-speed value map}, \qquad [K^U_{\text{ext}}] = \frac{(\text{km/s})^2}{\text{kpc}\cdot(\text{km/s})} = \frac{\text{km/s}}{\text{kpc}}.$$

Literature note. SPARC benchmark: SPARC database [33], Lelli et al. [14].

Data-support class. The SPARC five-galaxy lane instantiates a G0 radial shell data-support class. G0 active input fields are $G_i, r_{ij}, V^{\text{obs}}_{ij}, \sigma_{V,ij}, V^{\text{gas}}_{ij}, V^{\text{disk}}_{ij}, V^{\text{bulge}}_{ij}$. The report coordinate is $U = (V^{\text{obs}})^2$, $\sigma_U = 2V^{\text{obs}}\sigma_V$, and $\Delta U = U_{A\leftrightarrow\Omega} - U^{\text{obs}}$. G1/G2/G3 packages carry angular, vertical, or time-flow fields through declared source or proxy rows.

10.7 SPARC five-galaxy scored records

The five scored records are

$$\{\text{NGC2403}, \text{NGC3198}, \text{NGC6503}, \text{NGC2841}, \text{DDO154}\}.$$

Data files, scored outputs, and the regeneration script are included in the source package:

`manual/data/sparc`,
`manual/data/derived/sparc_five_galaxy_per_bin.csv`,
`manual/data/derived/sparc_five_galaxy_per_galaxy.csv`,
`manual/scripts/score_sparc_field_dynamics.py`.

Remark. The SPARC five-galaxy scored records are generated from the included rotation-curve files and the frozen declaration constants listed in this section.

$$\Delta U_{g,j} = U_{g,j,A\leftrightarrow\Omega} - U^{\mathrm{obs}}_{g,j}, \qquad z^{U}_{g,j} = \frac{\Delta U_{g,j}}{\sigma_{U,g,j}}.$$

$$\chi^2_{U,g} = \sum_j (z^{U}_{g,j})^2, \quad MAE_{U,g} = \frac{1}{N_g}\sum_j |\Delta U_{g,j}|, \quad MSE_{U,g} = \frac{1}{N_g}\sum_j (\Delta U_{g,j})^2.$$

$$MAPE_{U,g} = \frac{100}{N_g}\sum_j \left|\frac{\Delta U_{g,j}}{U^{\mathrm{obs}}_{g,j}}\right|, \qquad U^{\mathrm{obs}}_{g,j} > 0.$$

Bins are excluded only by a predeclared data-quality rule or by a predeclared denominator floor.

Table 31: SPARC five-galaxy scored records. Data-support class is G0 for this radial shell projection lane; the uncertainty policy is observation-only in this release.

G	Class	σ policy	N	χ^2_U	MAE_U	MSE_U	$MAPE_U$ [%]
DDO154	G0	obs-only	12	10893.83	697.2	602049.7	41.42
NGC2403	G0	obs-only	73	22038.64	2486.2	7212253.2	23.31
NGC2841	G0	obs-only	50	3235.74	23910.7	674670051.2	26.47
NGC3198	G0	obs-only	43	6352.96	5478.2	62935093.1	37.51
NGC6503	G0	obs-only	31	718.93	1697.4	3689898.7	13.50

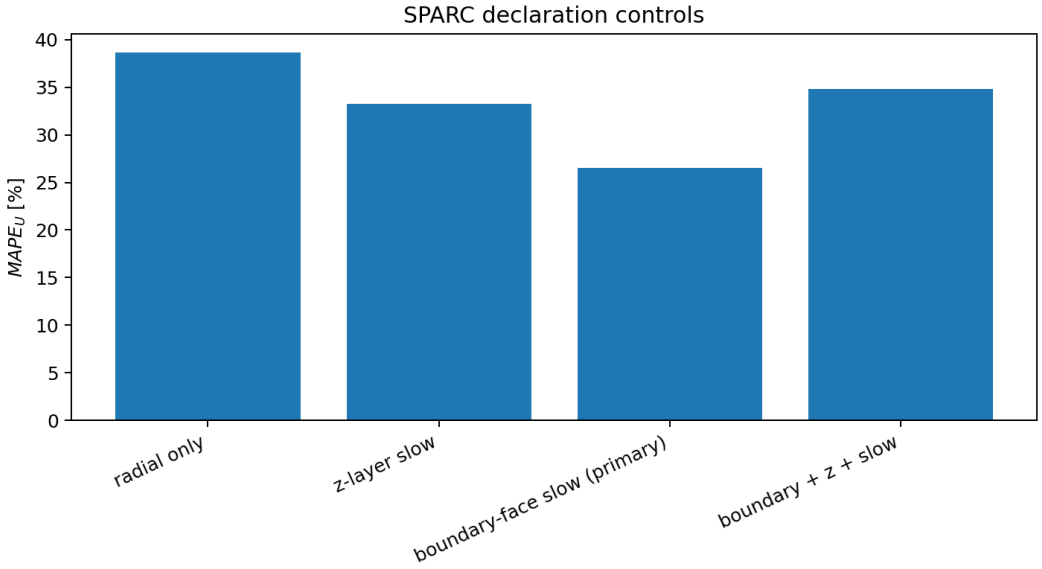


Figure 13: SPARC carried declaration-control scores. The four values in `sparc_declaration_controls.csv` are frozen comparison records; their variant recipes and aggregation rule are not supplied. Primary per-bin and per-galaxy scores are computed separately by the declared map.

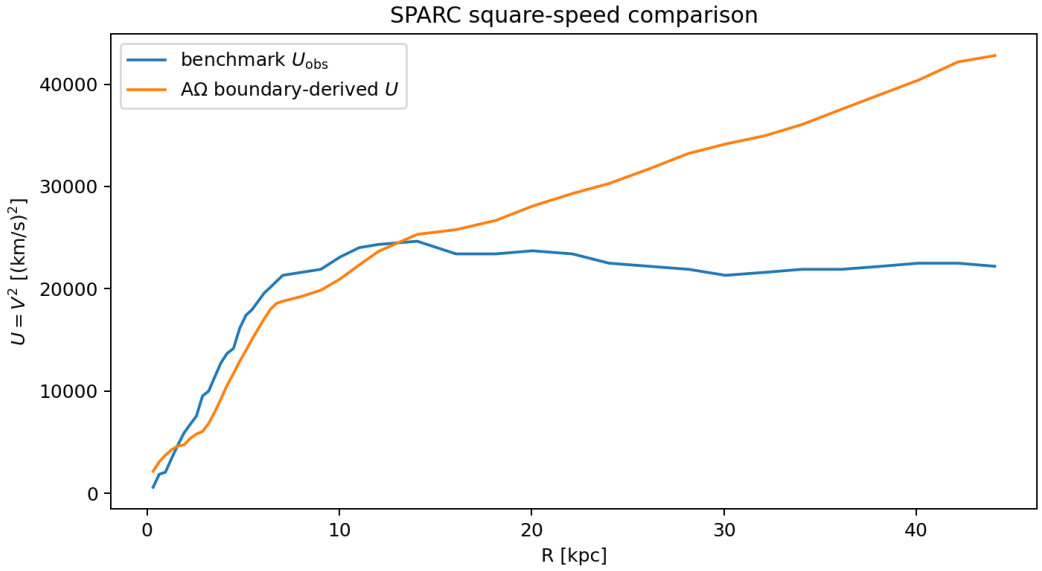


Figure 14: SPARC square-speed comparison for the NGC3198 display record. The scored coordinate is $U = V^2$.

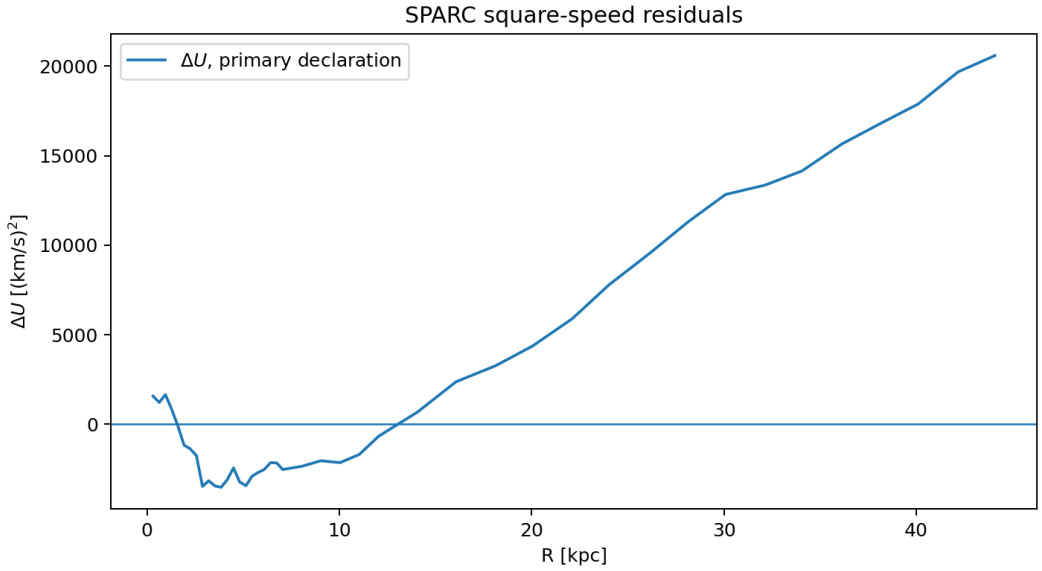


Figure 15: SPARC square-speed residuals for the NGC3198 display record. The residual is $\Delta U = U_{A\leftrightarrow\Omega} - U_{\mathrm{obs}}$.

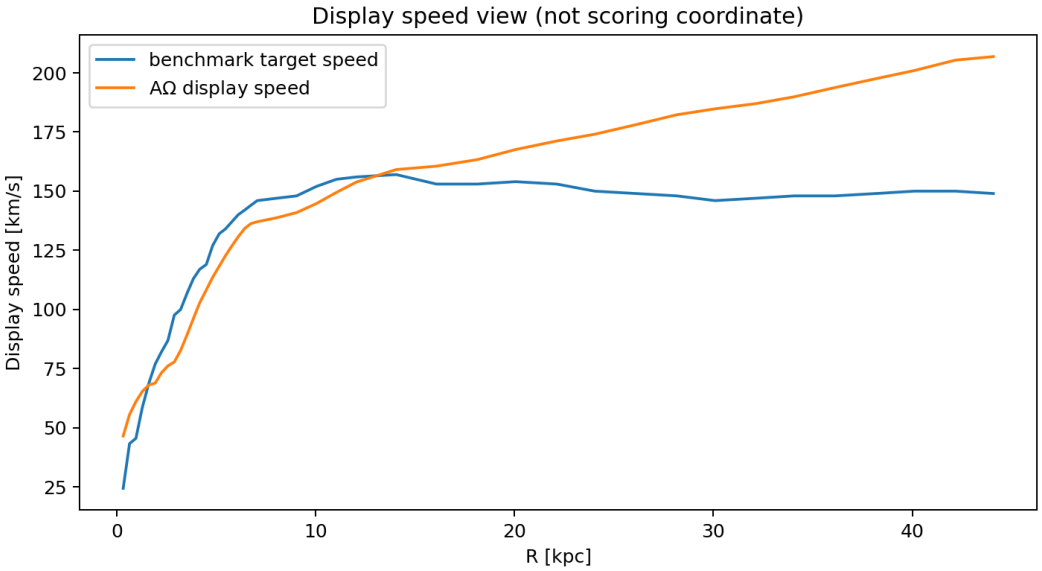


Figure 16: Display speed view. This presentation view uses $V_{\mathrm{disp}} = \sqrt{U}$ after scoring; the score coordinate is U.

10.8 Tau ring diagnostic fixture

The Tau ring fixture carries frozen S_τ, $S_{\tau,\mathrm{ring}}$, missing-fraction, and winner-margin records from `manual/data/derived/tau_ring_diagnostic.csv`.

$$\mathcal{K}_\tau = \{1{:}2{:}8, 1{:}2{:}9_{\mathrm{yro}}, 1{:}2{:}10, 3{:}3{:}8\}.$$

$$K_\tau^\star = \arg\min_{K\in\mathcal{K}_\tau} |S_{\tau,\mathrm{ring}}(K)|.$$

Winner lookup in the supplied diagnostic selects

$$\boxed{K_\tau^\star = 1{:}2{:}9_{\mathrm{yro}} = \text{Dimonnanyro.}}$$

For the primary lane,

$$1{:}2{:}9: \qquad S_\tau = -0.036807, \qquad S_{\tau,\mathrm{ring}} = -0.105804.$$

The missing-burden diagnostic records the exact ratio

$$\mathrm{miss}_\tau(1{:}2{:}9_{\mathrm{yro}}) = \frac{31}{50}, \qquad \mathrm{missing_fraction_display} = 0.62.$$

Precision note. Support identities and candidate declarations are exact. Plotted scores are presentation decimals carried by the frozen diagnostic CSV.

Source artifacts. The depth-3 rows in `manual/data/tau/aod_tau_fractal_ring_summary.csv` preserve this winner. The originating calculation recipe is not supplied. Compact missing fractions and depth-indexed source fractions remain separate records; their adapter is not supplied.

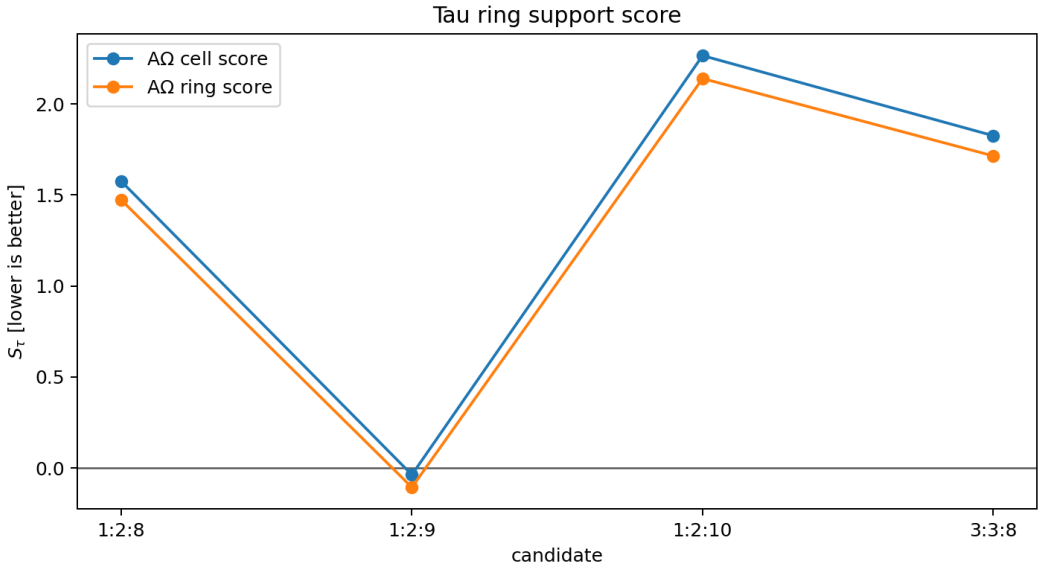


Figure 17: Tau ring support score. The ring diagnostic preserves $1{:}2{:}9_{\mathrm{yro}}$ / Dimonnanyro as the closest candidate.

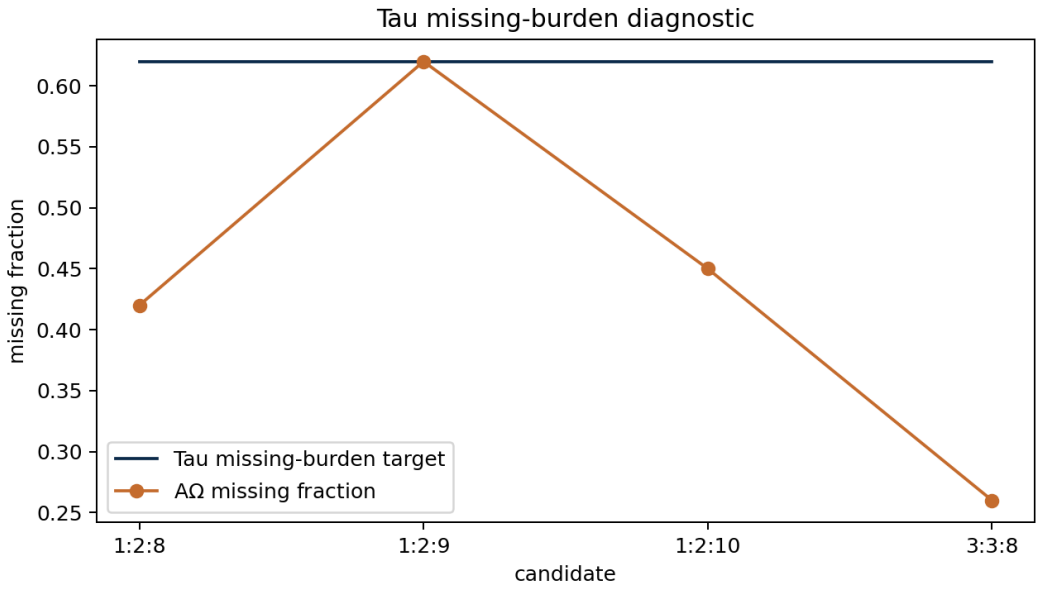


Figure 18: Tau missing-burden diagnostic. The plot renders the exact missing-burden ratio 31/50 as the presentation value 0.62 for the 1:2:9$_{\mathrm{yro}}$ / Dimonnanyro row.

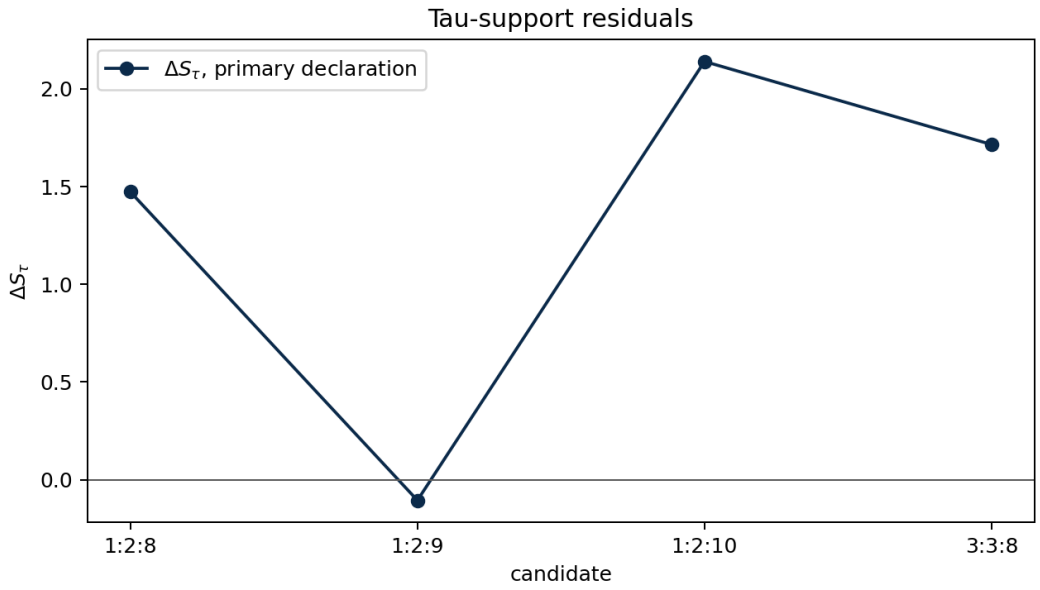


Figure 19: Tau-support residuals against the declared support-score target.

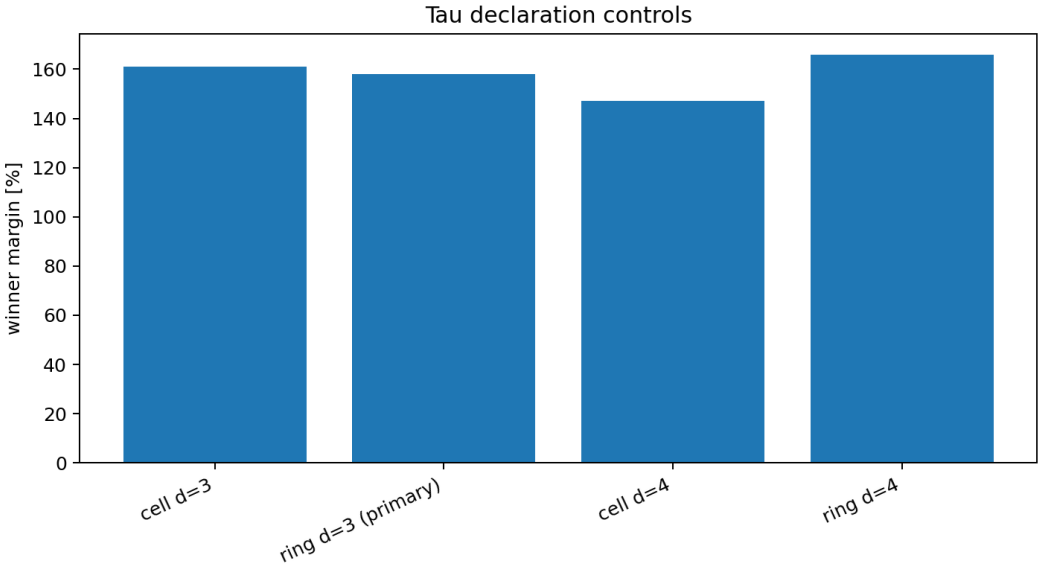


Figure 20: Tau declaration controls. Labels are declaration controls for the internal diagnostic lane.

10.9 Milky Way / Gaia follow-up declaration

The Sol bobbing record is recorded as a field dynamics diagnostic.

$$B_{\mathrm{Sol}}|t = (\partial_T R_{\mathrm{Sol}}(t), \omega_{\mathrm{Sol}}|t).$$

$$\Phi^D_{\partial_T R,\mathrm{mov}}(t) = \Phi^D_{\mathrm{cross}}(t) - \Phi^D_{\mathrm{sweep}}(t) + \Phi^D_{\mathrm{shedding}}(t).$$

A Gaia follow-up comparison is recorded after a benchmark target record is supplied.

10.10 Orbital-retention field-dynamics fixture plan

Data-support class: G2/G3 retention-first field-dynamics fixture. Its sequence is

medium setup → flip/fizz classification → range-prioritized SADAR,
capacity seating → orbital retention → δ_3 class tally,
optional U diagnostic.

The fixture declares the flip classifier inside the manual/application layer:

$$\mathrm{FlipClass}_{e,c,t} \in \{\mathrm{early_coupling}, \mathrm{fizz}, \mathrm{pending}, \mathrm{sheddic_route}\}.$$

The early-coupling indicator is

$$G^{\mathrm{early}}_{e,c,t} = \mathbf{1}[\rho^D_{\omega,e,c,t} > 0]\, \mathbf{1}[Q^D_{e,c,t} < Q_{\max}]\, C^{\theta}_{e,c,t}\, Z^{\mathrm{imp}}_{e,c,t}\, \mathbf{1}[\Delta A_{e,c,t} \neq 0].$$

When $G^{\mathrm{early}}_{e,c,t} > 0$, the fixture records early coupling. Fizz records successor motion / background-continuation jitter. Pending records retained return carried over the current window. The class sheddic_route records a flip carried through a declared sheddic path.

Orbital-retention input provenance gate. Before a scored orbital-retention run, the fixture records the data provenance of every field component used by the retained calculation. The input package records radial support rows; three-dimensional density fields; gas/stellar velocity fields; momentum fields; time-window/slosh fields; capacity rule; coupling family; sheddic-route declaration; and external value map when used. SPARC radial rows may populate the radial-support component. Three-dimensional density, velocity, current, and time-flow components are populated by their own declared source rows or proxy rows before scoring.

Table 32: Orbital-retention input provenance gate.

Field component	Source	Support status	Uncertainty status	Scored
stellar radial support	SPARC radial model	SPARC-supported radial	radial source row	diagnostic
stars, full medium	3D stellar density	external 3D	measured / proxy	if declared
gas, full medium	H I / Hα / IFU field	external 3D	channel / velocity uncertainty	yes, if declared
core / background	model or survey	latent / proxy	proxy recorded	conditional
current-cycle family	AΩD range edges	declared proxy	declaration audit	internal
time-flow / slosh	kinematic window	external 3D / latent	cadence / window	conditional
external value map	quarantined map	declared proxy	frozen before scoring	external only

Remark. A scored orbital-retention run records source, support status, uncertainty status, and whether each component is used for scoring before residuals are interpreted.

Table 33: Orbital-retention field-dynamics fixture spine.

No.	Sheet	Purpose
00	Declarations	Freeze declarations before scoring.
01	Cell grid 3D	Declare the retained 3D cell grid.
02	Input density fields	Declare star/gas/core/background/current density inputs and source/status records.
03	Fractal range / max field support	Compute fractal range and maximum supported field count.
04	Field regime	Declare gas/fluid/fizz regime.
05	Chiral flip: early coupling vs fizz	Classify flips as early coupling, fizz, pending, or sheddic route.
06	Range edge sets	Build range-prioritized coupling families.
07	SADAR coupling / pressure / velocity	Compute SADAR coupling, pressure, and motion quantities.
08	Capacity seating	Seat capacity by range and closure state.
09	Hinge shift / partial coupling	Record hinge shift and partial coupling.
10	Slosh / sheddic / pending	Record slosh, sheddic routing, and pending return.
11	Orbital retention class	Classify orbital retention before projection.
12	Delta3 retention tallies	Record exact δ_3 retention tallies.
13	U projection diagnostic	Optional square-speed diagnostic after retention scoring.
14	Audit	Check declaration, retention, projection, and error layers.

Remark. The SPARC square-speed lane is a G0 radial projection diagnostic. The orbital-retention fixture is the G2/G3 retention-first simulation path. Projection to U follows retention, capacity seating, and δ_3 tallies.

10.11 Completion requirements

A SPARC scored record is complete when the data file, declaration constants, square-speed target, residuals, and score metrics are present. A Tau ring fixture is complete when the candidate set, scores, missing fraction, and precision note are present. Display speed is a presentation view only. The orbital-retention field-dynamics fixture is complete when it records medium setup, flip/fizz classification, range-prioritized SADAR, capacity seating, retention class, and δ_3 tallies.

11 Galactic Lensing Diagnostic Plan

Scope. This section applies the current AΩD field-ring and δ_3 notation to a staged galactic-lensing diagnostic plan. It is a manual/application plan with an explicit L3 data-package gate.

Provenance. Main §5.10; Main §5.11; §5; §12.

Literature note. SWELLS target lane: Treu et al. [35], Brewer et al. [6].

11.1 Lensing lanes

Table 34: Galactic lensing diagnostic lanes.

Lane	Class	Active fields
Toy 1	L0	exact internal δ_3 fixture
SPARC5	L1	radial lens-medium data-support class; lens target fields are carried by SWELLS or L3 rows
SWELLS K0	L2	target acquisition
SWELLS K1	L1/L2	retained baseline
SWELLS K2/K3	L1/L2	audit diagnostics
K4	L3	higher-resolution lensing input-manifest row
L3 lensing input package	L3	L3 input-manifest row; required fields: independent 3D medium fields, time-flow inputs, radiation-current gates, target-side quarantine, null controls, and audit

11.2 Field-ring spine

The lensing fixture uses a discrete exoshedding field-ring calculation. The value table records the evaluated gate values.

$$B_{\mathrm{lens}} = (\partial_T F_{\mathrm{exo}}, \omega_{\mathrm{lens}}).$$

$$T_k^0 = \sum_i S_k^{(i)}.$$

$$T_k^{\times} = T_k^0 + \sum_{(i,j)\in E_{\mathrm{return}}} C_{ij,k}^{\times}(B_{\mathrm{lens}}).$$

$$T_k^{\times} - T_k^0 = \sum_{(i,j)\in E_{\mathrm{return}}} C_{ij,k}^{\times}(B_{\mathrm{lens}}).$$

$$\Delta_k^{\mathbb{Z}} = T_k^{\times} - O_k, \qquad \delta_{3,k}(T^{\times}, O) = (s_k^{(3)}, m_k), \qquad m_k = |\Delta_k^{\mathbb{Z}}|.$$

$$\delta_3(T^{\times}, O) = (\delta_{3,k})_{k\in K}.$$

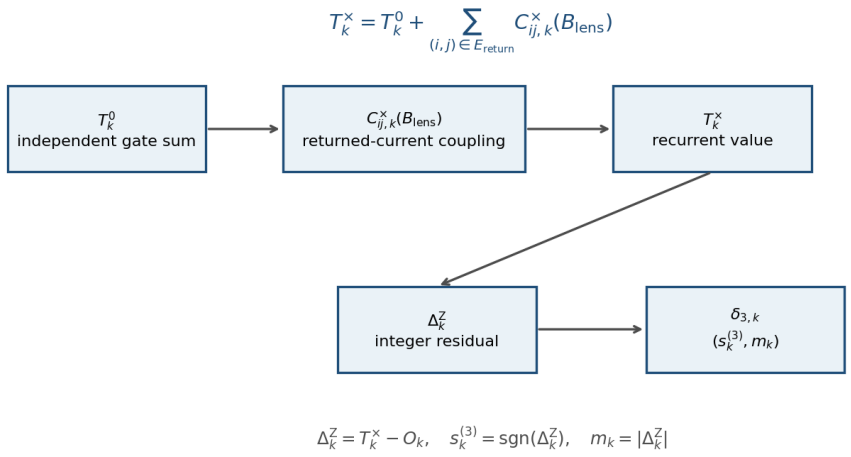


Figure 21: FRG value identity and δ_3 audit. Gate values T^0, $C^\times$, and $T^\times$ are computed before the fixture comparator joins.

11.3 Toy fixture lane

Toy 1 is the exact internal lensing toy. It belongs to the manual fixture lane and registry.

Table 35: Toy 1 exact internal δ_3 fixture summary.

Quantity	Value
gates	16
target bins	13
events	18
caustic bins	0, 2
δ_3 total m	0
checks passed	11/11

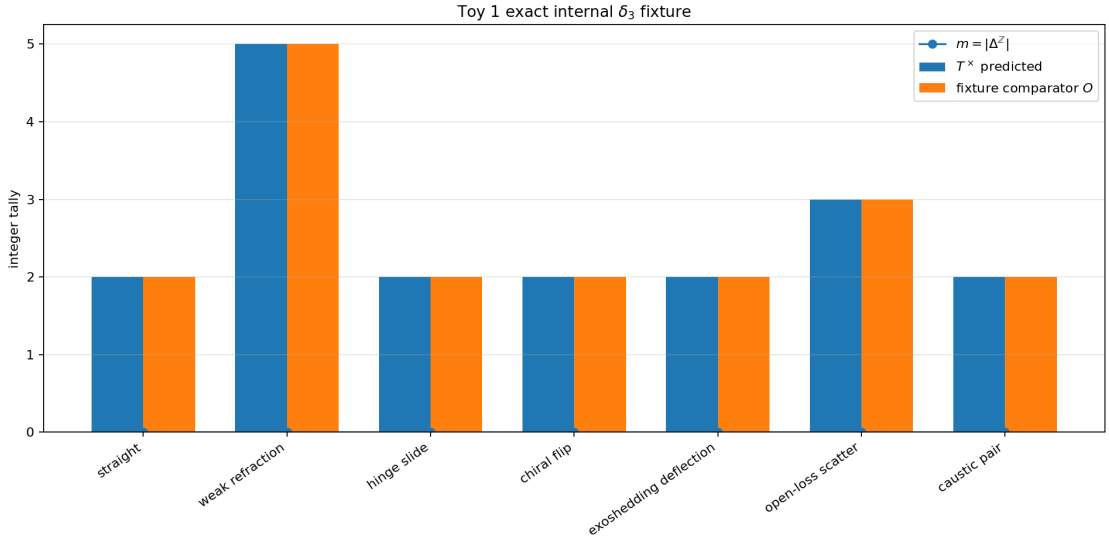


Figure 22: Toy 1 exact internal δ_3 class tallies.

11.4 SPARC5 lens-medium diagnostic

SPARC5 instantiates a radial lens-medium data-support class. It supplies radial field-medium inputs for diagnostic runs. Image-count, Einstein-radius, image-position, and shear target fields are target-side quantities carried by SWELLS or an L3 input package.

Table 36: SPARC5 lens-medium summary.

Galaxy	Bins	Gates	Events	Caustic	Median $\lvert\Delta A_\gamma\rvert$	Max $\lvert\Delta A_\gamma\rvert$	Dominant class	Status
NGC2403	73	32	44	12	1.1667	1.9	exoshedding deflection	lens-medium
NGC3198	43	32	44	12	1.2721	2.0	exoshedding deflection	lens-medium
NGC6503	31	32	44	12	1.3801	2.4706	exoshedding deflection	lens-medium
NGC2841	50	32	41	9	1.2416	1.7647	exoshedding deflection	lens-medium
DDO154	12	32	46	14	1.2941	2.625	exoshedding deflection	lens-medium

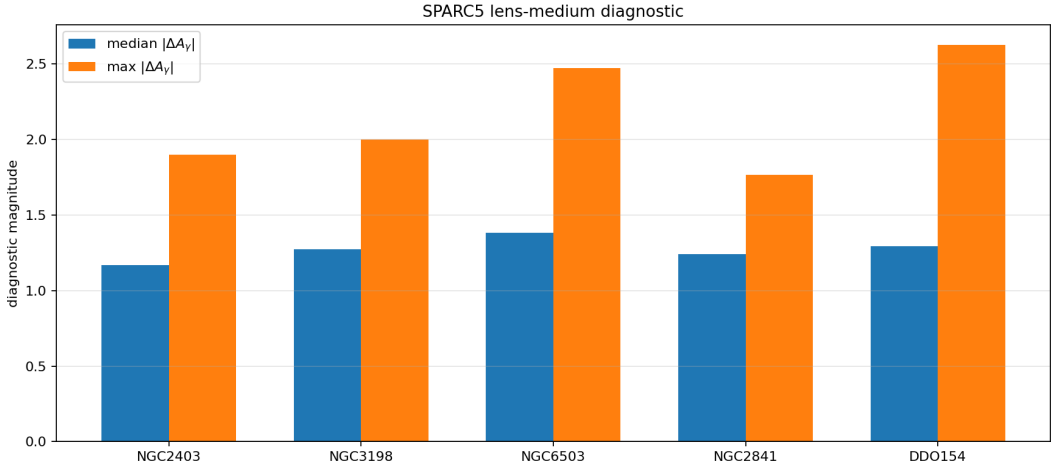


Figure 23: SPARC5 radial lens-medium data-support class. The plotted quantities are diagnostic medium proxies; lensing targets are carried by SWELLS or a higher-resolution class.

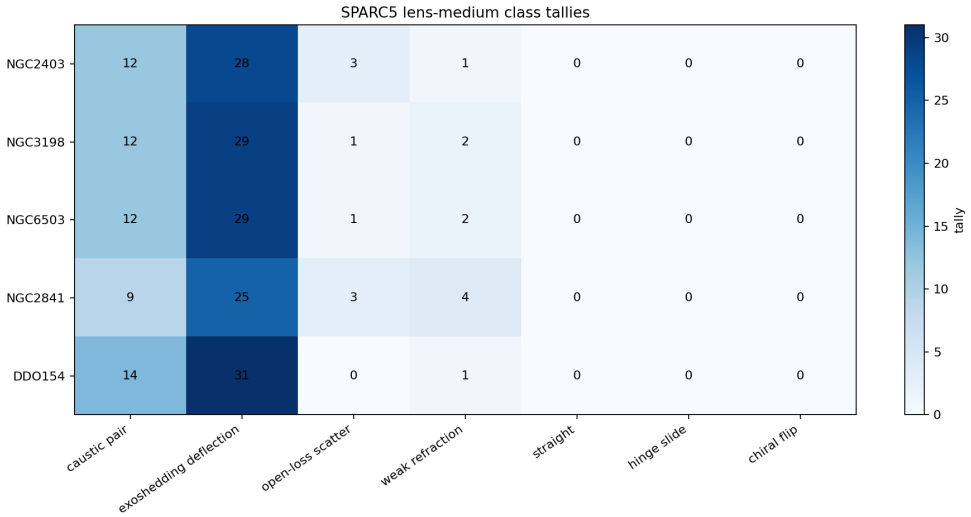


Figure 24: SPARC5 lens-medium class tallies. The dominant diagnostic route class is recorded for audit.

11.5 SWELLS target lane

SWELLS supplies target-side acquisition fields for the lensing lane.

Table 37: SWELLS K0 target-side acquisition fields.

Target family	Class k	O tally
strong-lens presence	modelled lens	20
grade	A	19
grade	B	1
Einstein radius bin	$\theta_E < 0.75$	3
Einstein radius bin	$0.75 \leq \theta_E < 1.00$	9
Einstein radius bin	$\theta_E \geq 1.00$	8
σ_{SIE} bin	$\sigma < 200$	8
σ_{SIE} bin	$200 \leq \sigma < 230$	7
σ_{SIE} bin	$\sigma \geq 230$	5

Table 38: SWELLS K1/K2/K3 diagnostic comparison.

Run	Input basis	θ_E δ_3 m_{tot}	θ_E match [%]	σ δ_3 m_{tot}	Decision
K1	σ_{SDSS} + Hα + axis ratio	8	40.0	4	retained baseline
K2	K1 + redshift geometry context	18	40.0	4	audit diagnostic
K3	Paper I Table 2 bulge/disk medium fields	10	33.3333	12	audit diagnostic

K1 reason. Lowest retained δ_3 residual.
K2 reason. Over-amplified high-θ_E class.
K3 reason. Under-resolved observable map with current medium data. K1 is retained as the baseline. K2 and K3 are audit diagnostics.

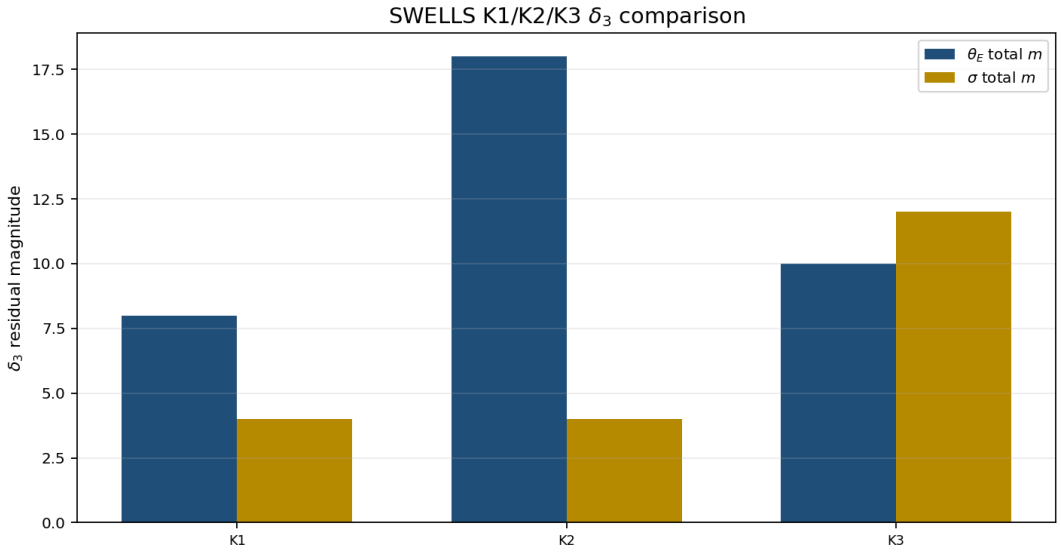


Figure 25: SWELLS K1/K2/K3 δ_3 comparison. K2/K3 are audit diagnostics with δ_3 comparison tallies.

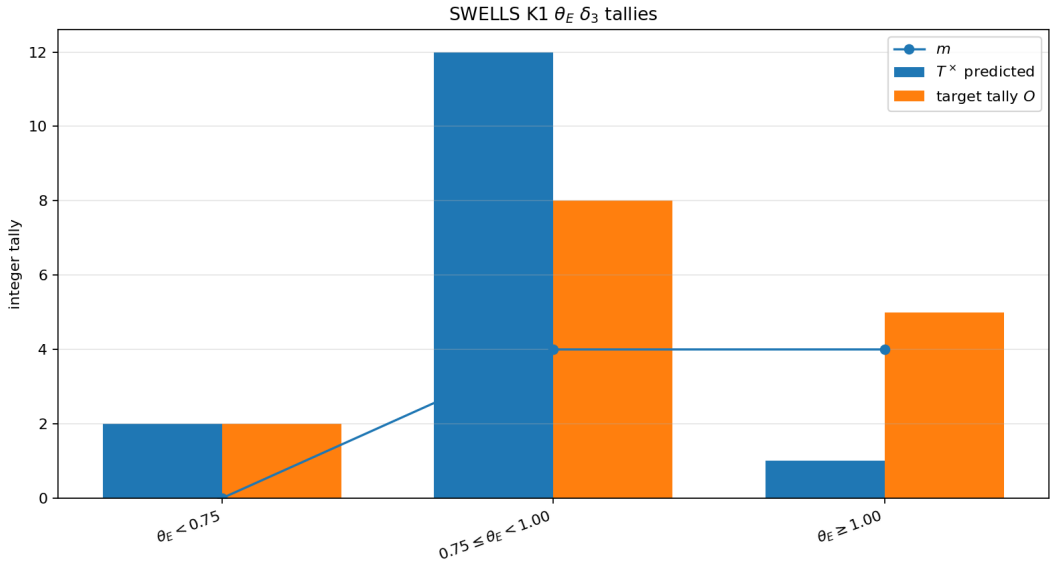


Figure 26: SWELLS K1 retained-baseline θ_E δ_3 tallies.

11.6 K4 input gate

K4 is the higher-resolution lensing input-manifest row for independent 3D and time-flow inputs.

Table 39: K4 independent 3D + time-flow input gate.

Input group	Required field	Source status	Gate
3D density fields	stellar density $\rho_\star(R, z, \theta)$	external / 3D proxy	required
3D density fields	gas density $\rho_{\rm gas}(R, z, \theta)$	external / 3D proxy	required
velocity and flow fields	gas and stellar velocity vector field	IFU / H I / Hα	required
velocity and flow fields	line-of-sight flow and bobbing/slosh phase	time / kinematic proxy	required
momentum fields	angular momentum and radial/linear momentum	independent kinematics	required
time-window state	duration/cadence windows for flow transfer	declared pre-score	required
lensing target side	image-position bins	quarantined target	required
lensing target side	Einstein-radius bins and image count classes	quarantined target	required
audit controls	null / permutation controls	declared pre-score	required

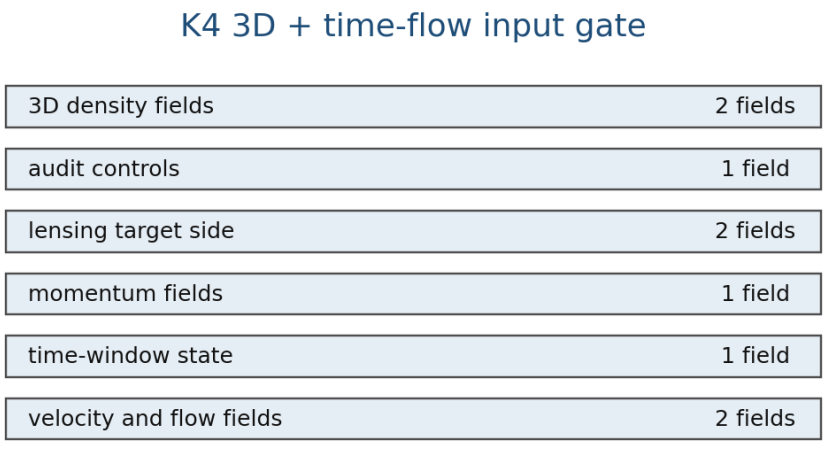


Figure 27: K4 higher-resolution lensing data-support gate. The L3 input package records independent 3D + time-flow medium data before the target side joins.

11.7 Target-side quarantine and release gate

The AΩ input side carries medium fields, gate values, and declared prediction values. The target side carries image-position, Einstein-radius, and image-count targets after prediction values freeze.

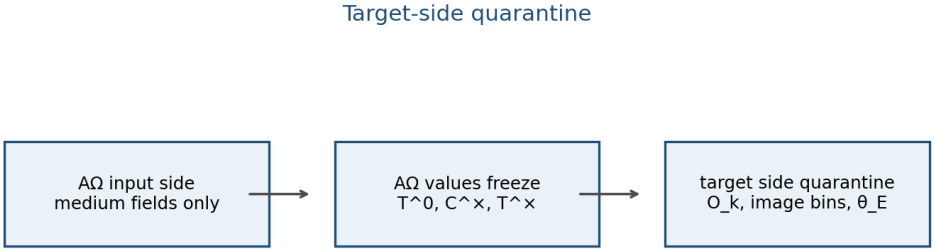


Figure 28: Target-side quarantine. AΩ values freeze before target tallies join.

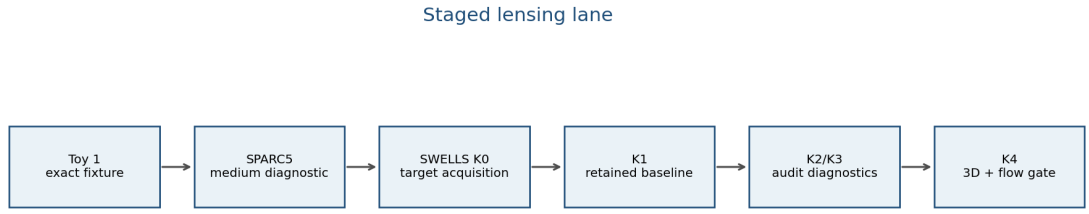


Figure 29: Staged galactic lensing lane. Toy fixtures, SPARC5 diagnostics, SWELLS acquisition, retained baseline, K2/K3 audit diagnostics, and K4 gate are kept distinct.

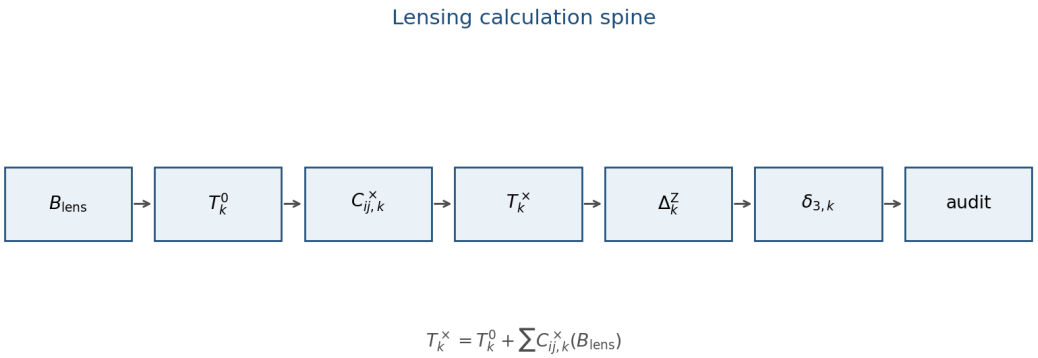


Figure 30: Galactic lensing calculation spine. The U-projection and lensing targets remain downstream of declared field-ring values.

A completed L3 lensing record carries declared medium fields, declared radiation-current gates, declared target-side quarantine, computed T^0, $C^{\times}$, and $T^{\times}$, declared λ_{lens}, δ_3 residual table, null controls, audit, and lens residuals after AΩ values freeze.

12 Prediction-Test Fixture Registry

Scope. This registry records completion fields for manual fixtures and comparisons. Rows carry their data-support class, active fields, required AΩ calculation, and audit artifact.

Table 40: Prediction-test fixture registry.

Record	Class	Active fields	Required AΩ calculation	Artifact
Hydrogen shell radii	O0	target rows	shell-ratio record for S_n	radius-ratio table
Hydrogen shell energies	O0	target rows	$\Delta\widehat{P}^D$, ratio, target	Lyman / Balmer / Paschen tables
Short-window 1:2:7	D0	integer fixture fields	$P^D, \Delta^{\mathrm{close}}, X_{\mathrm{shedding}}$	closure table
Short-window 1:2:9	D0	integer fixture fields	$P^D, C^{\mathrm{close}}, X_{\mathrm{shedding}}$	surplus table

Record	Class	Active fields	Required AΩ calculation	Artifact
Toy 1 field-ring fixture	D0	integer comparator bins	$T^0, C^\times, T^\times, O, \Delta^{\mathbb{Z}}, \delta_3$	δ_3 fixture table
Neuronal-ring benchmark acquisition	L2	acquisition fields	$\mathcal{S}_{\text{neuro}}, O_k$	acquisition declaration
Electron	O0	measured target rows	$\nu_{A\leftrightarrow\Omega}, E_{\text{ext}}$ for 1:2:6	external target declared
Muon	O0	measured target rows	$\nu_{A\leftrightarrow\Omega}, E_{\text{ext}}$ for 1:2:8	external target declared
Tau	O0/D0	measured target rows	$\nu_{A\leftrightarrow\Omega}, E_{\text{ext}}$ for $1{:}2{:}9_{\text{yro}}$ / Dimonnanyro	surplus fixture + external target declared
Proton	O0	measured target rows	$\nu_{A\leftrightarrow\Omega}, E_{\text{ext}}$ for 3:3:6	closed fixture + external target declared
Neutron	O0	measured target rows	$\nu_{A\leftrightarrow\Omega}, E_{\text{ext}}$ for 3:4:6	closed fixture + external target declared
B / B_s / B_c	O0	measured target rows	$\nu_{A\leftrightarrow\Omega}, E_{\text{ext}}$ for Ditrion records	debris fixture + external target declared
Boundary current	D0	declared boundary/current fields	$\Phi, J, P^D, \mathbf{A}_F$	substitution table
Field tunnelling hinge-slide fixture	D0	integer fixture fields	$\text{adm}(e_0; B\|t) = 0$, slide kernel, ρ_ω^D, T^D_{tunnel}	D.E.C. ledger table
3D Brownian phase-cycle fixture	G3/D0	3D coordinate track fields	$z_i \in \mathbb{Z}^3$, $\Delta z_i^{(n)}$, $Q_i^{(n)}$, phase counts, octant counts, δ_3	3D coordinate phase-cycle CSVs (Brownian application aliases), integer-ratio audit, and App. C
Run-tumble tracer-current fixture	G3/D0	mode m_t, selected channel j_t, sign σ_t, accumulated trace $h_t \in \mathbb{Z}^4$, projection $\pi_3(h_t) \in \mathbb{Z}^3$, Q^2_{3D}, Q^2_4, phase counts, octant counts, δ_3	$h_t = h_{t-1} + \sigma_t \mathbf{e}_{j_t}$, $z_t = \pi_3(h_t)$, Q^2_{3D} and Q^2_4 audit calculations, phase-cycle and octant δ_3 tallies	run-tumble trace CSV, phase-cycle δ_3, octant δ_3, integer-ratio audit, generated trace table, and App. C
Mercury perihelion	O0	observable-map target rows	observable-map value, target, error	solar table
Solar light bending	O0	observable-map target rows	observable-map value, target, error	solar table
Solar redshift	O0	observable-map target rows	observable-map value, target, error	solar table
Earth–Moon orbital shell	O0	target data manifest	Moon distance, speed, period records require numeric AΩ values	NASA target record
Saturn ring-shell	O0	target data manifest	ring-boundary and Cassini-division records require numeric AΩ values	NASA / ring-boundary target record
SPARC five-galaxy square-speed diagnostic	G0	SPARC radial rows	$U_{A\leftrightarrow\Omega}(r)$, ΔU, z^U, χ^2_U, MAE_U, MSE_U, $MAPE_U$	per-bin CSV, per-galaxy CSV, score table, and figures
SPARC full-sample / extended benchmark	G0/G1	target data manifest	full-sample $U_{A\leftrightarrow\Omega}(r)$, ΔU, z^U, χ^2_U, MAE_U, MSE_U, $MAPE_U$	extended benchmark package
Orbital-retention field-dynamics fixture	G2/G3	G2/G3 input package	medium setup, input-provenance gate, flip/fizz classifier, range SADAR, capacity seating, retention class, δ_3 tallies	workbook spine and provenance tables
Toy 1 lensing field-ring fixture	L0	integer comparator bins	$T^0, C^\times, T^\times, O, \Delta^{\mathbb{Z}}, \delta_3$	manual galactic lensing plan
SPARC5 lens-medium diagnostic	L1	radial lens-medium fields	radial field-medium class tallies	plan data and figures
SWELLS K0 target acquisition	L2	target acquisition fields	target tallies before prediction join	target-side quarantine table
SWELLS K1 retained baseline	L1/L2	baseline target fields	δ_3 comparison tallies	K1 diagnostic table
SWELLS K2/K3	L1/L2	audit diagnostics	δ_3 comparison tallies	K2/K3 audit table
K4 galactic lensing gate	L3	higher-resolution input manifest	independent medium data and target quarantine	K4 input-gate table
L3 lensing input package	L3	L3 input manifest	independent 3D medium fields, time-flow inputs, radiation-current gates, target-side quarantine, null controls	lens residuals after AΩ values freeze

Record	Class	Active fields	Required AΩ calculation	Artifact
Tritrioseptyro Higgs-support fixture	D0/O0	AFC trace rows + route detector	candidate: $3{:}3{:}7_{\mathrm{yro}}$; frozen map: $RD\,2^{L+4}$; mapped mass: 124.928 GeV; comparison layer: declared Higgs mass targets	next layer: production-channel and decay-channel residual tables

Data-support registry note. Registry rows are read together with data-support class. A row carries active fields, an AΩ calculation or report map, and an artifact. Residuals are scoped to the declared class.

Literature note. Earth–Moon orbital target: NASA [17]; Saturn ring/Cassini target: NASA [18]; SPARC five-galaxy diagnostic: SPARC database and Lelli et al. [33, 14].

12.1 Neuronal-ring benchmark acquisition

This registry subsection supplies a measured acquisition declaration for a field-ring benchmark record. For a biological benchmark acquisition, declare

$$\mathcal{S}_{\mathrm{neuro}} = (\mathcal{M}, \Delta t, \mathcal{B}_{\mathrm{spike}}, \kappa_k, \Theta_{\mathrm{stim}}, \Omega_{\mathrm{meas}}).$$

Here

$$\mathcal{M} \in \{\text{spike counts},\ \text{calcium bins},\ \text{voltage bins},\ \text{LFP bins}\}$$

is the measurement mode, Δt is the bin width, $\mathcal{B}_{\mathrm{spike}}$ is the bin extraction rule, Θ_{stim} is the gate-stimulus declaration, and Ω_{meas} is the measurement window. The fixture comparator is

$$O_k = \mathcal{S}_{\mathrm{neuro}}(k; \Omega_{\mathrm{meas}}).$$

Artifact. $\mathcal{S}_{\mathrm{neuro}}, T_k^0, C_{ab,k}^{\times}(B), T_k^{\times}, O_k, \Delta_k^{\mathbb{Z}}, \delta_{3,k}$. If O_k is continuous or averaged, write $O_k = x_k \pm \sigma_k$ and treat the comparison as a measured acquisition value map with a decimal precision record.

Completion rule. The neuronal-ring benchmark record is complete when benchmark acquisition is declared before bin comparison, the independent gate-sum value is computed as $T_k^0 = S_k^{(a)} + S_k^{(b)}$, and the cross-return row records a declared $C_{ab,k}^{\times}(B)$.

Remark. This subsection is a benchmark-acquisition record. The AΩ calculation is the recurrent ring value; the measured acquisition declaration is supplied before comparison.

12.2 SPARC benchmark declaration

The SPARC five-galaxy diagnostic is carried as a current field-dynamics square-speed diagnostic lane. The full protocol and scored records are given in §10.

$$\mathcal{D}_{\mathrm{SPARC}} = \{G_i, r_{ij}, V_{ij}^{\mathrm{obs}}, \sigma_{V,ij}, V_{ij}^{\mathrm{gas}}, V_{ij}^{\mathrm{disk}}, V_{ij}^{\mathrm{bulge}}, L_{3.6,i}, \ldots\}.$$

The active comparison coordinate is square speed:

$$U_{ij}^{\mathrm{obs}} = (V_{ij}^{\mathrm{obs}})^2, \qquad U_{ij,A\leftrightarrow\Omega} = \mathfrak{S}_{B_i}^{U}(C_{i,j}).$$

For each radial bin, the completed record carries

$$U_{ij,A\leftrightarrow\Omega}, \qquad \Delta U_{ij} = U_{ij,A\leftrightarrow\Omega} - U_{ij}^{\mathrm{obs}}, \qquad z_{ij}^{U} = \Delta U_{ij}/\sigma_{U,ij}.$$

For each galaxy, the completed record carries

$$\chi_U^2, \qquad \overline{\Delta U}, \qquad MAE_U, \qquad MSE_U, \qquad MAPE_U.$$

Literature note. SPARC benchmark: SPARC database [33], Lelli et al. [14].

Completion rule. External-comparison rows record numeric map value, target, residual, percent error, and audit hook. Observable-map fixture rows record map value, target, residual, percent error, citation, fixture status, and audit hook. Benchmark-dataset rows record benchmark data table, AΩ calculation row, report coordinate, residual, and audit hook. Cross-return rows record declared $C^{\times}_{ab,k}(B)$.

Registry note. Decimal precision records accompany manual artifacts that use decimal target/value-map values.

A Simulation and Prediction-Test Fixtures

A.1 Scope

This manual appendix preserves simulation fixtures and prediction-test placement context downstream from the main note.

A.2 Statement(s)

Required outputs are

$$RD, \qquad \rho^D, \qquad \rho^D_\omega, \qquad Q^D, \qquad p^D, \qquad \phi, \qquad P^D_F(B), \qquad \mathbf{A}_F(B), \qquad \mathcal{E}(R,B).$$

Fixtures include

$$1{:}2{:}6, \quad 1{:}2{:}7, \quad 1{:}2{:}8, \quad 1{:}2{:}9, \quad 1{:}2{:}10, \quad 3{:}3{:}6, \quad 3{:}4{:}6,$$
$$3{:}4_\alpha, \qquad 2 \times 3{:}4_\alpha, \qquad 3{:}4_\alpha + 3{:}4_\omega.$$

Separate-scope fixtures include $3{:}4_\alpha$ and $3{:}4_\omega$. The fixture set also includes short-window duration-clipping fixture, pending overhang, current split, and AFC Stokes residual.

The nine-window surplus assertion is

$$1{:}2{:}9: \qquad P^D = 27, \qquad C^{\mathrm{close}} = 24, \qquad X_{\mathrm{shedding}} = 3.$$

The 1:2:7 fixture has $P^D = 21$ and closure-dependent sign.

A.3 Four-step arithmetic fixtures

Specializing Main App. I.2 gives

$$6 = 4(1) + 2, \qquad 7 = 4(1) + 3, \qquad 8 = 4(2) + 0,$$
$$9 = 4(2) + 1, \qquad 10 = 4(2) + 2.$$

A.4 Demonstration

The model is Main App. D.3. The retained `wavelet_shedding_simulation.csv` fixture supplies 90 ticks. Specialize the model to retention coefficient $\eta_{\mathrm{ret}} = 41/50$, span gain $\gamma_{\mathrm{span}} = 11/20$, reclosure fraction $\lambda_{\mathrm{reclose}} = 19/50$, and initial burden $P_{-1} = 0$. The generator `manual/scripts/generate_cycle_shedding_fixture.py` validates the stored recurrences and summary before plotting the supplied lead and lag streams. All stored states are open; the local fraction R_t does not certify a return.

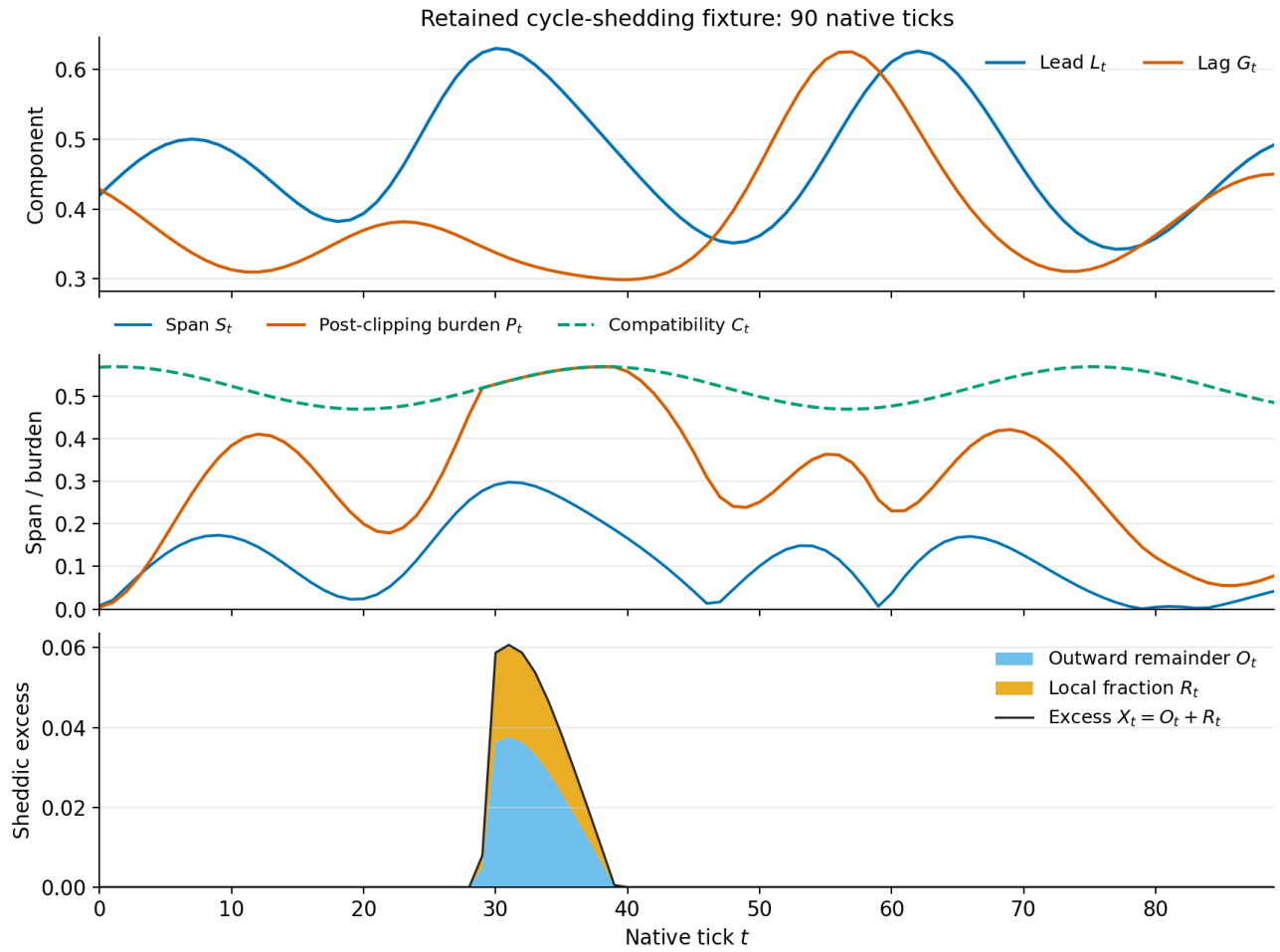


Figure 31: Retained temporal-cycle fixture. The burden P_t is post-clipping, with $P_t \leq C_t$; the stacked outward and local fractions sum to X_t. Figure and summary use the same CSV. The summary label "Total exoshedding" denotes $\sum_t O_t$ in this fixture.

Ticks	90
Maximum lead-lag span	0.298
Maximum temporal SADAR burden	0.570
Maximum sheddic excess	0.061
Total exoshedding	0.238
Total local reclosure	0.146
Sheddic path trigger ticks	0
First reclosure tick	none

A.5 Falsification

If the computed values violate the stated recurrences on the declared ticks, the demonstration fails on that run.

A.6 Data-support class reference

Manual fixtures and comparisons are read together with their declared data-support class. The compact class list is:

$D0$	exact internal fixture,
$O0$	observable-map fixture,
$G0$	radial shell projection,
$G1$	radial-medium projection,
$G2$	three-dimensional field-dynamics retention,
$G3$	time-resolved tracer-current simulation,
$L0$	exact lensing toy fixture,
$L1$	radial lens-medium diagnostic,
$L2$	target-acquisition lane,
$L3$	full lensing release class.

Errors and residuals are scoped to the class: δ_3 for exact internal comparators, observable-map residuals for O0, radial square-speed residuals for G0/G1, retention and field residuals for G2/G3, and target-side residuals for L2/L3 after prediction values freeze.

A.7 Audit

Fixture audit records signed direction, scalar pressure, scope cancellation inside its declared boundary/window, and $\mathcal{E}(R, B)$ for a declared Stokes-valid scope.

Provenance. Main (79); Main (98); Main (154); Main App. G; Main App. H.

A.8 Prediction-test placement

Prediction tests and any collision-window comparison context belong in this manual scaffold. Application labels and external comparison records are carried by this manual scaffold. The main AΩD note supplies the AF/AFC/AΩD calculus and internal form names.

B Quarantined External Comparison

B.1 Scope

This appendix compares AΩ outputs against external comparison coordinates. It does not import external units, constants, or symbols into the AΩ calculation. The active calculation produces a candidate such as

$$\nu_{x,A\leftrightarrow\Omega}.$$

Only after that value is declared may an external comparison record use it:

$$\mathsf{Q}_{\mathrm{ext}}(\nu_{x,A\leftrightarrow\Omega}; \mathcal{T}_x).$$

B.2 Presentation-numeric policy

AΩ-internal arithmetic remains integer/rational. The decimals in this appendix are external comparison presentation values. Measured external values are carried with a declared uncertainty; exact defining constants are marked as exact. The external normalizer η_{ext} is measured, so uncertainty propagation for normalized rates may include both the target-energy uncertainty and the uncertainty of η_{ext}.

B.3 External comparison symbols

External constants and measured-sector units are named only inside this quarantine. The active AΩ notation is unaffected by them.

B.3.1 Distance comparison

If a bip duration Δt_{bip} is declared, an external distance coordinate may be recorded as

$$d_{\mathrm{ext}} = c\, n\, \Delta t_{\mathrm{bip}}, \qquad n \in \mathbb{Z}_{\geq 0}.$$

Here c is an external defining constant in the comparison coordinate system.

B.3.2 Speed comparison

The active calculation may carry the exact integer pair

$$(\Delta q, \Delta n), \qquad 0 \leq \Delta q \leq \Delta n, \qquad \Delta n > 0.$$

The external speed coordinate is allowed only after the integer pair is declared:

$$\beta_{\mathrm{ext}} = \frac{\Delta q}{\Delta n}, \qquad v_{\mathrm{ext}} = \beta_{\mathrm{ext}} c.$$

B.3.3 Direction comparison

A direction comparison may record cycle count

$$N_\theta = \frac{\Theta}{2\pi}, \qquad 2\pi \mapsto 1 \text{ cycle}, \qquad 4\pi \mapsto 2 \text{ cycles},$$

with an orientation sign

$$s_\theta \in \{-1, +1\}.$$

B.3.4 Charge comparison

The active calculation carries integer orientation. The external charge coordinate may be recorded as

$$Q_{\mathrm{ext}} = ze, \qquad z \in \mathbb{Z},$$

where e is an external defining constant in the comparison coordinate system.

B.4 Choice: external rate normalizer

Define the quarantined external rate normalizer

$$\eta_{\mathrm{ext}} := \alpha_{\mathrm{fs}}^{-1}.$$

Use

$$\alpha_{\mathrm{fs}}^{-1} = 137.035999177(21)$$

inside this appendix only. This symbol is external and does not set active AΩ notation.

Define the external comparison coefficient

$$K_{\mathrm{ext}} = \frac{h_{\mathrm{ext}} \eta_{\mathrm{ext}}}{10^6 e} = 5.6673535511200 \times 10^{-19}\ \mathrm{MeV\,s}.$$

The uncertainty inherited from η_{ext} is

$$\sigma_{K,\eta} = 8.68490216354 \times 10^{-29}\ \text{MeV s}.$$

Here h_{ext} and e are external exact defining constants, while η_{ext} is measured. The active AΩ calculation does not use them.

Given a numeric AΩ candidate $\nu_{x,A\leftrightarrow\Omega}$, the external energy-coordinate comparison is

$$E_{x,\text{ext}}[\text{MeV}] = K_{\text{ext}}\nu_{x,A\leftrightarrow\Omega}.$$

B.5 Tau target record

The measured-comparison record in §6.6, Table 14, uses the measured target

$$E_\tau^{\text{tar}} = 1776.86\ \text{MeV}, \qquad \sigma_E = 0.12\ \text{MeV}.$$

The target in the external joule coordinate is

$$E_{\tau,J}^{\text{tar}} = 1776.86 \times 10^6 e = 2.84684357388924 \times 10^{-10}\ \text{J}.$$

The external frequency coordinate is

$$f_\tau = \frac{E_{\tau,J}^{\text{tar}}}{h_{\text{ext}}} = 4.2964283646910 \times 10^{23}\ \text{s}^{-1}.$$

The quarantined external target rate is

$$\nu_{\tau,\text{tar}} = \frac{f_\tau}{\eta_{\text{ext}}} = 3.1352552544544 \times 10^{21}\ \text{s}^{-1}.$$

The target-energy contribution to the external target-rate uncertainty is

$$\sigma_{\nu,E} = \frac{0.12 \times 10^6 e}{h_{\text{ext}}\eta_{\text{ext}}} = 2.11739039955 \times 10^{17}\ \text{s}^{-1}.$$

The contribution from the measured external normalizer is

$$\sigma_{\nu,\eta} = \nu_{\tau,\text{tar}}\frac{0.000000021}{137.035999177} = 4.80460322389 \times 10^{11}\ \text{s}^{-1}.$$

The combined external target-rate uncertainty is

$$\sigma_{\nu,\text{comb}} = \sqrt{\sigma_{\nu,E}^2 + \sigma_{\nu,\eta}^2} = 2.11739039956 \times 10^{17}\ \text{s}^{-1}.$$

B.6 Audit

F1. External-comparison audit records the active $\nu_{x,A\leftrightarrow\Omega}$ candidate before the external comparison.

F2. Speed-comparison audit records integer Δq, integer positive Δn, and $\Delta q \leq \Delta n$.

F3. Direction-cycle audit records direction relative to $2\pi = 1$ cycle.

F4. Charge-comparison audit records integer orientation before the external charge comparison.

F5. Quarantine audit records external constants, units, and measured-sector symbols only after the active AΩ calculation.

Provenance. Main (79); Main (98); §1.7; §1.8.

Literature note. SI defining constants: BIPM [7], inverse fine-structure constant: NIST CODATA [21], tau target: Zhang/PDG [38], DESY/Belle [9].

Remark. The quarantined external-comparison appendix compares against AΩ outputs. It does not import external constants or symbols into the AΩ calculation.

Table 41: Tau target record in the quarantined external-comparison coordinate.

x	def(x)	val(x)	$u(x)$
E_τ^{tar}	declared measured target	1776.86	MeV
σ_E	declared uncertainty	0.12	MeV
$E_{\tau,J}^{\mathrm{tar}}$	$1776.86 \times 10^6 e$	$2.84684357388924 \times 10^{-10}$	J
f_τ	$E_{\tau,J}^{\mathrm{tar}}/h_{\mathrm{ext}}$	$4.2964283646910 \times 10^{23}$	s^{-1}
η_{ext}	$\alpha_{\mathrm{fs}}^{-1}$	137.035999177(21)	1
$\nu_{\tau,\mathrm{tar}}$	$f_\tau/\eta_{\mathrm{ext}}$	$3.1352552544544 \times 10^{21}$	s^{-1}
$\sigma_{\nu,E}$	energy uncertainty contribution	$2.11739039955 \times 10^{17}$	s^{-1}
$\sigma_{\nu,\eta}$	normalizer uncertainty contribution	$4.80460322389 \times 10^{11}$	s^{-1}
$\sigma_{\nu,\mathrm{comb}}$	combined uncertainty	$2.11739039956 \times 10^{17}$	s^{-1}
K_{ext}	$h_{\mathrm{ext}}\eta_{\mathrm{ext}}/(10^6 e)$	$5.6673535511200 \times 10^{-19}$	MeV s
$\sigma_{K,\eta}$	normalizer contribution to K_{ext}	$8.68490216354 \times 10^{-29}$	MeV s

C 3D coordinate phase-cycle δ_3 fixture specification

This appendix records the active integer 3D coordinate fixture used by time-ordered trajectory records. It gives the coordinate-level specification: integerized trajectory bins, squared displacement counts, phase-cycle counts, octant counts, integer ratios, and exact δ_3 residuals. Continuous-motion summaries are downstream presentation maps and do not enter the exact fixture. The run-tumble tracer-current fixture in §10.5.1 is an application of this coordinate phase-cycle specification after its mode, accumulated trace, and projection declarations are fixed.

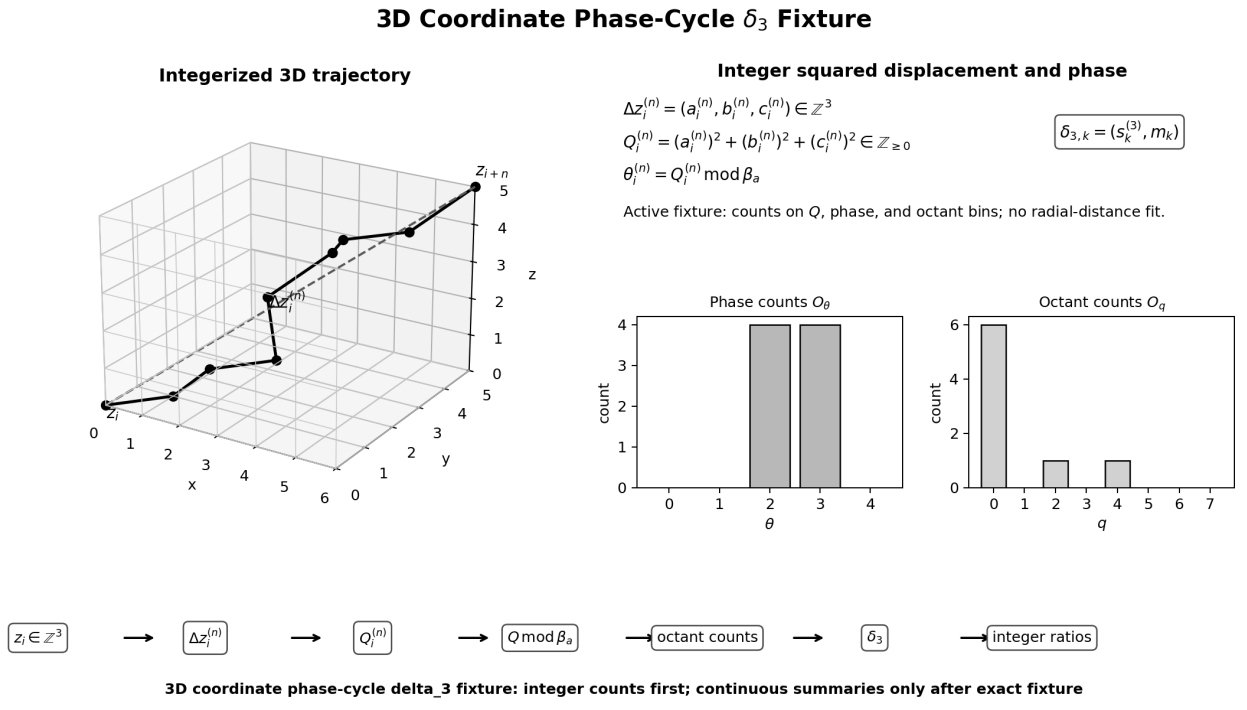


Figure 32: 3D coordinate phase-cycle δ_3 fixture. The coordinate fixture maps trajectory positions into $\mathbb{Z}^3$, forms integer squared displacement $Q = a^2 + b^2 + c^2$, records phase residues $Q \bmod \beta_a$, counts octants, and audits integer residuals by δ_3. Continuous summaries belong after the exact fixture as presentation or external maps.

C.1 Active data form

Declare a grid

$$G = (h_x, h_y, h_z, \Delta t).$$

Map measured positions into integer lattice coordinates

$$z_i = (X_i, Y_i, Z_i) \in \mathbb{Z}^3.$$

For lag n, write

$$\Delta z_i^{(n)} = z_{i+n} - z_i = (a_i^{(n)}, b_i^{(n)}, c_i^{(n)}).$$

The active integer displacement is

$$Q_i^{(n)} = (a_i^{(n)})^2 + (b_i^{(n)})^2 + (c_i^{(n)})^2 \in \mathbb{Z}_{\geq 0}.$$

This replaces radial-distance fitting in the fixture layer.

C.2 Integer displacement counts

Declare integer bins

$$I_\ell = [q_\ell, q_{\ell+1}) \cap \mathbb{Z}.$$

The observed displacement-bin count is

$$O_\ell^{(n)} = \#\{i : Q_i^{(n)} \in I_\ell\}.$$

The comparator count is an integer family

$$\widehat{O}_\ell^{(n)} \in \mathbb{Z}_{\geq 0}, \qquad \sum_\ell \widehat{O}_\ell^{(n)} = \sum_\ell O_\ell^{(n)}.$$

Record exact residuals by

$$\Delta_\ell^{\mathbb{Z},(n)} = \widehat{O}_\ell^{(n)} - O_\ell^{(n)}, \qquad \delta_{3,\ell}^{(n)} = \left(s_\ell^{(3),(n)}, m_\ell^{(n)}\right),$$

where

$$s_\ell^{(3),(n)} = \operatorname{sgn}(\Delta_\ell^{\mathbb{Z},(n)}), \qquad m_\ell^{(n)} = |\Delta_\ell^{\mathbb{Z},(n)}|.$$

The conservation check and total exact mismatch are

$$\sum_\ell \Delta_\ell^{\mathbb{Z},(n)} = 0, \qquad M_{\delta_3}^{(n)} = \sum_\ell m_\ell^{(n)}.$$

C.3 Phase-cycle fixture

Let the candidate monon-beat scale from mass/size be

$$\beta_a \in \mathbb{Z}_{>0}.$$

Define the phase residue

$$\theta_i^{(n)} = Q_i^{(n)} \bmod \beta_a.$$

The phase-cycle count is

$$O_\theta^{(n)} = \#\{i : \theta_i^{(n)} = \theta\}, \qquad \theta \in \{0, 1, \ldots, \beta_a - 1\}.$$

Against an integer comparator $\widehat{O}_\theta^{(n)}$, record

$$\Delta_\theta^{\mathbb{Z},(n)} = \widehat{O}_\theta^{(n)} - O_\theta^{(n)}, \qquad \delta_{3,\theta}^{(n)} = \left(\operatorname{sgn}(\Delta_\theta^{\mathbb{Z},(n)}), |\Delta_\theta^{\mathbb{Z},(n)}|\right).$$

The phase-cycle fixture is read as: mass/size data may declare β_a, and the trajectory tests the phase-cycle counts.

C.4 Octant direction counts

For each displacement vector, declare

$$q_i^{(n)} = \text{octant}(a_i^{(n)}, b_i^{(n)}, c_i^{(n)}) \in \{0, \dots, 7\}.$$

The octant count is

$$O_q^{(n)} = \#\{i : q_i^{(n)} = q\}.$$

The octant-neutral comparator is an integer family

$$\widehat{O}_q^{(n)} = \text{IntBalance}(N_n, 8), \qquad \sum_q \widehat{O}_q^{(n)} = \sum_q O_q^{(n)}.$$

Then

$$\delta_{3,q}^{(n)} = \left(\text{sgn}(\widehat{O}_q^{(n)} - O_q^{(n)}), |\widehat{O}_q^{(n)} - O_q^{(n)}|\right).$$

C.5 Integer-ratio motion summaries

Use integer ratios:

$$R_Q^{(n)} = \left(\sum_i Q_i^{(n)} : N_n\right).$$

Axis ratios are

$$R_x^{(n)} = \left(\sum_i (a_i^{(n)})^2 : \sum_i Q_i^{(n)}\right),$$

$$R_y^{(n)} = \left(\sum_i (b_i^{(n)})^2 : \sum_i Q_i^{(n)}\right),$$

$$R_z^{(n)} = \left(\sum_i (c_i^{(n)})^2 : \sum_i Q_i^{(n)}\right).$$

A three-axis balanced motion fixture records $R_x^{(n)} : R_y^{(n)} : R_z^{(n)}$ against the declared integer target. Phase and octant ratios are

$$R_\theta^{(n)} = (O_0^{(n)} : O_1^{(n)} : \cdots : O_{\beta_a - 1}^{(n)}), \qquad R_{\text{oct}}^{(n)} = (O_0^{(n)} : O_1^{(n)} : \cdots : O_7^{(n)}).$$

C.6 Comparator choices

Use integer comparators. The phase-neutral comparator is

$$\widehat{O}_\theta^{(n)} = \text{IntBalance}(N_n, \beta_a),$$

which distributes N_n counts across β_a phase classes by a declared integer balancing rule. The octant-neutral comparator is $\widehat{O}_q^{(n)} = \text{IntBalance}(N_n, 8)$. Shuffle controls may be declared by time-shuffled increments, sign-shuffled increments, axis-permuted increments, or phase-rotated increments. Each control produces its own δ_3 and M_{δ_3}.

C.7 AFC/AΩD mapping

The active coordinate-motion boundary is

$$B_{\mathrm{mot}} = (\partial_T F_{\mathrm{track}}, \omega_{\mathrm{track}}).$$

The exact fixture path is

$$z_i \in \mathbb{Z}^3 \to \Delta z_i^{(n)} \to Q_i^{(n)} \to \theta_i^{(n)} \to O_\theta^{(n)} \to \delta_3.$$

The retained motion trace is

$$\Delta z_i^{(n)} \to HT(h_i; B_{\mathrm{mot}}) \to RD_{h_i} \to RCD_{h_i} \to \rho^D_{\omega,h_i} \to \mathfrak{S}_B.$$

Moving-boundary current remains

$$\Phi^D_{\partial_T R,\mathrm{mov}}(t) = \Phi^D_{\mathrm{cross}}(t) - \Phi^D_{\mathrm{sweep}}(t) + \Phi^D_{\mathrm{shedding}}(t).$$

C.8 Script target and outputs

The generic coordinate script target is

`aod_3d_coordinate_phase_delta3.py`.

Inputs are

`track_id`, `t`, `x`, `y`, `z`.

Optional fields are

`ux`, `uy`, `uz`, `mass`, `size`, `beta`.

Output files are

`displacement_q_counts.csv`,
`phase_cycle_delta3.csv`,
`octant_delta3.csv`,
`integer_motion_ratios.csv`.

C.9 Final lock

The 3D coordinate phase-cycle fixture uses integer phase-cycle ratios. It uses $Q = a^2 + b^2 + c^2$, phase residues Q mod β_a, octant counts for three-dimensional direction, and δ_3 on integer counts. External continuous-motion summaries live after the exact fixture.

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